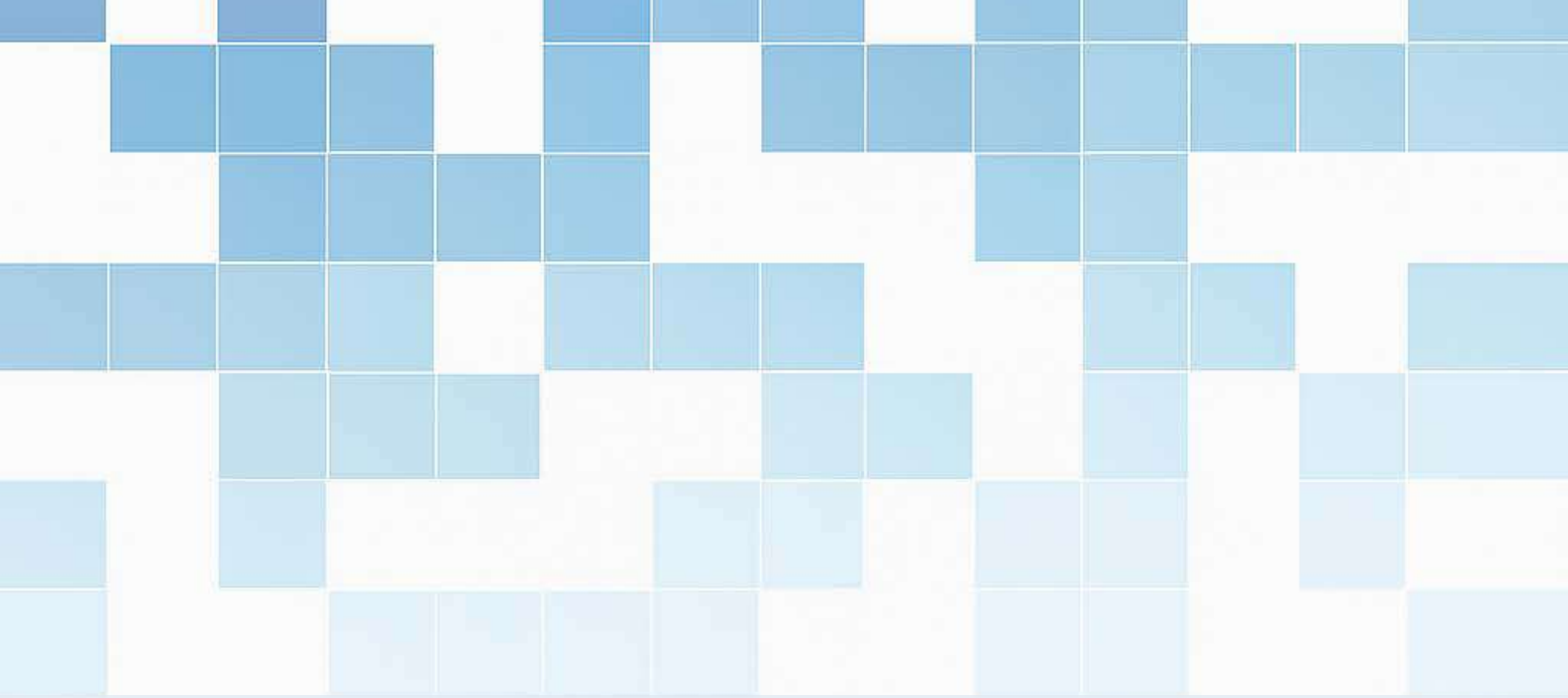




ASTRAL TRAILS



Astral Trails

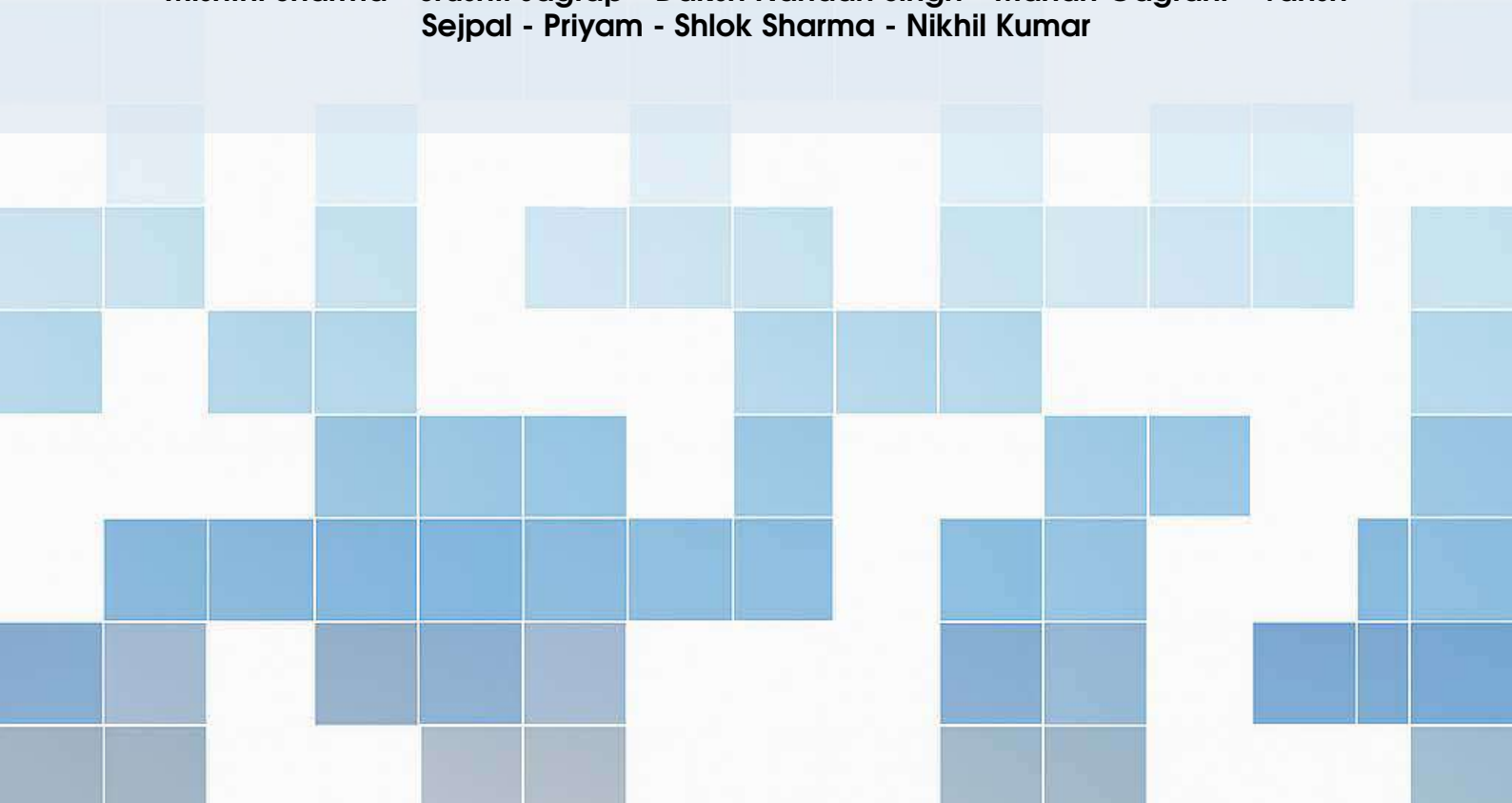
Project Report

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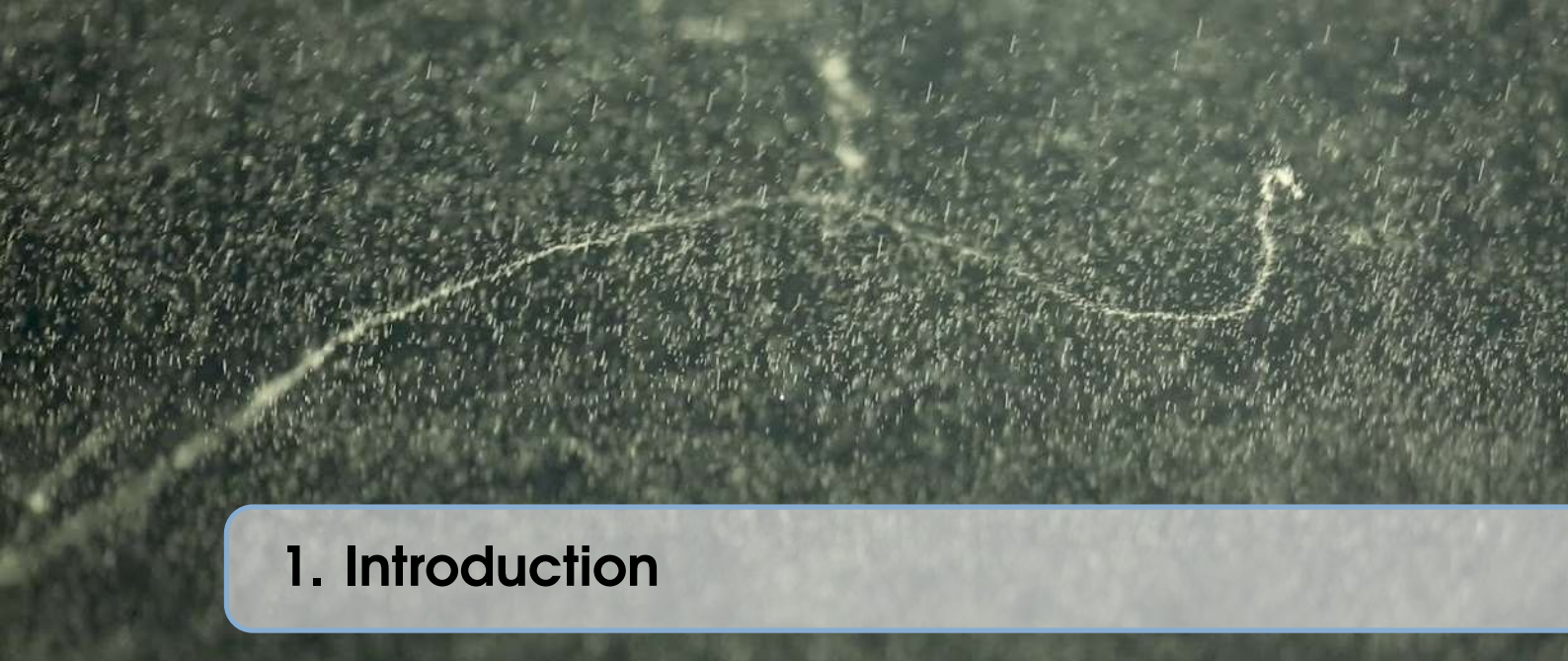
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1. Introduction

1.1 Motivation

The motivation for the Astral Trails project stems from the need to analyze the interactions and potential biological effects of cosmic rays. This is especially relevant given the growing accessibility of space travel and the advent of long-term human missions beyond Earth's protective atmosphere. Unlike Earth, spacecraft and planetary bases offer limited shielding against high-energy cosmic radiation, making it a significant concern for astronaut health and mission safety.

Understanding cosmic rays is essential not only for mitigating radiation risks in space but also for advancing our knowledge of high-energy astrophysical phenomena. Moreover, studying cosmic radiation can inform the development of better shielding materials, early-warning systems for solar particle events, and models of radiation exposure. By constructing a cloud chamber to visualize these particles, we aim to contribute to the broader effort of preparing for the challenges posed by space environments.

1.2 Objectives

The primary objectives of the Astral Trails Project are as follows:

- **Understanding Cosmic Rays:** Learn about the origin of cosmic rays, their constituent particles, and their effects on DNA and Computers.
- **Cloud Chamber Design:** Develop a simple yet accurate model of a Diffusion Cloud Chamber to make visible trails.
- **Computer Vision:** Analyse the tracks using Computer Vision (OpenCV) to automatically classify cosmic rays based on their constituent particles.
- **DNA Mutation:** Model the effects of Cosmic Rays on DNA to learn more about the impact of long-term space missions on the human body.

1.3 Scope

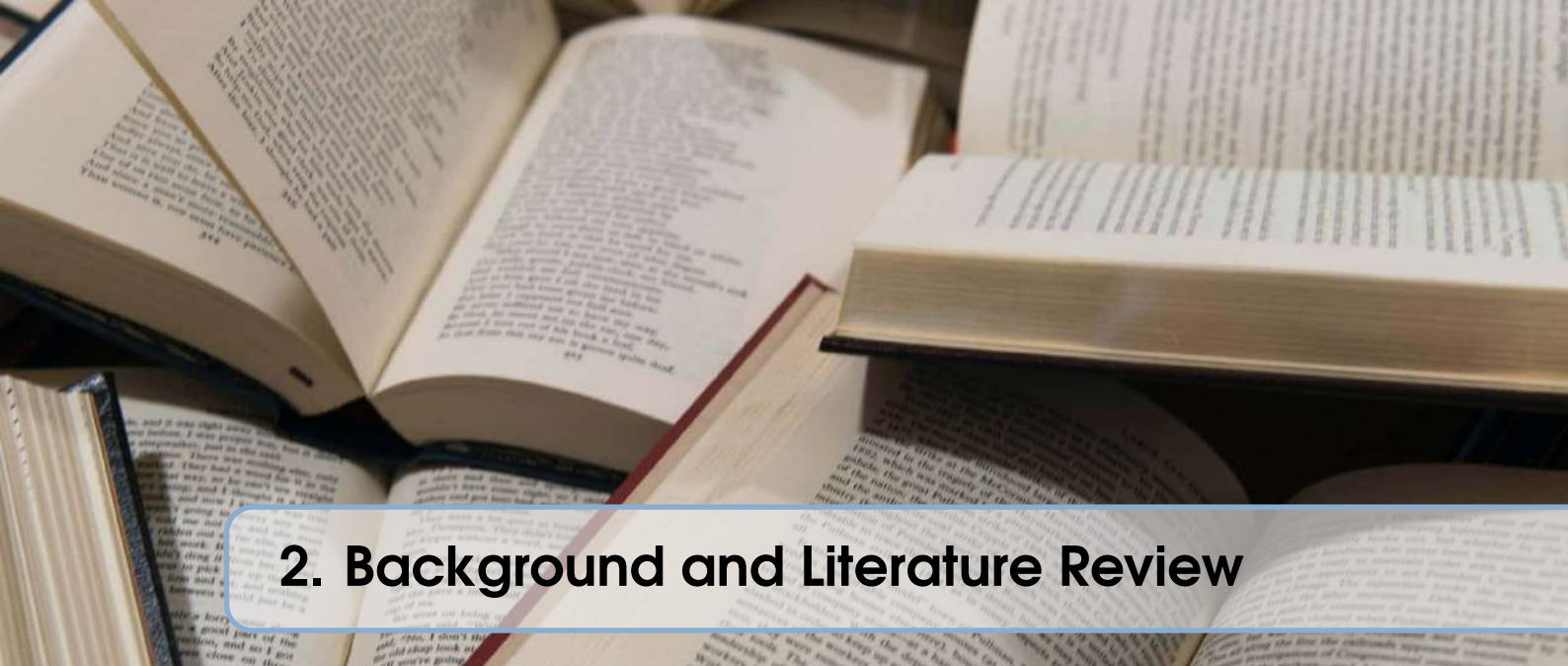
The scope of this project includes the following key areas:

- **Structural Design:** Develop a stable and compact cloud chamber that is structurally sound, cost-effective, and suitable for sustained observation.
- **Particle Detection:** Enable real-time visualization of ionizing particles by optimizing conditions for track formation, such as temperature gradients and alcohol vapor saturation.

- **Material Selection:** Identify and integrate appropriate materials (e.g., acrylic sheets and felt) to ensure durability, visibility, and effective condensation.
- **Thermal System Engineering:** Design a cooling system (using silicon gel cooling pads) capable of maintaining the sub-zero temperatures required for supersaturation inside the chamber.
- **Biological Impact Assessment:** Explore the implications of cosmic ray exposure on biological systems by studying existing research on DNA damage, mutations, and carcinogenesis caused by ionizing radiation.

1.4 Significance

The significance of the Astral Trails project lies in its contribution to understanding the fundamental nature of cosmic radiation and its potential effects on human health, especially in the context of long-duration space missions. By enabling the direct observation of cosmic ray interactions through a cloud chamber, the project provides an accessible platform for studying high-energy particles that are otherwise invisible. This research not only supports the future of safe space travel but also fosters public interest in particle physics and astrophysics. Additionally, insights gained from radiation exposure studies may have broader applications in medical imaging, radiation therapy, and radiation shielding technologies on Earth.



2. Background and Literature Review

2.1 History of Cosmic Rays

Cosmic rays are messengers from the vast extents of space, carrying valuable information about their sources and the space they travel through. The term was coined by physicist Robert Millikan in 1925. Victor Hess discovered cosmic rays for the first time in 1912 during his balloon experiments.

B. Rossi and A. Piccard (1930s) worked on cosmic rays, paving the way for later discoveries of sub-atomic particles within cosmic ray showers. Walter Baade and Fritz Zwicky (1934) as well as Enrico Fermi (1949), hypothesized the origin of these rays, suggesting origin from supernovae and other extra-galactic sources. The origin of UHECRs remains an active area of research in Astrophysics.

2.2 History of the Cloud Chamber

The cloud chamber is an indispensable tool in the field of nuclear and particle physics, allowing scientists to observe phenomena that were previously completely theoretical directly. Particles such as electrons, positrons, alpha particles, and so on produce visible paths inside a cloud chamber.

Wilson's (1911) original design of an expansion chamber was intended to visualize cloud formation, which led to the accidental discovery. Wilson went on to improve and perfect the design of his cloud chamber. American physicist Carl Anderson (1932-1936) discovered the positron (1932) and muon (1936) using cloud chambers.

2.3 Referenced Literature

The various types of cloud chambers are as follows:

- Measurement of Cloud Properties Using a Self-Designed Cloud Chamber - Ridhesh Goti¹, Bhashin Thakore¹, and Rohit Srivastava
- The Cloud Chamber Experiment - Ken Clark January 3, 2019
- Tronic Cloud Chamber - refrigeration based miniaturized cloud chamber - Sanjay Lakshminarayana



Figure 2.2.1: Victor Hess Balloon Experiment

3. Astronomy 101

3.1 History

Pythagoras

- Believed that everything was related to mathematics and could be predicted in rhythmic patterns or cycles.
- Placed astronomy as one of the four mathematical arts, alongside arithmetic, geometry, and music.

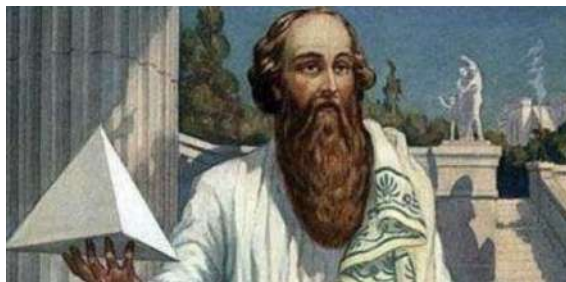


Figure 3.1.1: Pythagoras

Aristotle

- Proposed the Geocentric Model: Earth at the center with the Sun, Moon, stars, and planets orbiting in separate spheres.
- Believed motion was caused by a supernatural being.
- Suggested Earth was spherical (eclipses showed its shadow) and stationary (falling objects drop straight down).



Figure 3.1.2: Retrograde motion over time



Figure 3.1.3: Retrograde Motion

Ptolemy

- Refined the geocentric model using epicycles—planets fixed on small spheres moving on larger spheres.
- Model persisted for 1,000 years until its inaccuracies became apparent.



Figure 3.1.4: Geocentric model of the Universe

Copernicus

- Introduced the Heliocentric model with the Sun at the center.
- Overthrew Ptolemaic astronomy.



Figure 3.1.5: Nicolaus Copernicus

Tycho Brahe

- Corrected serious errors in astronomical tables.
- Discovered a supernova in Cassiopeia.
- Provided the most accurate pre-telescopic observations.

Johannes Kepler

- Determined planetary orbits are elliptical.
- Formulated three laws of planetary motion:
 1. Planets move in ellipses with the Sun at a focus.
 2. Planets move faster when closer to the Sun.
 3. The square of a planet's period $\propto$ the cube of its orbit radius.

Galileo Galilei

- Provided telescopic observations confirming the Copernican model.
- Challenged Aristotle's universe with evidence like Jupiter's moons and sunspots.

Sir Isaac Newton

- Unified celestial and terrestrial physics under the same laws of motion.
- Formulated three laws of motion and universal gravitation.
- Explained and refined Kepler's laws.

Edwin Hubble

- Measured distances to galaxies beyond the Milky Way.
- Discovered the expansion of the universe.

Georges Lemaître

- Proposed the Universe is expanding from a dense initial state.

- Developed the “Primeval Atom” hypothesis, later called the Big Bang Theory.

3.2 Domains of Astronomy

Theoretical Astronomy

Uses physics and chemistry to model astronomical phenomena with analytical and computational simulations.

Computational Astronomy

- Processes massive datasets and time-series data.
- Performs simulations and image preprocessing.
- Builds models for analysis and prediction.

Radio Astronomy

- Studies celestial objects via radio waves.
- Converts radio telescope signals into visual maps of the sky.

Observational Astronomy

Focused on acquiring and analyzing data from telescopic or sensor-based observations.

3.3 Coordinate Systems

Celestial coordinate systems specify positions of stars, planets, satellites, and galaxies.

Equatorial System

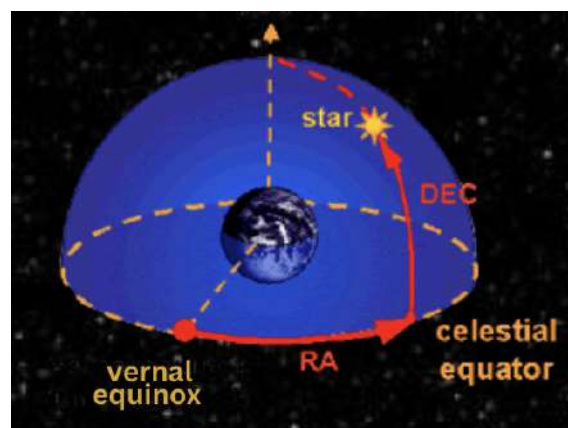


Figure 3.3.1: Equatorial coordinate system

Altitude-Azimuth System

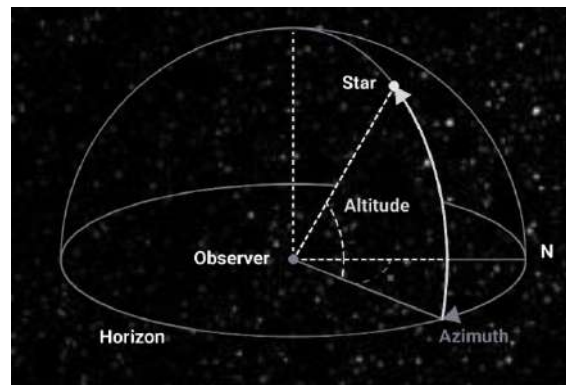


Figure 3.3.2: Altitude-Azimuth coordinate system

Ecliptic System

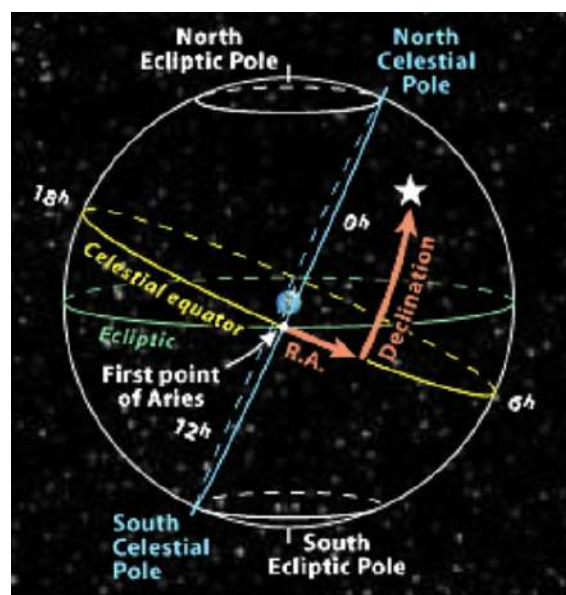


Figure 3.3.3: Ecliptic coordinate system

Other systems include Galactic and Supergalactic coordinates.

3.4 Stellar Life Cycle

Introduction

Stars are massive spheres of hydrogen and helium undergoing nuclear fusion. The initial mass determines whether a star ends as a white dwarf, neutron star, or black hole.

Formation and Main Sequence

- Stars form in nebulae that collapse under gravity.
- Heating triggers hydrogen fusion; star enters the main sequence.

Red Giant / Supergiant Phase

- Low-to-medium mass stars: Expand to red giants.
- High-mass stars: Become supergiants and fuse heavier elements.

Final Stages

Low-to-medium mass stars:

- Planetary Nebula → White Dwarf → (hypothetical) Black Dwarf.

High-mass stars:

- Supernova → Neutron Star (if $<3M_{\odot}$) or Black Hole ($>3M_{\odot}$).

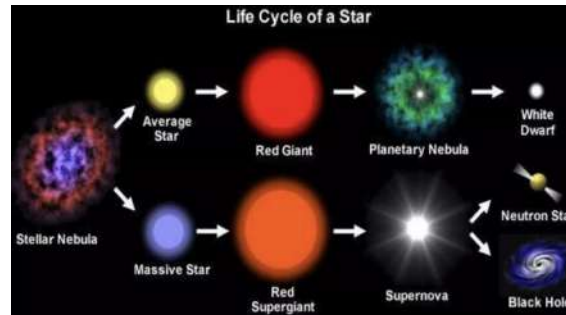
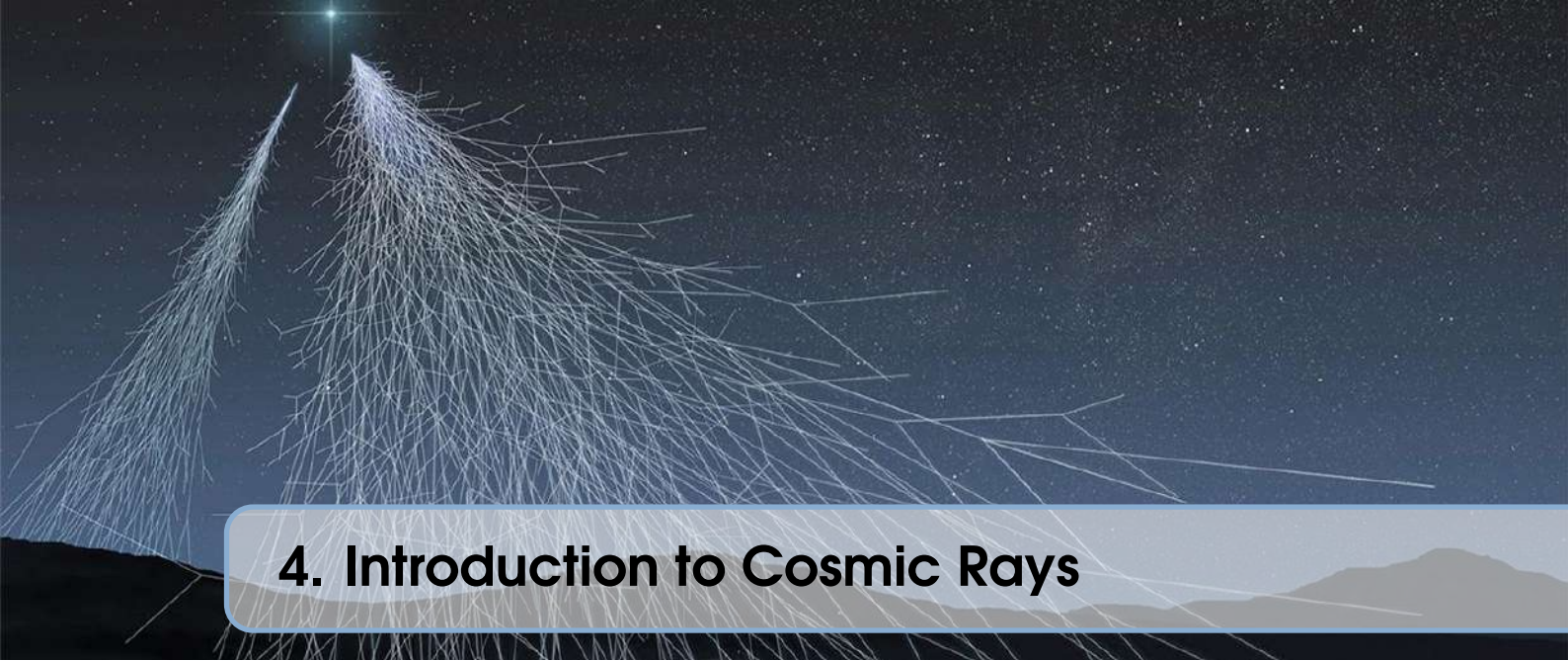


Figure 3.4.1: Stellar Life Cycle

Other Celestial Objects

- **Pulsars:** Rapidly rotating neutron stars emitting EM beams.
- **Magnetars:** Neutron stars with extreme magnetic fields producing X-ray and gamma bursts.



4. Introduction to Cosmic Rays

4.1 What are Cosmic Rays?

Cosmic rays are high-energy particles originating from outer space that travel through the universe at relativistic speeds. Originally, scientists thought these were electromagnetic rays, hence the name “cosmic rays,” but they are actually particles—predominantly protons and atomic nuclei.

Primary Cosmic Rays

Primary cosmic rays travel through space before entering Earth’s atmosphere. They are composed approximately of:

- ~90% protons
- ~9% alpha particles (helium nuclei)
- ~1% heavier nuclei (C, O, Fe)
- Trace amounts of electrons, positrons, and antiprotons

Secondary Cosmic Rays

When primary cosmic rays collide with atoms in Earth’s atmosphere, they produce a cascade of *secondary* particles:

- Pions, muons, electrons, positrons, neutrinos, gamma rays, and neutrons
- Muons are highly penetrating and reach Earth’s surface; roughly 150 muons strike each square meter per second

Ultra-High Energy Cosmic Rays (UHECRs)

UHECRs have energies exceeding 10^{18} eV, the highest energy particles known in nature. A single UHECR can carry the energy of a baseball moving at 100 km/h, compressed into a subatomic particle.

Key Characteristics:

- Minimal deflection by galactic/intergalactic magnetic fields due to extreme energy
- Potential to trace back to their cosmic sources

Potential Sources:

- Colliding galaxies
- Magnetars (super-magnetized neutron stars)
- Gamma-ray bursts (GRBs)
- Rapidly spinning supermassive black holes

Challenges:

- Extremely rare: flux above 10^{19} eV ≈ 1 particle / km² / century
- Require massive detector arrays spanning thousands of km²
- Energy loss over long distances due to the GZK effect

4.2 What Can We Learn from Them?

Cosmic rays serve as natural messengers of the universe and help answer fundamental questions:

- What elements and isotopes are present in the universe?
- How do extreme astrophysical processes accelerate particles?
- What role do cosmic rays play in the chemical evolution of galaxies?

Cosmic Ray Spallation

- Cosmic rays collide with interstellar atoms, breaking them into lighter nuclei.
- Produces rare elements like lithium, beryllium, and boron.

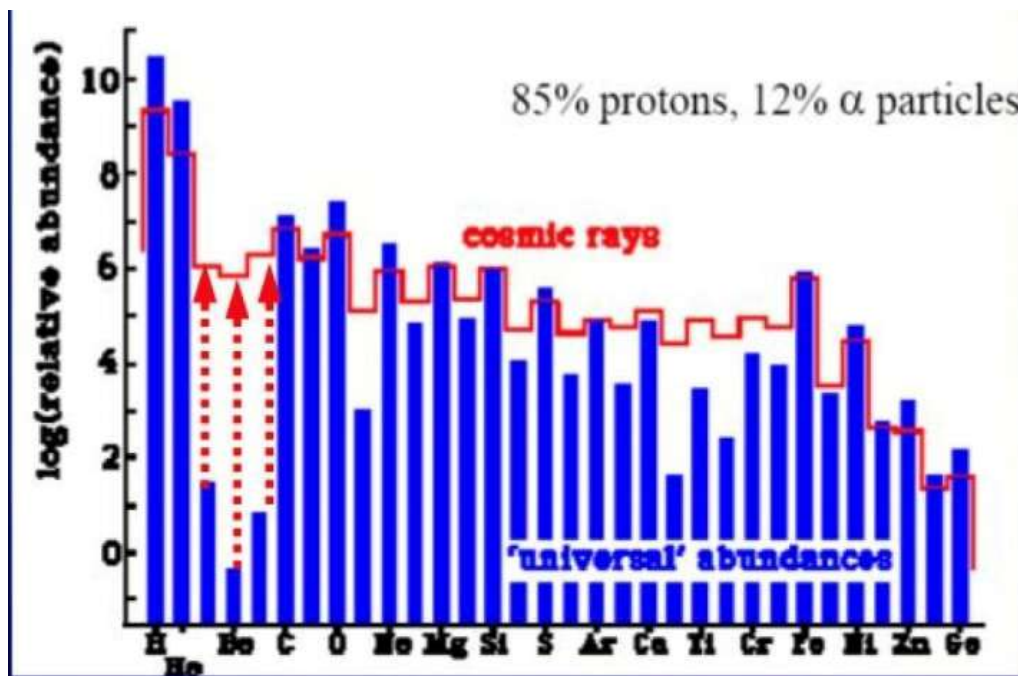


Figure 4.2.1: Abundance of cosmic ray elements vs universal abundances. Light elements (Li, Be, B) are enhanced due to spallation.

4.3 Where Do They Come From?

Cosmic rays originate across a broad energy spectrum from multiple astrophysical phenomena:

Low-Energy Cosmic Rays

- Produced by stars, including solar wind from the Sun
- Primarily protons and alpha particles

Medium-Energy Cosmic Rays

- Produced by supernova explosions
- Particles accelerated via shock waves in the interstellar medium

Ultra-High Energy Cosmic Rays (UHECRs)

- $E > 10^{18}$ eV, rarely detected
- Can point to extragalactic sources due to minimal deflection

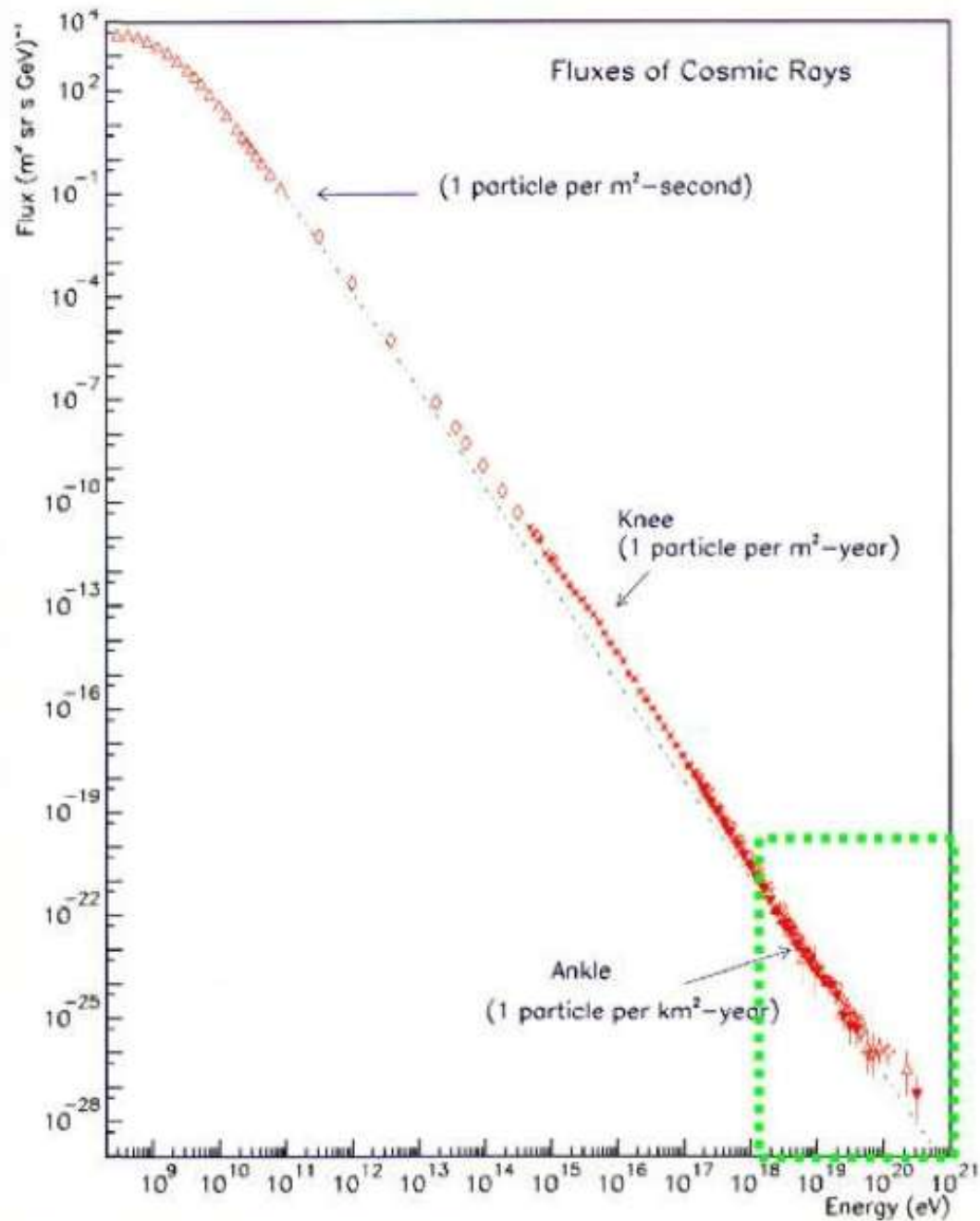


Figure 4.3.1: Energy spectrum of cosmic rays with possible sources and spectral features.

The GZK Cutoff and Paradox

- **Greisen-Zatsepin-Kuzmin (GZK) Effect:**
 - UHECRs above 6×10^{19} eV interact with the cosmic microwave background (CMB) photons
 - Limits their propagation to ~ 50 Mpc

- **GZK Paradox:**
 - Observations detect events above the GZK threshold
 - No confirmed astrophysical mechanism for $> 10^{20}$ eV particle acceleration



5. Cosmic Rays and Their Detection Methods

5.1 Detection of Primary Cosmic Rays

To observe primary cosmic rays before they undergo any interactions with Earth's atmosphere, specialized instruments must be placed above the atmospheric layer—either on high-altitude balloons or satellites in space. These methods allow for direct detection, providing unaltered information about the energy, charge, and composition of cosmic rays.

5.1.1 Balloon-borne Experiments

Instruments are carried to the stratosphere (~ 40 km) via manned or unmanned balloons. Victor Hess, the physicist who discovered cosmic rays, flew a manned hot air balloon for their study. Another example is BESS (Balloon-borne Experiment with a Superconducting Spectrometer).

5.1.2 Satellite-Based Detectors

Detectors and instruments are mounted on artificial satellites and operate in orbit by detecting cosmic rays directly. The AMS-02 (Alpha Magnetic Spectrometer) mounted on the ISS is the most prominent example of such a detector.

The AMS can detect a variety of particles such as electrons, alpha and beta particles, muons, nuclei of heavier elements, etc. It has special sensors that detect antimatter particles as well. By gathering comprehensive data of energies, masses, momenta, fluxes, charges, etc. AMS seeks to unravel profound mysteries, including the potential existence of primordial antimatter from the early universe, indirect evidence of dark matter through anomalies in cosmic ray spectra, and the presence of any exotic forms of matter.

5.2 Detection of Secondary Cosmic Rays

When primary cosmic rays—high-energy particles originating from outer space—enter Earth's atmosphere, they collide with atmospheric nuclei and produce a cascade of secondary particles known as an extensive air shower. These secondary cosmic rays can be detected at the Earth's surface using ground-based observatories.

5.3 Detection of UHECRs

When it comes to cosmic ray (CR) detectors, bigger is better. Larger detectors have several advantages:

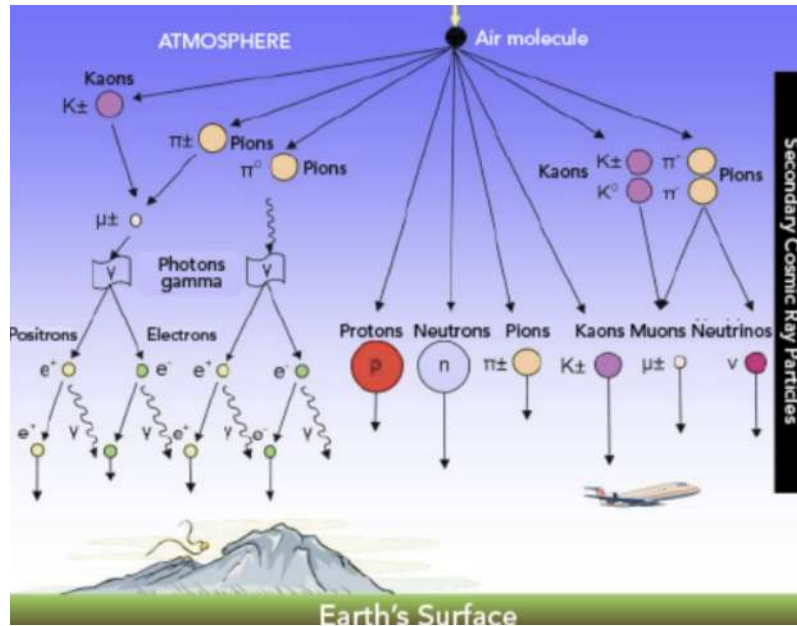


Figure 5.2.1: Secondary Cosmic Ray Particles

- They catch more cosmic rays overall.
- They can detect more of the rare events, such as ultra-high-energy cosmic rays (UHECRs) with energies greater than 10^{18} eV, which occur at a rate of only about one per square kilometer per century.
- A larger detection area allows for the observation of bigger particle showers, which result from higher-energy cosmic rays.
- Site selection is also important for optimal detection. For example, the West Desert is an ideal location for fluorescence observations due to its altitude of approximately 4,500 feet, distance from population centers (over 30 miles away), natural shielding from light pollution by surrounding mountains, and excellent visibility—better than 10 miles for 347 days per year

5.4 Detecting an Extensive Air Shower

Two different methods can be used to detect the passage of an extensive air shower: one can look for the particles in the pancake directly, or one can look for the Cherenkov light generated by the particles in the atmosphere. The figure below illustrates both techniques.

The first is an air Cherenkov telescope (ACT). These are large mirrors that focus the Cherenkov light generated by the air shower onto an array of PMTs, which form an image of the air shower. Properties of the image are used to distinguish between air showers generated by gamma-ray primaries and nuclear primaries. Though very few particles may survive to the ground, the Cherenkov light will reach the ground. Thus, air Cherenkov telescopes can detect lower energy cosmic rays than extensive air shower arrays. However, since they are optical instruments they can only operate on clear moonless nights and they can only view a small piece of the sky at a time.

The second method is to use an EAS array. Figure 5.2 shows an extensive air shower array (EAS array). An EAS array has traditionally been composed of a sparse array of plastic scintillators. The scintillators detect the passage of charged particles that travel through them. They are very inefficient detectors of the gamma rays in the EAS. Since gamma rays outnumber electrons and

positrons by a ratio of roughly 4:1 and the scintillator covers less than 1% of the total area of the array, traditional EAS arrays have rather high energy thresholds. Unlike ACTs EAS arrays can operate under all conditions, night or day, and can view the entire overhead sky continuously. By using buried counters they can detect the muons in air showers generated by cosmic-ray nuclei. However, this method of distinguishing between gamma rays and nuclear cosmic rays is not as efficient as the imaging method used by ACTs.

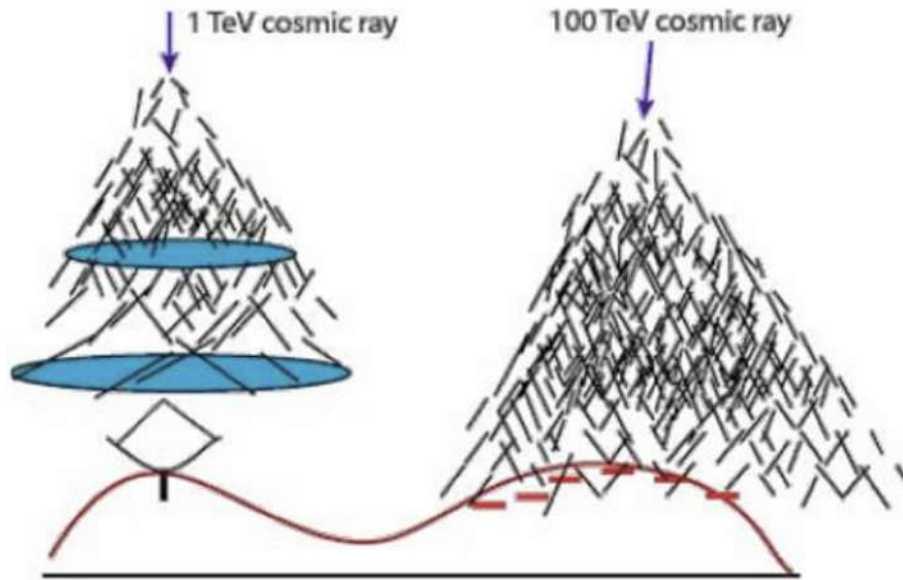


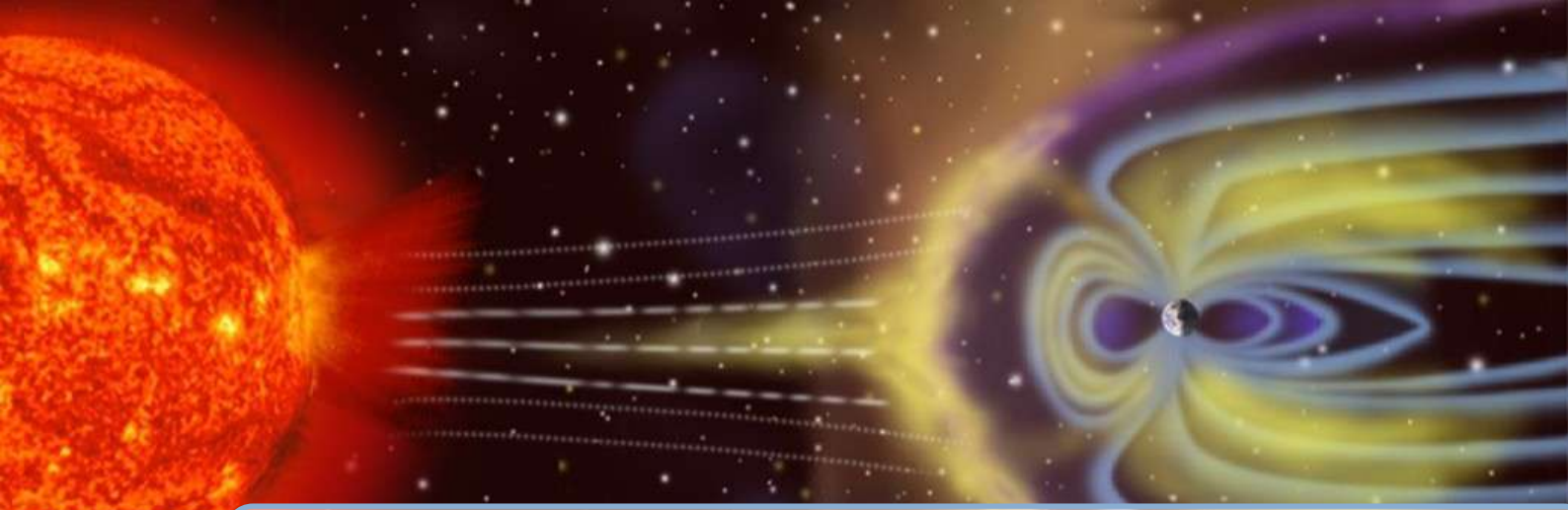
Figure 5.4.1: An EAS Array



Figure 5.4.2: Pierre Auger Observatory

5.5 Pierre Auger Observatory

- Located in Mendoza Province, Argentina
- Contains 1600 water Cherenkov detectors on a 1.5 km grid
- 4 fluorescence eyes—a total of 30 telescopes, each with a $30^\circ \times 30^\circ$ field of view (FOV)
- Auger will detect the shower in two ways. Twenty four hours a day, an array of over 1600 particle detectors will measure shower particles as they hit the ground, which will allow a reconstruction of the shower providing measures of the original cosmic ray's energy, arrival direction, and mass.
- During clear, moonless nights, the showers will be viewed as they traverse the atmosphere. The passage of the showers will cause the atmosphere to fluoresce, and the faint UV light is detected by arrays of large mirrors equipped with fast photomultiplier image arrays



6. Cosmic Rays and Associated Real Life Phenomena

6.1 Air Fluorescence

The passage of charged particles in an extensive air shower through the atmosphere results in the ionization and excitation of the gas molecules (mostly nitrogen). Some of this excitation energy is emitted in the form of visible and UV radiation. This is luminescence, but is referred to as air Fluorescence.

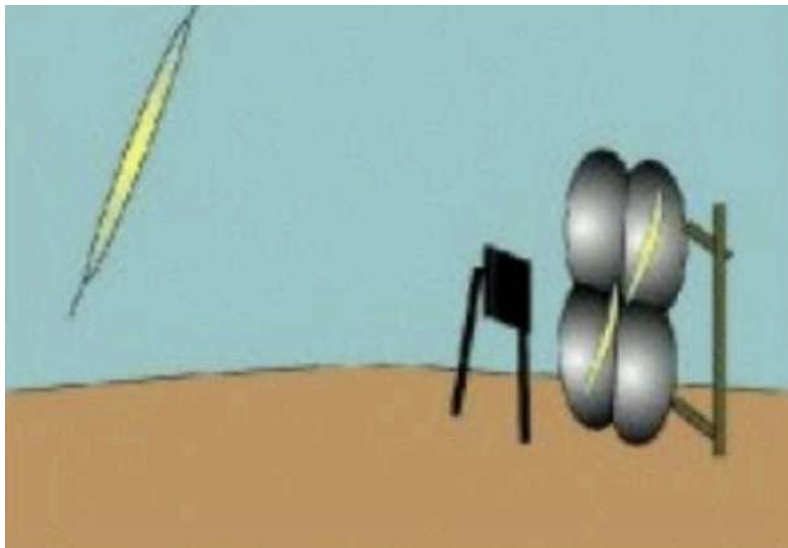


Figure 6.1.1: Fluorescence Air Shower Detector

This figure shows a schematic of a fluorescence air shower detector. The scintillation light is collected using a lens or a mirror and imaged onto a camera located at the focal plane. The camera pixelizes the

image and records the time of arrival of light along with the amount of light collected at each pixel element. This technique can be made to work on clear, moonless nights, using very fast camera elements to record light flashes of a few microseconds in duration.

Many charged particles are expelled from a nuclear explosion, and these particles will also produce scintillation light as they pass through the air. The amount of light collected can then be used to estimate the total energy released from the device.

6.2 Cosmic rays and the weather

While low-energy cosmic rays such as the solar wind cause ionization in the upper atmosphere, muons cause most of the ionization in the lower atmosphere. When a muon ionizes a gas molecule, it strips away an electron, making that molecule into a positive ion. The electron is soon captured, either by another gas molecule forming it into a negative ion, or it may find an already ionized positive ion and neutralize it (this is called recombination).

There is a balance between ionization and recombination, and so there is a fairly constant density of positive and negative ions in the atmosphere. But there is a difference between the types of molecules that become negative ions and the ones that are positive. On average, the negative ions are more "mobile" than the positive ones, and this results in the fact that there is an electric field in the atmosphere.

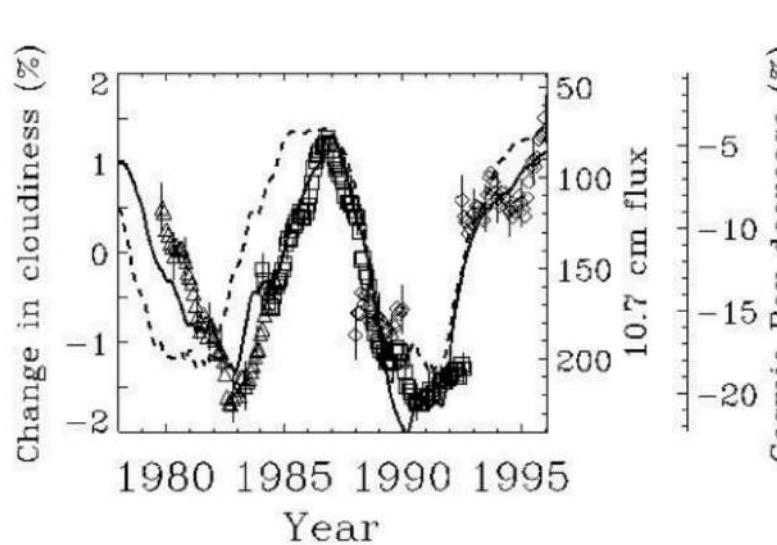


Figure 6.2.1: Cloudiness is affected by the amount of cosmic rays present

On a normal, quiet day, this electric field is about 100 Volts per meter. When a thunder shower forms, there is an as yet not completely understood mechanism that tends to lift the negative ions up while pushing the positive ones down. This changes the electric field strength to tens of thousands of Volts/meter. When the field strength becomes too high, a discharge occurs: lightning. Clearly, without ionization, thunder and lightning would not happen, so cosmic rays have a direct influence on the types of weather we can have on Earth.

6.3 An Air Show Caused by a Cosmic Ray

When a high-energy cosmic ray enters the atmosphere, it loses its energy via interactions with the nuclei that make up the air. At high energies, these interactions create particles. These new particles go on to create more particles, etc. This multiplication process is known as a particle cascade. This process continues until the average energy per particle drops below about 80 MeV (million electron-volts). At this point, the interactions lead to the absorption of particles, and the cascade begins to die. This altitude is known as the shower maximum. The particle cascade looks like a pancake of relativistic particles traveling through the atmosphere at the speed of light. Though the number of particles in the pancake may be decreasing, the size of the pancake always grows as the interactions cause the particles to diffuse away from each other. When the pancake reaches the ground, it is roughly 100 meters across and 1-2 meters thick. If the primary cosmic ray were a photon, the pancake would contain electrons, positrons, and gamma rays. If the primary cosmic ray were a nucleus, the pancake will also contain muons, neutrinos, and hadrons (protons, neutrons, and pions). The number of particles left in the pancake depends upon the energy of the primary cosmic ray, the observation altitude, and fluctuations in the development of the shower. This particle pancake is known as an extensive air shower (or simply an air shower).

6.4 Van Allen Radiation Belts

The Van Allen radiation belts were first observed by the satellite Explorer I in 1958, and their existence was shortly confirmed by Explorer 111 and Sputnik 111 observations. Professor James Van Allen had conducted both experiments on Explorers I and I11 and determined that the observed data indicated the existence of charged particles trapped

by the magnetic field.

Over the years, the Van Allen belt has been mapped and refinements have been made in the various zones until finally the picture of an inner and outer zone emerged with particles of energies listed in Table I. Table I is given in the B-L coordinate system, where B is the magnitude of the magnetic induction, and L is the equatorial distance to the dipole field line about which the particles are spiraling. In addition to the naturally occurring radiation zones, high-altitude nuclear explosions have ejected high-energy particles that distort the existing zones.

TABLE 1
VAN ALLEN RADIATION BELT

<u>Heart of Inner Zone (L ~ 1.4, B ≈ 0.12, alt = 3600 km)</u>		
<u>Particle</u>	<u>Energy</u>	<u>Flux</u>
Protons	(E > 30 Mev)	~ 3 x 10 ⁴ /cm ² sec
Electrons	(E > 600 Kev)	~ 2 x 10 ⁸ /cm ² sec
Electrons	(E > 40 Kev)	~ 10 ⁸ /cm ² sec
<u>Heart of Outer Zone (L ~ 3.5, alt ≈ 25,000 km)</u>		
Electrons	(E > 40 Kev)	~ 10 ⁷ /cm ² sec
Electrons	(1.5 < E < 5 Mev)	~ 10 ⁴ /cm ² sec
Protons	(0.1 < E < 5 Mev)	~ 10 ³ /cm ² sec
Protons	(E > 1 Mev)	~ 10 ⁷ /cm ² sec
Protons	(E > 75 Mev)	~ 0.1/cm ² sec

Figure 6.4.1: Table 1

Fortunately, however, the artificially created belts decay rather rapidly in the outer zones, leaving a more lasting nucleus of high energy particles contained within the limits of the natural radiation zone. It has been found that vehicles in a relatively low circular earth orbit (200-600 km) will receive an insignificant amount of radiation from the Van Allen zones; however, a vehicle with an eccentric orbit or one that has a relatively high circular orbit receives a much higher flux. For example, a satellite in a synchronous orbit over the equator will be very close to the center of the outer Van Allen zone. Methods are now under development to determine the optimum trajectories (in terms of dose rates) to be used for various mission profiles. If this method proves successful, the mission may be made more complicated due to the specification of a path to be followed through the Van Allen zones. The Van Allen zones vary diurnally as the magnetosphere.

Since on the sunward side of the Earth the magnetosphere is flattened and condensed, the particles which are trapped by the magnetic lines of force are also flattened. At the nadir, however, the magnetosphere is elongated; therefore, the zones of trapped particles are also stretched. It has been discovered that the boundary of the magnetosphere itself contains a great number of charged particles, apparently due to the flow of the solar wind. There have been a number of theories proposed regarding how the Van Allen belts are maintained and how they lose particles. It has been fairly well established that protons are lost to the belts by their interaction with atmospheric constituents.

TABLE 2
RANGES OF PROTONS IN ALUMINUM

<u>Energy (Mev)</u>	<u>Range (gm/cm²)</u>
10^2	10.5
10^1	1.7×10^{-1}
10^0	3.5×10^{-3}

Figure 6.4.2: Table 2

Table 2 shows the range that protons have in aluminum. Electrons in the Van Allen belts are probably lost by scattering due to coulomb forces. Several mechanisms have been investigated to determine how the particles are maintained in the belts. One theory suggests that solar particles may be able to enter the magnetosphere at the polar regions. Such a theory could be acceptable for the outer belt, but the inner belt requires a different explanation. One possibility is that the inner belt is due to the decay of neutrons, but these may only account for the relatively high energy particles. Since the atmospheric density in the exosphere varies markedly with the solar cycle, it should not be surprising to find the particle flux varying inversely with the solar cycle. The Van Allen zones should be of considerable consequence as a radiation hazard to space vehicles.

6.5 Galactic Cosmic Radiation

The radiation with the highest energy per particle encountered in outer space will undoubtedly be the galactic cosmic radiation. Since the flux of these particles is relatively low, the radiation problem will not be as significant as for particles with lower energies and higher flux. The energies of these particles will usually exceed one billion electron volts (1 Bev) and, as a result, may produce secondaries which will be of more concern than the primary radiation itself. Galactic cosmic radiation exhibits a tendency to vary inversely with the solar cycle. This has been attributed to the solar magnetic field lines, which are carried out from the sun farther during higher solar activity periods.

Since the cosmic ray particles are charged, they are deflected by these magnetic lines of force. As the solar activity period decreases, the "interplanetary lines of magnetic force" become weaker, and the resulting deflection of particles becomes less. Thus, fewer particles are excluded from the solar system. A variation very similar to this is the 27-day variation with the solar cycle. This is due to the reappearance of sunspots that have long lines. These interplanetary magnetic lines of force exhibit variations with the sunspot groups. Since the mean rotational period of the sunspot region is approximately twenty-seven days, the 27-day period exhibited by the cosmic rays is considered to be a direct result. The final variation noted in the flux of cosmic rays reaching the Earth is known as the Forbush decrease. The Forbush decrease is a significant decrease in the flux of cosmic rays incident on the Earth following a solar flare. A decrease of approximately 30% of the flux has been noted, and the recovery time following the decrease is usually from one to two days. This phenomenon is considered to be caused by the ejection of a vast plasma cloud. This cloud, in turn, carries 5 magnetic lines of force away from the surface of the sun.

When the cloud reaches the earth, the cosmic rays suddenly decrease due to the presence of the magnetic lines of force. At about the same time, the beginning of a magnetic storm takes place as well as the other associated geomagnetic disturbances. Particles reaching the earth will penetrate the magnetic field to a certain depth depending upon the rigidity, R , exhibited by the particles.

Rigidity is simply the ratio of the particle's momentum to its charge,

TABLE 3
COMPOSITION OF GALACTIC COSMIC PARTICLES

<u>Particles</u>	<u>Flux</u> ($R \geq 4.5$ bv)	<u>Percent</u>
Protons	$\sim .77/\text{cm}^2\text{-sec}$	$\sim 84\%$
Alpha Particles	$\sim .11/\text{cm}^2\text{-sec}$	$\sim 14\%$
Carbon, Nitrogen, Oxygen*	$\sim .009/\text{cm}^2\text{-sec}$	$\sim 1\%$
$Z > 10$	$\sim .002/\text{cm}^2\text{-sec}$	$\sim 0.25\%$
Gamma Radiation and Electrons**	---	$\sim 1\%$

Figure 6.5.1: Table 3

$$R = \frac{pc}{Ze}$$

where p is the momentum (in units of Bev/c), c is the velocity of light, Z is the atomic number, e is the unit charge, and the resulting R is in billion volts. If we examine the formula and assume a uniform velocity for particles of any mass, we see that R increases as we go up the atomic scale, since the mass increases faster than does Ze . Then, higher mass particles will penetrate deeper into the magnetic field of the earth than will the lower mass particles. A generalised table of composition for galactic cosmic radiation is found in Table 3. The energy range for these particles ranges from a few MeV up to at least 10^{12} MeV, with the average energy in the vicinity of 4×10^3 Mev. Despite the high energies that the galactic cosmic rays exhibit, there will be little biological hazard during short-term missions due to this radiation. On longer missions the situation may change due to the constant accumulation of radiation.

6.6 Solar Wind

In interplanetary space, the surrounding plasma will have some kinetic velocity due to the high temperatures in the corona. At present, there is some evidence to suggest that the corona exhibits a "breathing" effect which varies directly with the eleven-year solar cycle period. This seems to be borne out when one observes the density and velocity variations of the plasma flow during minimum and maximum sunspot activity. During the minimum sunspot activity at the Earth's distance

from the sun, the mean density of the solar wind is approximately 10^2 particles per cm^3 traveling at a velocity of 500 kilometers per second. This results in an average particle flux of about 5×10^9 particles/ cm^2 sec.

At sunspot maximum the density of particles is expected to increase to approximately 10^3 particles per cm^3 , with a velocity in the vicinity of 1500 kilometers per second. This results in a particle flux at the earth's orbit of about 1.5×10^{12} particles/ cm^2 sec. The kinetic temperature in the earth's vicinity is approximately $220,000^\circ \text{C}$. However, this will have a very small effect on the heating of vehicles when one considers that the solar radiation flux will amount to about 1.4×10^6 ergs/ cm^2 sec. (September-December 1962) data obtained by a plasma probe, plasma energy is approximately 4.4×10^{-9} ergs/ cm^2 .

From an analysis of Mariner II, the solar wind is considered to encompass the Earth's magnetosphere and to flow around the outer boundaries much like the wake of a bullet as it passes through air. The magnetosphere is thus flattened on the sunward side of the earth and elongated on the dark side. Although the flux of particles in the solar wind is high, their energies are relatively low. For this reason, It is expected that the biological radiation hazard behind a small amount of shielding will be insignificant.

6.7 Solar Flares

Solar flares will undoubtedly be the source of radiation that should concern us most during interplanetary travel. The "great solar bursts" or type 3+ flares emit particles with near-relativistic velocities. These particles have energies comparable to the galactic cosmic rays, but their flux is much greater. Solar flares are associated with the dark spots noted on the photosphere of the sun. The dark spots are local magnetic fields of great intensity. These high magnetic fields inhibit the outward flow of the hot gases from within, and thus cause cooler and darker patches on the 7 photosphere of the sun. If these areas "collapse" due to the buildup of strong perturbing fields by the ionized gas flowing around the sunspots, then great volumes of ionized gases are spewed out into the chromosphere. This phenomenon is known as the solar flare.

Much of the data presently available, which tells the story of the solar

flare, has been accumulated by studies of the various radio wave emissions accompanying the solar flare. The refractive index of an ionized medium is based upon the physical characteristics of the medium itself. The refractive index, μ , of a medium (containing N number of particles with charge e and mass m) to a wave with frequency f is

$$\mu^2 = 1 - \frac{N e^2}{\pi \epsilon_0 m f^2} .$$

Assuming that electrons are the particles affecting the refractive index the most, we may substitute the mass and charge of the electron for m and e . The point at which $\mu^2=0$ is the point at which the radio waves will be refracted.

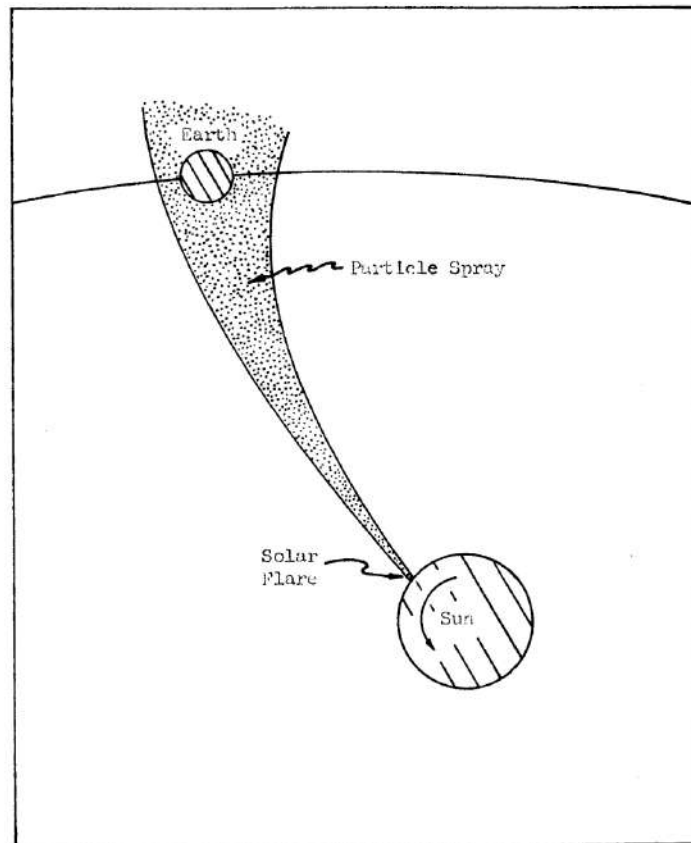


Figure 2. Path Along Which Particles From the Sun Following A Solar Flare Travel, As Seen From Above the Plane of the Ecliptic

TABLE 4
CHARACTERISTICS OF SOLAR NOISE

Type	Identifying Characteristics	Source Characteristics	Frequency Characteristics
I	Noise storms usually lasting from hours to days; or bursts of ~ 1 second duration.	Assumed to be of a non-thermal origin, associated with sunspots, "B centers," and sometimes flares.	Less than ~ 250 Mc/sec with bandwidth 1-10 Mc/sec for bursts and 10-100 Mc/sec for continuum. The intensity at 100 Mc/sec 10^{-21} to $10^{-19} \frac{\text{watts}}{\text{m}^2 \text{ c/sec}}$.
II	Bursts with slow drift of ~ 0.3 Mc/sec ² lasting from 5-10 minutes.	Source is due to plasma oscillations associated with flares. Occurrence begins about 7 minutes after flare. The source moves outward at ~ 1000 km/sec.	Mainly less than 150 Mc/sec with the bandwidth of about 2×10^{-4} of the observed frequency. Intensity at 100 Mc/sec is usually 10^{-20} to $10^{-19} \frac{\text{watts}}{\text{m}^2 \text{ (c/sec)}}$.
III	Bursts with fast drift of ~ 30 Mc/sec ² lasting singly 3-10 sec or in groups of 1-3 minutes.	Assumed to be associated with plasma oscillation associated 50-60% of the time with flares. The source has an outward velocity of ~ 10^5 km/sec.	Ranges from > 4000 Mc/sec to < 10 Mc/sec with a bandwidth almost equal to the frequency. The intensity usually is less than $10^{-20} \frac{\text{watts}}{\text{m}^2 \text{ (c/sec)}}$.

TABLE 4 (Cont'd)

Type	Identifying Characteristics	Source Characteristics	Frequency Characteristics
IV	Smooth continuum lasting from minutes to hours.	Source is due to synchrotron radiation. Occurring 70-80% of the time with flares at ~ 15 minutes after start. Initial source velocity ranges from $1-5 \times 10^3$ km/sec for about 10 minutes and then source becomes stationary.	Cover the complete radio band but vary from burst to burst. The bandwidth is frequently several octaves with intensities from 10^{-20} to $10^{-19} \frac{\text{watts}}{\text{m}^2 \text{ (c/sec)}}$.
V	Smooth continuum lasting from 1-2 minutes.	Synchrotron radiation occurring before the maximum of solar flares. The velocity of the source is ~ 5×10^3 km/sec.	Frequencies less than 200 Mc/sec with a bandwidth of several Mc/sec at 50-100 Mc/sec. Intensities 10^{-20} to $10^{-19} \frac{\text{watts}}{\text{m}^2 \text{ (c/sec)}}$.
Micro-wave	Continuum and bursts lasting 0.5 to 20 minutes.	Assumed to be of synchrotron and possibly thermal origin associated about 80% of the time with flares.	The frequency range is ~ 1000-20,000 Mc/sec with a bandwidth of several octaves. The intensity is usually ~ 5×10^{-20} to $5 \times 10^{-20} \frac{\text{watts}}{\text{m}^2 \text{ (c/sec)}}$ at 3000 Mc/sec.

This means, that (assuming N increases as we approach the sun) any frequency, f , must have originated at some distance from the sun equal to or greater than the distance of which

$f = 9 \times 10^{-3} \sqrt{N}$. For this reason frequencies ranging from about 300 to 30,000 megacycles/sec are assumed to originate in the chromosphere, whereas lower frequencies are assumed to originate in the corona. As the frequency becomes lower, it must have originated at greater and greater distances from the sun. Figure 1 illustrates when the specific radio bursts occur and in what frequency range during a solar flare. Table 4 is a summary of the various characteristics of radio emissions from the sun. The flux and energies of the emitted particles are of primary concern when considering the effects of solar flares. It is impossible to make any specific statement as to the flux of particles at the earth from any flare. This is because the flux of particles of a

particular energy depends not only upon the characteristics of the flare itself, but also upon the location of the flare on the surface of the sun. In the same sense, the flux of particles is dependent upon the location of the earth as it orbits around the sun. Anyone who has ever watched a rotating lawn sprinkler can visualise the path which solar particles take as they erupt from the sun (Fig. 2). To an observer in the path of the particles, however, the flux would appear to be traveling in a straight line from the sun. Table 5 gives the ranges over which the magnitude of the solar flares may be expected to extend. The wide range is necessitated by the variables which may exist.

TABLE 5 (Ref. 6)
PARTICLE RADIATION DUE TO SOLAR FLARES

Flux	Range from 3×10^3 to $1.2 \times 10^4/\text{cm}^2 \text{ sec-ster}$ distributed isotropically, and containing predominantly H^+ particles.
Energy	Range from $\sim 10 \text{ Mev}$ to $\sim 500 \text{ Mev}$.
Duration	From 10 to 100 hours.
Origin	Associated with class 3 or greater flares, which follow the solar cycle: $\sim 12/\text{yr}$ at solar maximum to $\sim 3/\text{yr}$ at solar minimum
Dose	Depending upon flux and energy ranges from 10^{-3} rad to 10^3 rad .

TABLE 6

Wavelength $\text{\AA}$	Percent of solar Electromagnetic Energy
< 2000	0.2
2000 - 3800	7.5
3800 - 7000	41.0
7000 - 10000	22.0
10000 - 20000	23.0
> 20000	6.0

6.8 Spectral Radiation

Of all the radiation hazards which will be encountered in space, the effects of spectral irradiation are the most difficult to analyse at the present time. This is true because of the difficulty in studying the solar photon flux in this most important range below $3000 \text{ \AA}$. Since radiation of wavelengths less than $3000 \text{ \AA}$ is strongly absorbed by our atmosphere, earth-bound studies in this range are mostly hypothetical. Artificial sources of photon flux in the X-ray and γ -ray range have been

studied and may inform us as to the hazards which may be encountered.

In the region from about $2200 \text{ \AA}$ to $2900 \text{ \AA}$, atmospheric ozone absorbs almost all the photon radiation. From about $9000 \text{ \AA}$ atmospheric oxygen above 75 km absorbs the radiation, and below 900 $\text{\AA}$ many atmospheric constituents share in absorbing radiation.

Table 6 summarizes the percent of solar electromagnetic energy radiated in various wavelength intervals. Figure 3 illustrates the wavelength spectrum from the sun, and Figure 4 is a corresponding frequency graph.

TABLE 7

	Date	2-8 $\text{\AA}$ ergs/cm ² sec	8-20 $\text{\AA}$ ergs/cm ² sec	44-60 $\text{\AA}$ ergs/cm ² sec
Background and Bright Surge	7-24-59	1.3×10^{-2}	9×10^{-2}	0.15
Background	8-7-59	0.1×10^{-2}	2.3×10^{-2}	----
Background	8-14-59	0.1×10^{-2}	2.3×10^{-2}	0.075
Background	8-29-59	0.33×10^{-2}	----	0.17
Flare 2+	8-24-59	2.6×10^{-2}	16×10^{-2}	< 0.41
Flare 2+	8-31-59	3×10^{-2}	90×10^{-2}	< 1.8
Flare 2+	9-1-59	$> 8.8 \times 10^{-2}$	$> 90 \times 10^{-2}$	----

The most hazardous region of the electromagnetic spectrum is the short wavelength X-ray range. Fortunately, the most predominant energy region is not in this range. The X- and y-radiations increase as solar flares occur. The following table lists the X-ray fluxes detected by rocket-borne equipment.

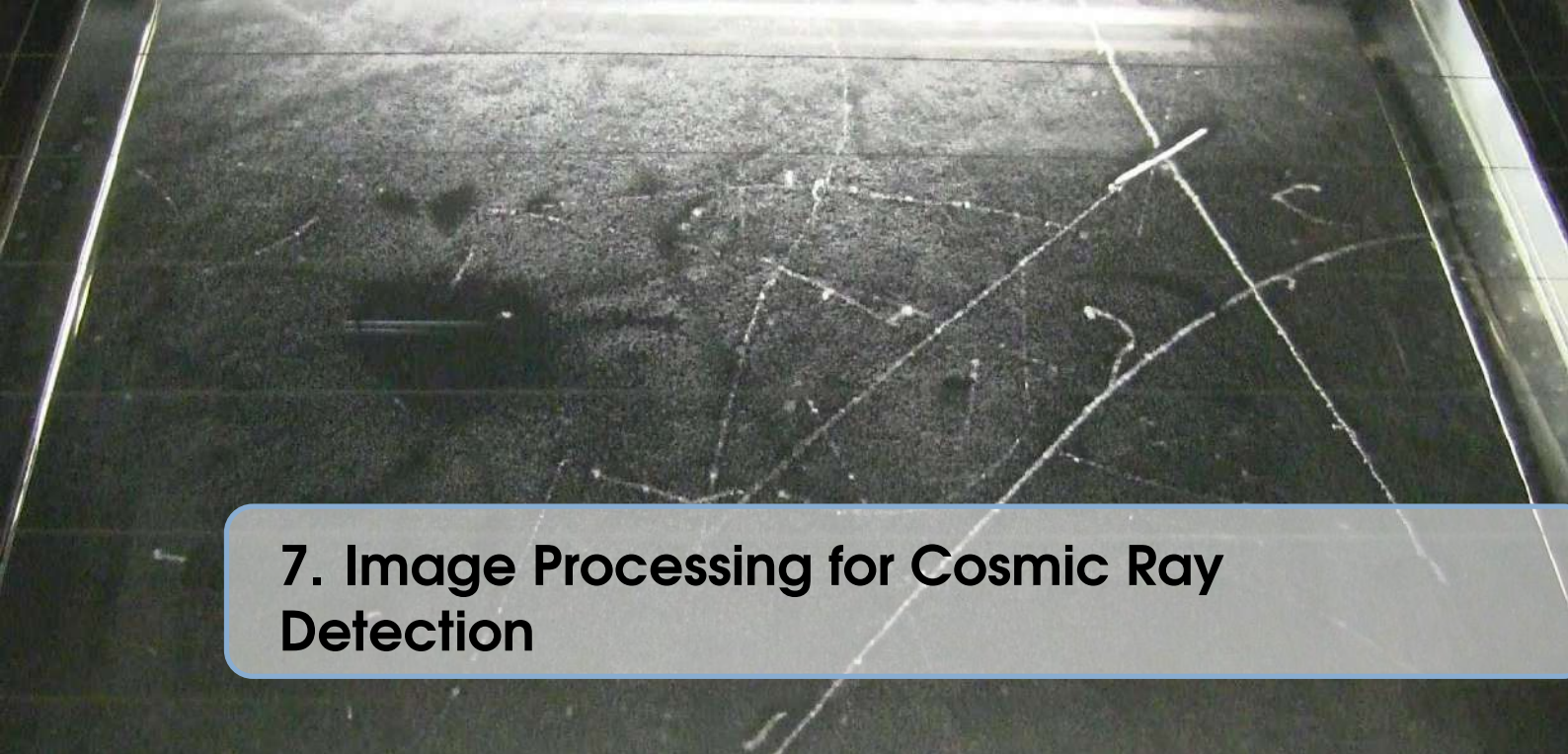
From most of the data presently available, it appears that the spectral component of radiation need not be considered. The effects of the electromagnetic radiation on the eyes are apparent.

6.9 Conclusion

From the preceding sections it is apparent that Van Allen and solar flare radiations need be of primary concern during space flight. A long duration earth orbit, as well as an extended extraterrestrial mission, must be adequately protected against the harmful radiations which

each will encounter. In the case of an extended earth orbit where the radiation dose may be accumulated over a period of time, the obvious solution is to have the personnel periodically replaced and sensitive instrumentation adequately protected. By a frequent replacement of personnel, the dose will not have time to accumulate to serious proportions. Such an arrangement would obviously be required for a manned orbiting space station or orbiting laboratory. This might become troublesome to the people working on a specialized problem to which they have become dedicated, or to persons working in intricate areas where several people would have to be trained in one specialized area. Other than these human factors, the problems involved do not appear too difficult.

On an extended extraterrestrial mission, personnel cannot be replaced as they acquire a specific minimal amount of radiation. One solution to this problem might be to fractionate the dose. Assuming complete shielding for the entire vehicle will not in itself be sufficient to cope with any situation, it may be more realistic to provide an area which has sufficient shielding to cope with any radiation hazard. Each member of the crew would be required to spend a certain portion of his time in this area. During solar flares perhaps all members of the crew could remain in this area for a short period of time (about three to four hours). Such an arrangement is obviously to be included for conservation of shielding weight. The engineering problems encountered may be a bit more complicated, but nothing which cannot be overcome. At the present time very little is known of the effects of radiation on the human body, especially the effects which exposure to radiation may impose upon future generations. In addition, sensitive vehicle instrumentation - especially items concerned with the guidance and control systems and the spacecraft life systems - must be adequately tested to establish limits of radiation exposure so that this equipment may be protected from possible excess.



7. Image Processing for Cosmic Ray Detection

7.1 Computational Tools Used

- Python
- OpenCV (CV2)
- Pandas
- Numpy
- Matplotlib

7.2 Introduction to Image Representation and Digitalization

In the realm of digital image processing, an image is fundamentally represented as a discrete grid of numerical values. This transformation from a continuous optical signal to a digital format is crucial for computational analysis. For grayscale images, each point in this grid, known as a pixel (picture element), holds a single numerical value representing its intensity, typically ranging from 0 (black) to 255 (white) in an 8-bit system.

For color images, the representation is slightly more complex. Each pixel is characterized by multiple values, one for each color channel. The most common color model is RGB (Red, Green, Blue), where each pixel has three intensity values corresponding to the red, green, and blue components of the light it emits or reflects. Therefore, an image can be thought of as a multi-dimensional array or matrix. A grayscale image is a 2D array, while a color image is a 3D array (width $\times$ height $\times$ channels).

For instance, a $W \times H$ grayscale image can be represented as a matrix

I , where $I(x, y)$ is the intensity of the pixel at coordinates (x, y) . A color image would be $I(x, y, c)$, where c denotes the color channel (e.g., R, G, B). This array representation allows us to apply mathematical operations to manipulate and extract information from images, which is fundamental to image processing. In the context of cosmic ray detection in a cloud chamber, the images captured, whether grayscale or color, are converted into such digital arrays, forming the basis for subsequent analysis.

7.3 Convolutions and Kernels

Convolution is a fundamental operation in image processing, particularly for feature extraction and filtering. It involves sliding a small matrix, known as a kernel or filter, over the image and computing the sum of element-wise products between the kernel and the underlying image pixels. Mathematically, the output pixel $G(x, y)$ at position (x, y) in the filtered image is given by:

$$G(x, y) = \sum_{i=-a}^a \sum_{j=-b}^b K(i, j) I(x - i, y - j)$$

where I is the input image, K is the kernel of size $(2a + 1) \times (2b + 1)$, and a, b define the kernel's dimensions.

Different kernels are designed to achieve specific effects:

7.3.1 Edge Detection Kernels

Edges represent significant changes in image intensity, often corresponding to boundaries of objects. Kernels like the Sobel, Prewitt, and Roberts operators are designed to highlight these transitions. For example, a common Sobel kernel for detecting horizontal edges is:

$$K_{\text{Sobel-horizontal}} = \begin{pmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{pmatrix}$$

And for vertical edges:

$$K_{\text{Sobel-vertical}} = \begin{pmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{pmatrix}$$

These kernels compute an approximation of the image gradient, with larger values indicating stronger edges. In cosmic ray detection, these

kernels can help identify the sharp boundaries of the condensation trails left by cosmic rays.

7.3.2 Gradient Kernels

Gradient kernels, like those used in edge detection, are designed to measure the rate of change of pixel intensities. They are essentially discrete approximations of derivatives. The magnitude of the gradient at a pixel indicates the strength of the edge, while its direction points in the direction of the greatest intensity change.

7.3.3 Blur Kernels (Smoothing)

Blurring kernels are used to smooth images by averaging pixel values within a neighborhood. This reduces noise and can help in feature extraction by removing fine details. A common example is the Gaussian kernel, which gives more weight to central pixels:

$$K_{\text{Gaussian}} = \frac{1}{16} \begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{pmatrix}$$

For cosmic ray images, blurring can pre-process the image to reduce spurious noise that might be mistaken for ray tracks.

7.3.4 Sharpening Kernels

Sharpening kernels enhance edges and fine details by increasing the contrast between adjacent pixels. They often work by subtracting a blurred version of the image from the original, effectively amplifying high-frequency components. A simple sharpening kernel might look like:

$$K_{\text{Sharpen}} = \begin{pmatrix} 0 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 0 \end{pmatrix}$$

7.4 Basic Morphological Operations

Morphological operations are a set of non-linear image processing techniques that process images based on shapes. They are typically applied to binary images (black and white), though they can be extended to grayscale images. They probe an image with a small shape or template called a structuring element.

7.4.1 Erosion

Erosion "shrinks" objects in a binary image. A pixel is set to foreground (white) only if all pixels under the structuring element are also foreground. Otherwise, it's set to background (black). This operation can be used to remove small, isolated noise points or to separate connected components.

7.4.2 Dilation

Dilation "grows" objects in a binary image. A pixel is set to foreground if at least one pixel under the structuring element is foreground. This is useful for filling small holes in objects or connecting broken components.

In the context of cosmic ray detection, morphological operations can refine the detected tracks, for example, by eroding away thin spurious lines and then dilating the true tracks to make them more robust.

7.5 Canny Edge Detection

The Canny edge detector is a multi-stage algorithm widely regarded as one of the optimal edge detection techniques due to its good performance in terms of low error rate, good localization, and minimal spurious responses. It is particularly effective for identifying the sharp, well-defined tracks of cosmic rays in a cloud chamber. The Canny algorithm involves the following steps:

7.5.1 Noise Reduction (Gaussian Smoothing)

The first step is to smooth the image to remove noise, which can otherwise lead to false edge detection. This is typically achieved by convolving the image with a Gaussian filter. This step is crucial for ensuring that the detected edges are true features rather than artifacts of noise. The standard deviation (σ) of the Gaussian kernel controls the degree of smoothing, influencing the scale at which edges are detected.

7.5.2 Gradient Calculation

After smoothing, the algorithm calculates the gradient intensity and direction for each pixel. This is commonly done using Sobel operators. Two gradient images, G_x (horizontal changes) and G_y (vertical changes), are computed. The gradient magnitude G and direction Θ

are then calculated as:

$$G = \sqrt{G_x^2 + G_y^2}$$

$$\Theta = \arctan\left(\frac{G_y}{G_x}\right)$$

The gradient magnitude provides information about the strength of a potential edge, while the gradient direction indicates the orientation of the edge.

7.5.3 Non-Maximum Suppression

This step thins the edges to a single-pixel width. For each pixel, the algorithm checks if its gradient magnitude is a local maximum in the direction of the gradient. If it's not a local maximum, the pixel is suppressed (its gradient magnitude is set to zero). This ensures that only the most prominent pixels along the edge are retained, producing thin, sharp edges. For example, if the gradient direction is horizontal, the pixel's magnitude is compared with its left and right neighbors.

7.5.4 Double Thresholding

After non-maximum suppression, many weak edge pixels may still remain. Double thresholding is applied to identify strong, weak, and non-edges. Two thresholds are used: a high threshold (T_H) and a low threshold (T_L).

- Pixels with gradient magnitude greater than T_H are classified as strong edge pixels.
- Pixels with gradient magnitude between T_L and T_H are classified as weak edge pixels.
- Pixels with gradient magnitude less than T_L are suppressed (set to zero).

Strong edge pixels are definitively considered edges and are included in the final output. Weak edge pixels are considered only if they are connected to strong edge pixels.

7.5.5 Edge Tracking by Hysteresis

This is the final step, where weak edge pixels are included in the final edge map only if they are connected to strong edge pixels. This process effectively traces out complete edges. Starting from strong edge pixels, the algorithm searches for connected weak edge pixels (usually in an 8-connected neighborhood). If a weak edge pixel is found, it is also

classified as an edge, and the process continues from that pixel. This prevents spurious edges caused by noise while ensuring that true, but slightly weaker, parts of an edge are not missed.

In the context of cosmic ray detection, Canny edge detection is highly effective because it can precisely locate the fine, often faint, tracks left by the particles, even in the presence of some background noise. The ability to produce thin, continuous edges is particularly valuable for accurately measuring the length and curvature of these tracks, which can provide information about the energy and nature of the cosmic rays.



Figure 7.5.1: Result of Canny Edge Detection applied to the cosmic ray image.



8. Introduction to Cloud Chambers

8.1 Overview of Clouds and Their Role in the Climate System

Clouds play a crucial role in Earth's climate system by regulating the planet's radiation budget. They reflect and scatter incoming solar radiation while absorbing and re-emitting infrared radiation from the Earth's surface. Additionally, clouds are fundamental to the hydrological cycle, as they facilitate precipitation.

The formation of clouds depends on **hygroscopic nuclei** (aerosols), which act as seeds for water vapor condensation. The microphysical properties of clouds—such as droplet size, concentration, and liquid water content—determine their radiative effects. Variations in aerosol composition, humidity, and turbulence influence cloud formation, making it a complex and dynamic process.

8.2 Studying Cloud Processes: Methods and Challenges

Understanding cloud dynamics requires a combination of:

- **Satellite observations** (limited by temporal resolution)
- **Ground-based measurements** (limited by spatial coverage)
- **Model simulations** (helpful but require validation)
- **Laboratory experiments** (controlled conditions for detailed analysis)

Among these, **cloud chambers** provide a unique experimental approach to simulate atmospheric processes under controlled conditions, bridging gaps between theoretical models and real-world observations.

8.3 Evolution of Cloud Chambers

The first cloud chamber was developed by **C.T.R. Wilson in 1911** to study particle tracks, but its potential for cloud physics research was soon recognized. Over time, cloud chambers evolved into sophisticated tools for atmospheric studies:

- **Expansion Cloud Chambers** – Early designs for nucleation studies
- **Continuous Flow Chambers** – Enabled long-duration experiments with aerosols
- **Turbulence-Based Chambers** – Simulated atmospheric mixing effects
- **Wind Tunnels** – Studied cloud dynamics under varying airflow conditions

Modern cloud chambers, such as the **Manchester Ice Cloud Chamber**, have been used to investigate ice particle scattering, droplet nucleation, and turbulence effects.

8.4 Key Applications of Cloud Chambers

Cloud chambers help researchers study:

- **Nucleation Mechanisms** – How aerosols trigger droplet formation
- **Turbulence Effects** – Impact on droplet distribution and precipitation
- **Light Attenuation** – How clouds scatter and absorb radiation
- **Artificial Precipitation** – Techniques like laser-induced condensation (as an alternative to traditional cloud seeding with silver iodide)

8.5 Objectives of Cloud Chamber Experiments

The primary goals of experimental studies using cloud chambers include:

- Observing **temperature gradients** and their effect on droplet formation
- Analyzing **droplet size distribution** under varying turbulence conditions
- Measuring **light attenuation** due to changing droplet concentrations
- Validating theoretical models with empirical data

8.6 Significance in Climate and Weather Research

Cloud chambers provide critical insights into:

- **Cloud-Climate Feedback Mechanisms** – How clouds influence global warming
- **Precipitation Efficiency** – Factors affecting rain and snowfall
- **Aerosol-Cloud Interactions** – The role of pollution in altering cloud properties

By replicating atmospheric conditions in a controlled setting, cloud chambers enhance our understanding of cloud microphysics, helping improve weather forecasting and climate models.

8.7 Future Prospects

Advancements in cloud chamber technology, such as **laser-based condensation techniques** and high-resolution sensors, will further refine our ability to simulate real-world cloud processes. Continued experimentation will address unresolved questions about cloud dynamics, turbulence effects, and aerosol-cloud interactions, contributing to more accurate climate predictions.

This chapter sets the foundation for understanding how cloud chambers serve as vital tools in atmospheric science, enabling researchers to explore complex cloud processes in a controlled laboratory environment.



9. Types of Cloud Chambers

9.1 Diffusion Cloud Chambers

Diffusion cloud chambers represent a significant evolution from expansion-type chambers, enabling continuous operation without mechanical pressure changes. This section details their operating principles, design characteristics, and experimental applications.

9.1.1 Operating Principle

Diffusion chambers create supersaturated vapor through vertical temperature stratification:

- **Cold base:** Maintained at -20°C to -40°C (liquid nitrogen/Peltier cooling)
- **Warm top:** Ambient temperature ($\sim 20^{\circ}\text{C}$)
- **Alcohol reservoir:** Ethanol/isopropanol-saturated felt liner

Alcohol vapor diffusing downward encounters decreasing temperatures, reaching supersaturation ($S = P_v/P_{\text{sat}}(T) > 1$) near the cold plate. Charged particles ionize vapor molecules, creating nucleation sites for visible droplet tracks ($10\text{-}100\ \mu\text{m}$).

9.1.2 Key Components

Table 9.1.1: Design Components of Diffusion Chambers

Component	Function
Metal Base Plate	High-thermal-conductivity (Cu/Al) cold surface
Transparent Enclosure	Glass/acrylic walls for observation
Felt Liner	Alcohol reservoir at chamber top
Source Port	Insertion point for radioactive materials
HV Electrode	Enhances track density (5-15 kV)
Insulation Layer	Maintains temperature gradient

9.1.3 Operational Characteristics

- **Sensitive Volume:** Typically $10\times 10\times 5\ \text{cm}$ (height-dependent)
- **Temperature Gradient:** $10\text{-}20^{\circ}\text{C}/\text{cm}$ vertical stratification
- **Startup Time:** 5-10 minutes for vapor equilibration
- **Operation Duration:** 2-4 hours per liquid nitrogen fill

- **Track Lifetime:** 0.1-1 second visibility window

9.1.4 Advantages over Expansion Chambers

- Continuous operation without expansion cycles
- No moving parts or pressure systems
- Lower maintenance requirements
- Atmospheric pressure operation
- Real-time particle tracking capability

9.1.5 Experimental Applications

Diffusion chambers enable investigations of:

- Particle track curvature under magnetic fields ($p = qBR$)
- Radon detection via alpha decay counting
- Rutherford-style scattering using metal foils
- Energy loss studies with material shielding
- Nucleation thresholds via corona discharge

9.2 Atmospheric Cloud Chambers

Atmospheric cloud chambers are specialized laboratory systems designed to simulate cloud micro-physical processes under controlled conditions. Unlike particle detection chambers, these focus on replicating atmospheric phenomena such as droplet formation, precipitation mechanisms, and aerosol-cloud interactions. The chamber described by Goti et al. exemplifies this category, enabling experimental studies of turbulence effects on cloud dynamics.

9.2.1 Design and Construction

- **Dimensions:** 0.55 m × 0.6 m × 1 m acrylic enclosure (5 mm thickness)
- **Temperature control:** Dry ice cooling system achieving -8°C at base (simulating mid-level clouds)
- **Humidity system:** Steam injection at 0.8 m height via boiler (70°C water vapor source)
- **Turbulence generation:** Variable-speed fan (2.8–14.4 m/s) for controlled mixing
- **Observation ports:** Laser windows and optical access points

9.2.2 Instrumentation

Table 9.2.1: Measurement Systems in Atmospheric Cloud Chambers

Instrument	Function
Hygrometer	Temperature/humidity profiling ($\pm 0.1^\circ\text{C}$, -40–100°C)
He-Ne Laser (632.8 nm)	Light attenuation studies
Lux Meter	Scattered light intensity measurement
DSLR Camera (18 MP)	Droplet imaging and size analysis
ImageJ Software	Droplet distribution quantification

9.2.3 Key Experimental Capabilities

- **Turbulence-droplet interactions:** Quantify collision-coalescence processes
- **Light attenuation studies:** Measure scattering dependence on droplet concentration
- **Temperature-humidity profiling:** Vertical gradient analysis (0–80 cm height)

- **Droplet size distribution:** Gaussian distribution fitting under varying turbulence

9.2.4 Characteristic Findings

9.2.4.1 Temperature-Humidity Profiles

- Near-linear temperature gradient from -6°C (base) to 23°C (top)
- Humidity $\geq 99\%$ throughout chamber during operation
- Gradient steepness inversely proportional to turbulence:

$$\Delta T_{\text{low}} = 29^\circ\text{C} (2.8\text{ m/s})$$

$$\Delta T_{\text{high}} = 27^\circ\text{C} (14.4\text{ m/s})$$

9.2.4.2 Light Attenuation

Turbulence reduces scattering efficiency:

Table 9.2.2: Turbulence Effects on Light Transmission

Turbulence (m/s)	Luminosity (lux)	Attenuation (%)
2.8	70	76.67
7.2	103	65.67
14.4	125	58.34

9.2.4.3 Droplet Size Distribution

- Gaussian distribution confirmed for all turbulence regimes
- Mean droplet size increases with turbulence:

$$r_{\text{low}} = 3.39 \pm 2.83\text{ mm}$$

$$r_{\text{high}} = 3.91 \pm 3.18\text{ mm}$$

- Enhanced collision-coalescence at higher turbulence

9.2.5 Applications and Limitations

9.2.5.1 Research Applications

- Precipitation efficiency studies
- Aerosol-cloud interaction mechanisms
- Cloud radiative properties analysis
- Turbulence-microphysics parameterization

9.3 TRONIC Cloud Chamber

The TRONIC cloud chamber represents an innovative refrigeration-based approach to particle detection, addressing limitations of traditional Wilson chambers through engineered cooling systems. Developed by Lakshminarayana (2014), this design eliminates dependency on dry ice while enabling operation in isolated field conditions ideal for cosmic ray studies.

9.3.1 Design Innovations

- **Compact dimensions:** 510 mm $\times$ 180 mm $\times$ 700 mm glass chamber
- **Refrigeration system:** Copper coils with anhydrous ammonia coolant (NH_3 aqueous solution)
- **Thermal management:** Aluminum heat sink plate + condenser with cooling fan

- **Cooling capacity:** Achieves -16°C (phase transition threshold)
- **Power system:** 12V DC pump for coolant circulation

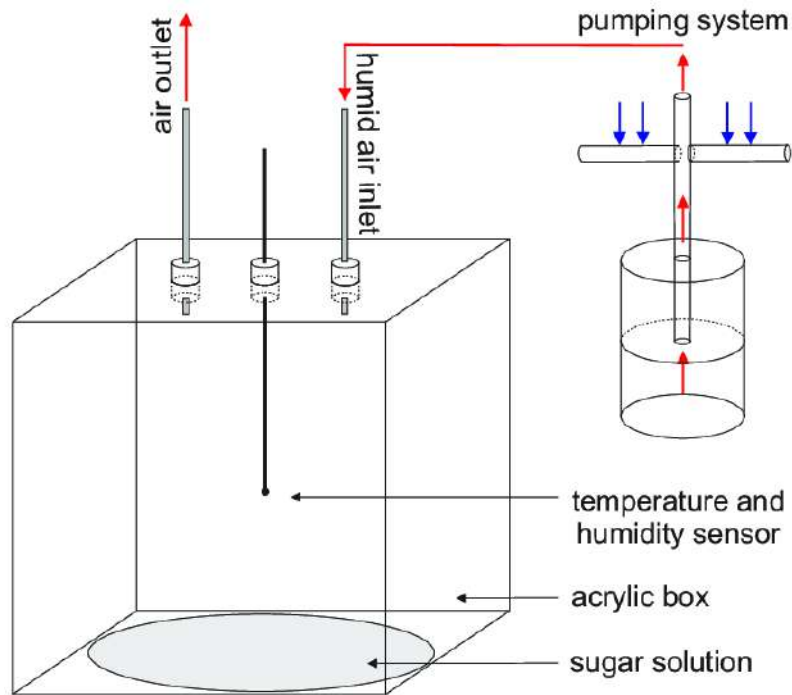


Figure 9.3.1: TRONIC chamber's refrigeration schematic (Lakshminarayana, 2014)

9.3.2 Operating Principle

The system creates supersaturation through thermodynamic cooling:

1. Coolant circulates through copper coils beneath aluminum plate
2. Heat transfers from ethanol-filled chamber to coolant
3. Vertical temperature gradient established (2°C to -16°C)
4. Cosmic rays ionize vapor, forming visible tracks

The cooling profile follows:

$$\frac{dT}{dt} < 0 \quad (\text{until } -16^{\circ}\text{C}) \rightarrow \frac{dT}{dt} \approx 0 \quad (\text{phase transition})$$

9.3.3 Particle Identification

Table 9.3.1: Track Characteristics in TRONIC Chamber

Track Feature	Particle Identification
Thick, short trails	High-energy alpha particles
Long, thin trails	Muons
Thick, extended trails	Beta particles
Branched trajectories	Delta rays (Rutherford scattering)

9.3.4 Performance Advantages

9.3.4.1 Over Dry Ice Chambers

- Eliminates rapid sublimation issues
- Maintains stable temperatures >4 hours
- Eco-friendly (no CO₂ release)

9.3.4.2 Over Electromagnetic Chambers

- No wire-induced particle interference
- Operates without continuous grid power
- Reduced experimental noise

9.3.5 Field Deployment Capabilities

- Portable design for high-altitude cosmic ray studies
- Optimal operation at 4115m altitude (validated)
- Magnetic curvature analysis of low-momentum muons
- Energy loss measurements via track radius variation

9.3.6 Technical Significance

This refrigeration-based approach demonstrates:

- Viable alternative to hazardous coolants
- Scalable miniaturization for field studies
- Sustainable operation in power-constrained environments
- Extended observation windows for rare cosmic events

9.4 Expansion Cloud Chambers

The expansion cloud chamber, pioneered by C.T.R. Wilson in 1911, represents the original design for visualizing ionizing radiation through controlled adiabatic expansion. As documented in the Mound Laboratory report (1949), this chamber type enables precise study of particle interactions through vapor trail formation.

9.4.1 Operating Principle

- **Supersaturation mechanism:** Adiabatic expansion reduces pressure, causing saturated vapor to become supersaturated
- **Ion visualization:** Charged particles create ion pairs that act as condensation nuclei
- **Track formation:** Vapor condenses into visible droplets along particle paths (0.1-1 second visibility)
- **Cyclic operation:** 15-20 second cycles for droplet clearance and equilibrium restoration

9.4.2 Core Components

9.4.3 Key Technical Parameters

- **Expansion ratio:** 1.2-1.4 (adjustable via rubber ring thickness)
- **Sensitive volume:** Typically 30-40 cm diameter × 30 cm depth
- **Gas media:** Hydrogen, helium, nitrogen, oxygen, argon, or methane
- **Operating pressure:** Variable for energy tuning (low for low-energy studies)
- **Magnetic fields:** 1000-2000 Gauss for momentum analysis

Table 9.4.1: Essential Elements of Expansion Chambers

Component	Function
Glass enclosure	Visible track observation
Rubber diaphragm	Pressure control mechanism
Circular glass rim	Gas/vapor containment
Magnetic valves	Synchronized pressure control
Colloidal graphite ring	Ion sweeping electrode
Flashing illumination	High-intensity track lighting (7000K)

9.4.4 Particle Measurement Capabilities

9.4.4.1 Specific Ionization

- Alpha/protons: Dense, straight tracks
- Electrons/positrons: Thin, wispy trails
- Neutrons/gamma: Indirect detection via secondary particles

9.4.4.2 Momentum Analysis

$p = 0.3BR$

Where p = momentum (GeV/c), B = magnetic field (Tesla), R = track curvature (meters). Momentum change measured via lead plate insertion.

9.4.4.3 Mass Determination

Combines measurements of:

- Specific ionization density
- Particle range
- Magnetic curvature
- Momentum change rate

9.4.5 Nuclear Reaction Studies

- Conservation law verification (energy/momentum)
- Resonance level analysis of excited nuclei
- Alpha/beta/gamma spectra characterization
- Neutron energy measurement via proton recoils

9.4.6 Design Innovations

Figure 9.4.1: Modular expansion chamber design with interchangeable components (Mound Lab, 1949)

- **Variable expansion:** Adjustable rubber ring compression
- **Gas interchange system:** Multiple gas inlets for different stopping powers
- **Stereoscopic imaging:** Prism-mirror camera system for 3D track reconstruction
- **Electronic control:** Automated expansion, lighting, and camera sequencing



10. Building your own Cloud Chamber

10.1 Materials Used

- Acrylic Sheets (12"x6")
- Acrylic Sheets (12"x12")
- Aluminium Tape
- Isopropyl Alcohol
- Heat Sink (150 x 60 mm)
- Heat Sink Gel
- Foam
- Glue
- Double Tape
- Camera

10.2 Basic Principles

The cloud chamber works by creating a supersaturated environment where isopropanol vapor is in a gaseous state at a temperature it normally cannot exist at. This supersaturation is achieved by evaporating isopropanol from a foam at the top of the chamber and allowing it to sink towards a very cold heat sink at the bottom, which is cooled by the heat sink gel. When an electrically charged particle crosses this supersaturated vapor, it ionizes the gas molecules along its path by tearing away electrons, leaving these molecules positively charged. These charged molecules then act as condensation nuclei, causing small droplets of alcohol to form along the particle's path. The visible accumulation of these droplets forms the "tracks" that can be observed.

10.3 Working of our model

To operate the cloud chamber, first, prepare the heat sink; wrap it with aluminium tape ensure a high-contrast background and conductivity for viewing white particle tracks. Next, attach the foam to the bottom of the acrylic container using split pins. Apply heat sink gel to the heat sink and let it cool down for sometime. With plastic gloves and safety goggles on, soak the felt inside the acrylic container with pure isopropanol; tilt the chamber to ensure the felt is fully saturated. Place the heat sink, ensuring the felt inside the acrylic container is already soaked. Finally, place the acrylic container upside down onto the cooled heat sink, fitting its walls into any grooves if present, which helps seal the box. After a few minutes for cooling and the formation of a stable sensitive area, a

rain-like mist of alcohol will be visible. Turn off room lights and use torch lights to illuminate the alcohol mist directly above the metal plate. Within minutes, tracks of high-energy particles should become visible, appearing like thin spider's threads. Capture the images of the trails formed with the camera. Additional alcohol can be added through holes in the top of the box if needed.

The cloud chamber design involves a clear container made with acrylic sheets with foam attached to its bottom. This container is inverted and placed on a Heat sink. The Heat sink is filled with the heat sink gel which is cooled down. The foam inside the acrylic container is soaked with isopropanol. A light source illuminates the sensitive area just above the metal plate to make the tracks visible against the black background. The design is based on the "continuously sensitive diffusion cloud chamber" concept, utilizing heat sink to cool down the bottom of the chamber.

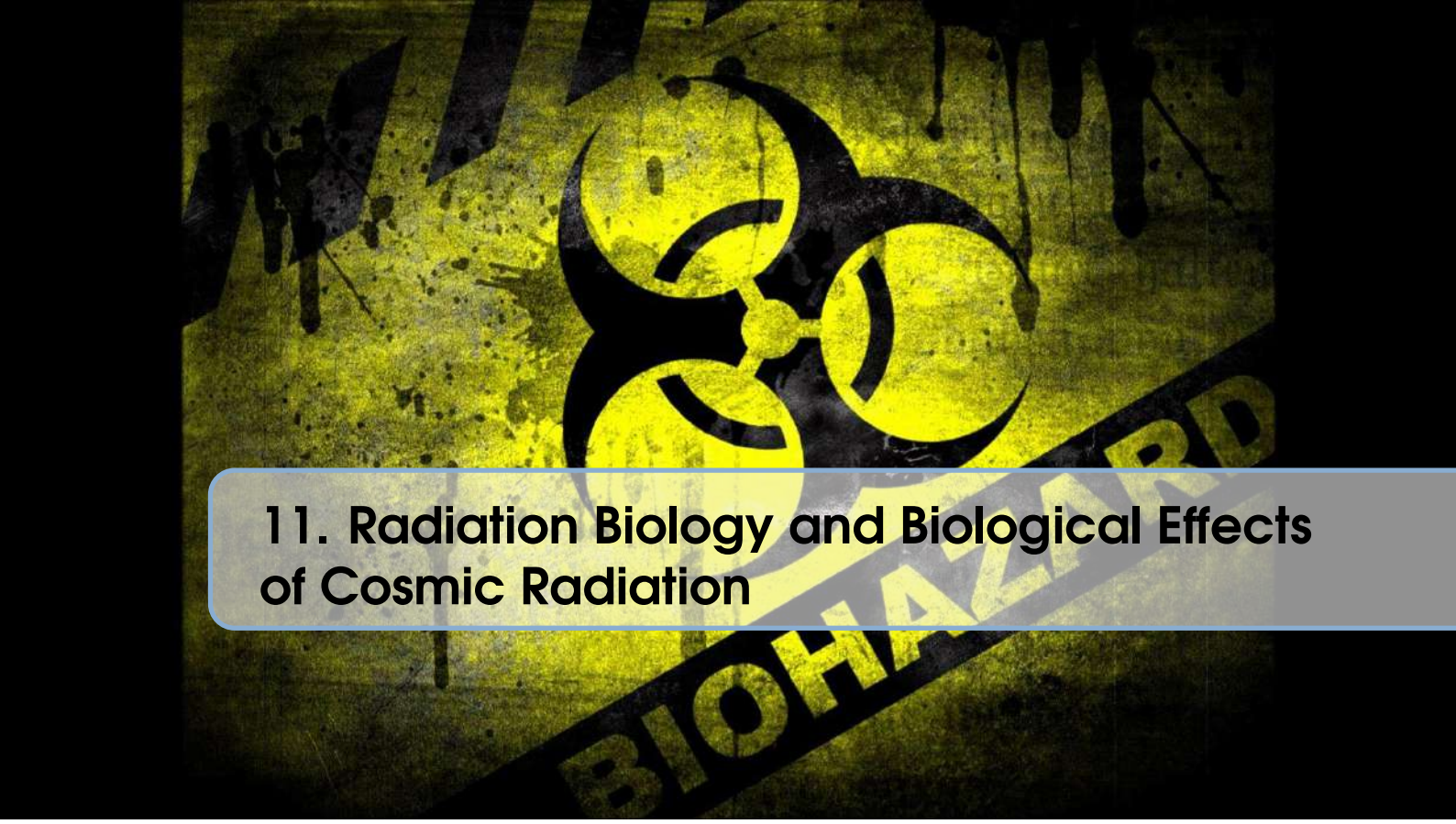
10.4 Gallery



Figure 10.4.1: Making of the Cloud Chamber



Figure 10.4.2: Finalised Acrylic structure



11. Radiation Biology and Biological Effects of Cosmic Radiation

11.1 Space Radiation Environment

The space radiation environment is primarily composed of:

- **Galactic Cosmic Rays (GCR):** High-energy protons and heavy nuclei (up to iron and beyond), originating outside the solar system. These particles exhibit energies from ~ 100 MeV to above 10 GeV and have high penetration capability.
- **Solar Energetic Particles (SEP):** Sporadic solar flares and coronal mass ejections release bursts of protons and heavier ions, which can dramatically increase radiation levels for hours to days.
- **Trapped Radiation Belts:** The Van Allen belts in low-Earth orbit, consisting of protons and electrons trapped by Earth's magnetic field, are relevant for ISS missions.

Radiation dose levels in deep space are significantly higher than those near Earth. Astronauts on a six-month deep-space mission can accumulate doses of ~ 1000 mSv, far exceeding typical terrestrial occupational limits.

11.2 Radiation Interactions with Biological Matter

Cosmic rays deposit energy in biological tissues via ionization and excitation events. Two major mechanisms dominate:

1. **Direct Damage:** Charged particles directly ionize DNA or other biomolecules, leading to strand breaks and base damage.
2. **Indirect Damage:** Ionization of water molecules produces reactive oxygen species (ROS), which chemically damage biomolecules.

High-energy heavy ions (HZE) have high linear energy transfer (LET), producing dense ionization tracks and complex DNA damage clusters. These clustered lesions are more difficult to repair and are more mutagenic than the sparse damage caused by low-LET X-rays or gamma radiation.

The **Relative Biological Effectiveness (RBE)** of HZE particles is significantly higher than that of low-LET radiation, with a peak in the LET-RBE curve corresponding to dense track structures.

11.3 Cellular and Molecular Damage Mechanisms

At the molecular level, ionizing radiation induces:

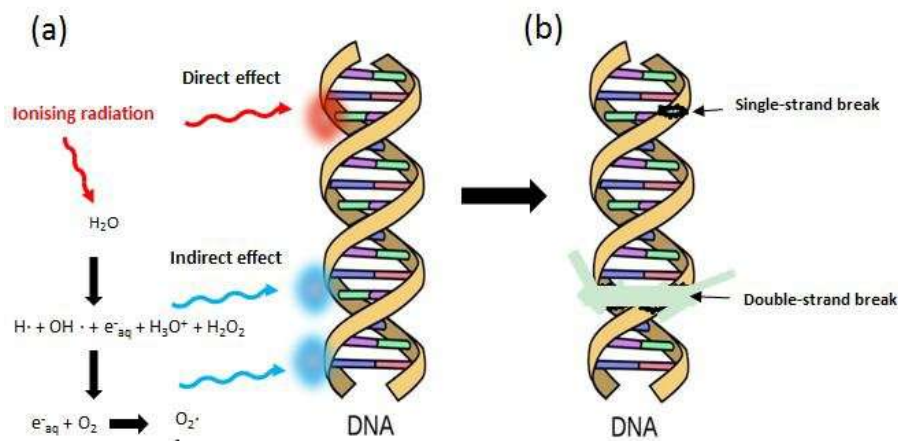


Figure 11.2.1: (a) Schematic of how ionising radiation utilised for radiotherapy can damage DNA. Radiation can directly damage DNA or indirectly damage it through the generation of reactive oxygen species (ROS). (b) DNA damage can occur as a result of single-strand breaks (SSB), double-strand breaks (DSB) or other interactions with DNA and proteins.

- DNA single-strand breaks (SSBs) and double-strand breaks (DSBs)
- Base damage and abasic sites
- DNA-protein crosslinks and clustered lesions

Cells rely on repair pathways such as non-homologous end joining (NHEJ), alternative end joining (alt-NHEJ), and homologous recombination (HR). Exposure to HZE particles often shifts repair towards error-prone pathways, promoting chromosomal rearrangements.

Non-targeted effects such as bystander signaling and delayed genomic instability have been observed. Irradiated cells may transmit damage signals to neighboring cells, enhancing tissue-level effects beyond the initial particle tracks.

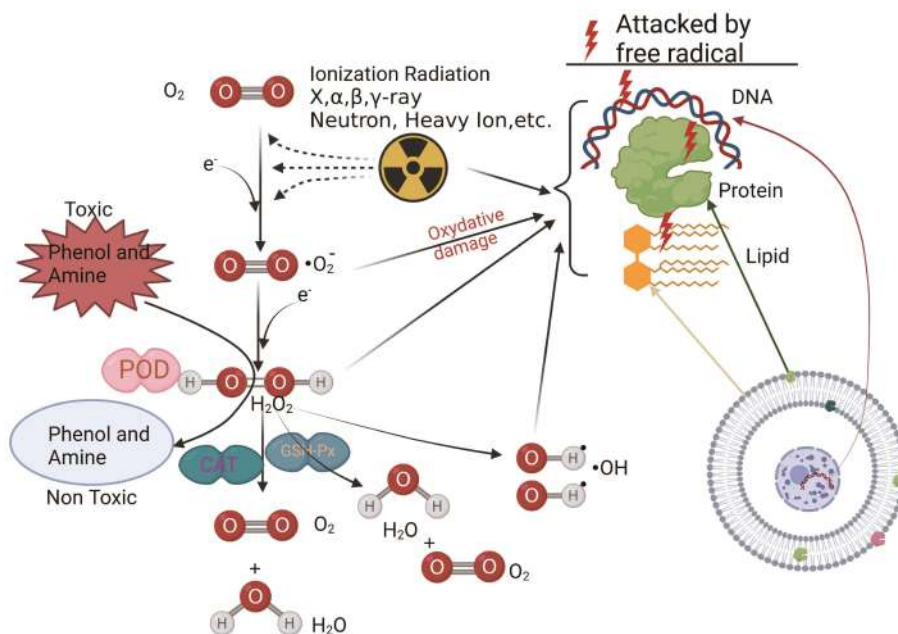


Figure 11.3.1: The Role of DNA Damage Induced by Low/High Dose Ionizing Radiation in Cell Carcinogenesis

11.4 Tissue and Organismal Effects

Radiation damage propagates from the cellular level to tissue and organismal effects:

- **Acute Effects:** Extremely high doses (rare in space) can cause acute radiation syndrome (ARS), with nausea, immune suppression, and hematopoietic depletion.
- **Cancer:** HZE particle radiation has strong carcinogenic potential, producing chromosomal aberrations and oncogenic fusion events (e.g., EML4–ALK in lung cells).
- **Central Nervous System (CNS):** Prolonged exposure to GCR can impair cognition, reduce neurogenesis, and possibly promote Alzheimer-like pathology.
- **Degenerative Conditions:** Chronic exposure accelerates aging phenotypes, promotes cardiovascular disease, induces cataracts, and can lead to bone and muscle loss.

Animal studies have shown that heavy-ion radiation leads to persistent oxidative stress, tissue senescence, and reduced organ functionality even months after exposure.

11.5 Experimental and Epidemiological Studies

Key insights into radiation biology come from:

1. **Astronaut Studies:** Personal dosimeters, blood samples, and the NASA Twins Study provide data on cumulative exposure and biomarker changes.
2. **Accelerator-Based Simulations:** Heavy-ion beams (e.g., ^{12}C , ^{28}Si , ^{56}Fe) simulate GCR components, measuring tumorigenesis, DNA damage, and cognitive impairment in rodents.
3. **In Vitro Experiments:** Human and animal cells irradiated with high-LET ions exhibit increased chromosomal translocations, complex DSBs, and bystander signaling.
4. **Epidemiology:** Studies of high-altitude pilots and residents in high natural background regions provide long-term cancer and cataract incidence data.

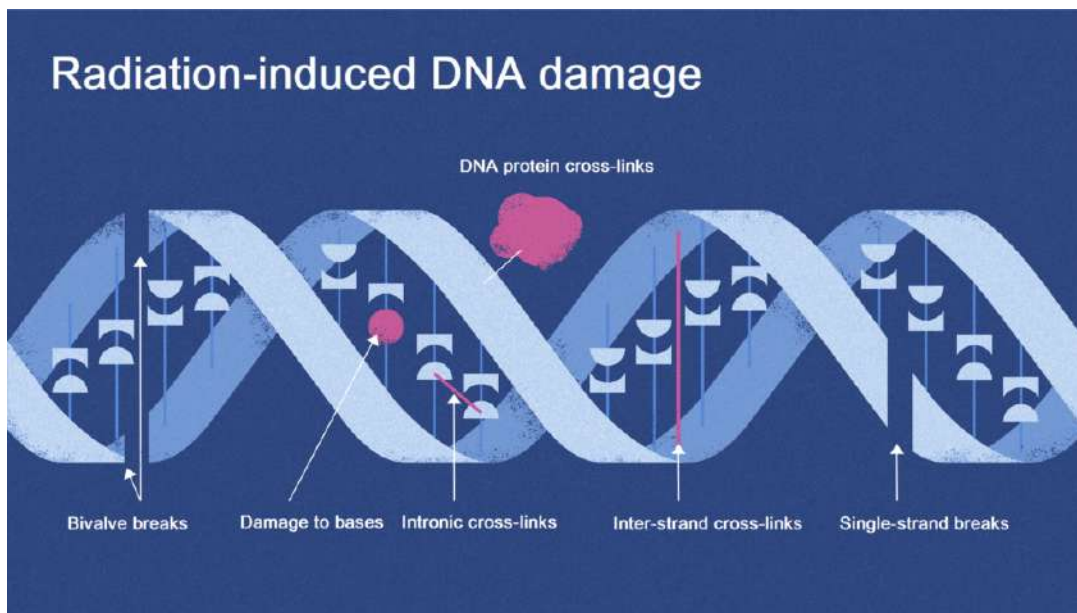


Figure 11.5.1: DNA Damage & Repair due to Radiation

11.6 Protection and Countermeasures

Radiation risk mitigation in deep-space missions involves:

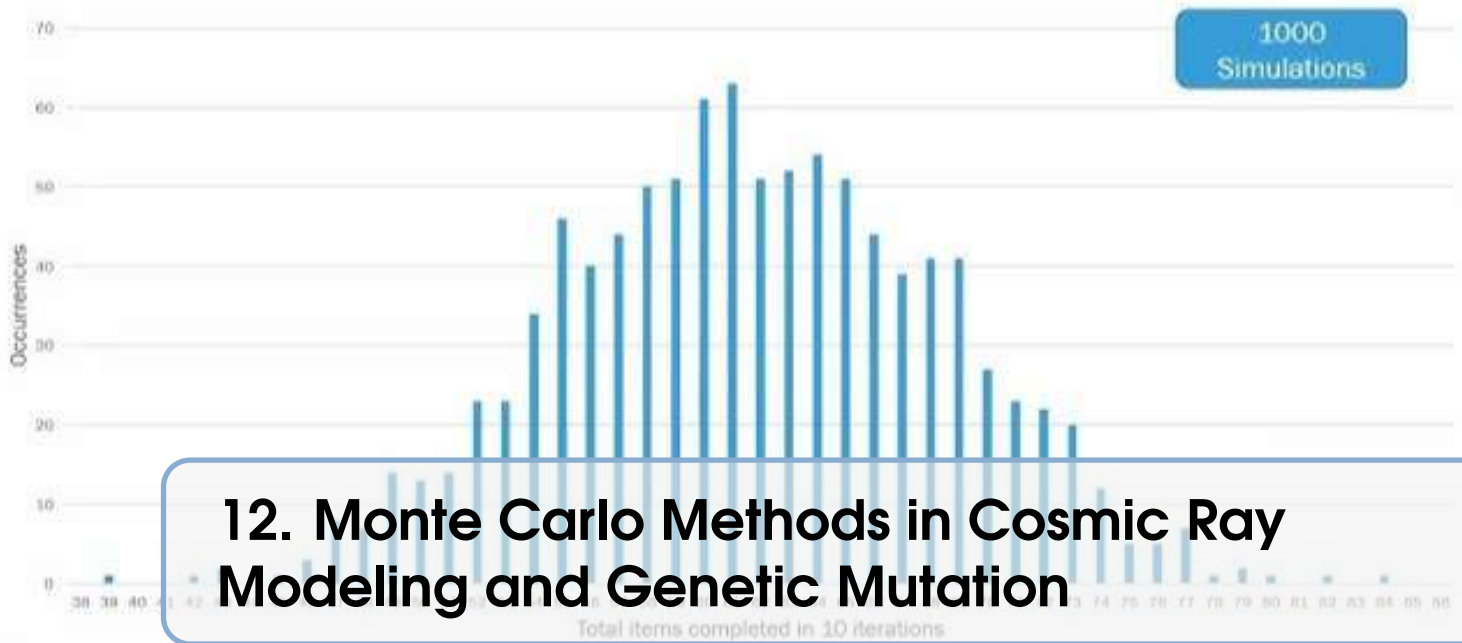
- **Physical Shielding:** Hydrogen-rich materials (e.g., polyethylene, water) reduce GCR and SEP exposure. Shielding is limited against high-energy HZE ions.
- **Operational Measures:** Mission planning around solar cycles, real-time dosimetry, and storm shelters during SEP events.
- **Pharmacological Approaches:** Radioprotectors and antioxidants (e.g., melatonin, immune modulators) under investigation to mitigate DNA and ROS damage.
- **Monitoring and Dosimetry:** Continuous measurement of absorbed dose and biological dosimetry (e.g., chromosomal assays, biomarkers) for real-time risk assessment.

Future approaches include **active magnetic shielding** concepts and gene-level biomedical countermeasures, although these remain experimental.

11.7 Summary

Space radiation biology bridges physics, molecular biology, and medicine. Cosmic rays induce complex, persistent biological damage that elevates risks of cancer, CNS impairment, and degenerative diseases. Understanding these processes through astronaut studies, accelerators, and in

vitro models informs the development of shielding, pharmacological countermeasures, and mission planning strategies. This knowledge forms the biological foundation for the Monte Carlo risk simulations in the subsequent chapter.



12.1 What is Monte Carlo Simulation?

Monte Carlo methods are a class of computational algorithms that rely on repeated random sampling to obtain numerical results. These techniques are especially useful for modeling systems with a high degree of uncertainty or a large number of variables. Rather than solving deterministic equations directly, Monte Carlo simulations approximate solutions by exploring a wide range of outcomes based on probabilistic rules.

- **Principle:** At the core of Monte Carlo simulations lies the Law of Large Numbers. As the number of trials increases, the average of the results converges to the expected value.
- **Usage:** Monte Carlo methods are employed in fields as diverse as finance, nuclear physics, engineering, and biology. They are indispensable in evaluating integrals, modeling stochastic processes, and simulating the effects of random variables.

12.2 Where Monte Carlo Methods Are Used

Monte Carlo simulations have demonstrated their versatility in a range of domains:

- **Estimating π :** Using random points in a square and checking how many fall inside a unit circle.
- **Radiation Transport:** Simulating how particles like neutrons or cosmic rays interact with matter.
- **Optimization:** In stochastic or probabilistic algorithms (e.g., Simulated Annealing).
- **Bayesian Inference:** Using MCMC (Markov Chain Monte Carlo) to sample from complex posterior distributions.
- **Quantifying Uncertainty:** In climate modeling, stock market simulations, and physical experiments.

12.3 Monte Carlo Simulation of Cosmic Ray Interactions

Cosmic rays, especially high-energy particles from space, interact with spacecraft materials and biological tissues in complex and random ways. Monte Carlo simulations allow us to model these interactions in silico:

- **Particle Propagation:** Simulations track the stochastic paths of incoming cosmic rays through shielding materials or electronics.
- **Energy Deposition:** Models estimate the distribution of deposited energy along the particle's

Chapter 12. Monte Carlo Methods in Cosmic Ray Modeling and Genetic Mutation

path, a critical step for predicting SEUs or radiation dose.

- **Spallation and Fragmentation:** Monte Carlo codes simulate secondary particle showers resulting from nuclear reactions between cosmic rays and matter.

Monte Carlo radiation transport codes like GEANT4 and FLUKA are extensively used to simulate cosmic ray effects on both spacecraft and astronaut health.

12.4 Genetic Mutation via Cosmic Rays: A Monte Carlo Perspective

Cosmic rays are known to cause DNA strand breaks and chromosomal aberrations, leading to potential mutations. Modeling this process deterministically is nearly impossible due to its randomness and complexity. Monte Carlo methods provide a powerful framework:

- **DNA Damage Simulation:** Tracks ionizing particles as they traverse biological cells, estimating the probability and location of single- or double-strand breaks.
- **Mutation Likelihood:** Probabilistic models can estimate the likelihood of genetic mutations over long-duration spaceflight or in extreme environments like Mars.
- **Space Biology Applications:** Such simulations help assess long-term risks to astronauts and support the design of protective measures.

12.5 The Potential of Monte Carlo in Space Biology

Monte Carlo methods offer profound potential in predicting and mitigating cosmic radiation damage:

- **Predictive Modeling:** Enables long-term forecasts of biological damage on lunar or Martian missions.
- **Personalized Radiation Risk:** Combining astronaut-specific genomics with stochastic models may help personalize shielding or mission duration.
- **Bioinformatics Integration:** Monte Carlo models can be fused with cellular repair simulations and omics data to understand systemic radiation responses.

Monte Carlo simulations, thus, bridge the gap between physics-based models of cosmic radiation and the biological effects they induce, opening avenues for risk assessment, biomedical countermeasures, and safer human exploration beyond Earth.

12.6 Canonical Monte Carlo Examples and Graphs

12.6.1 Estimating e from Probability

Drop uniform $[0,1]$ numbers until their sum exceeds 1. Average number of trials $\rightarrow$ estimate of $e \approx 2.718$.

12.6.2 Estimating Integrals

Monte Carlo can approximate definite integrals, especially in higher dimensions, by sampling points from the function domain.

12.6.3 Monty Hall Problem

Monte Carlo confirms the counter-intuitive truth that switching doors yields a $2/3$ win probability in the Monty Hall game.

12.6.4 The Drunkard's Walk (Random Walk)

Simulates a particle taking random steps in 1D or 2D. This helps visualize concepts in diffusion, probability, and motion.

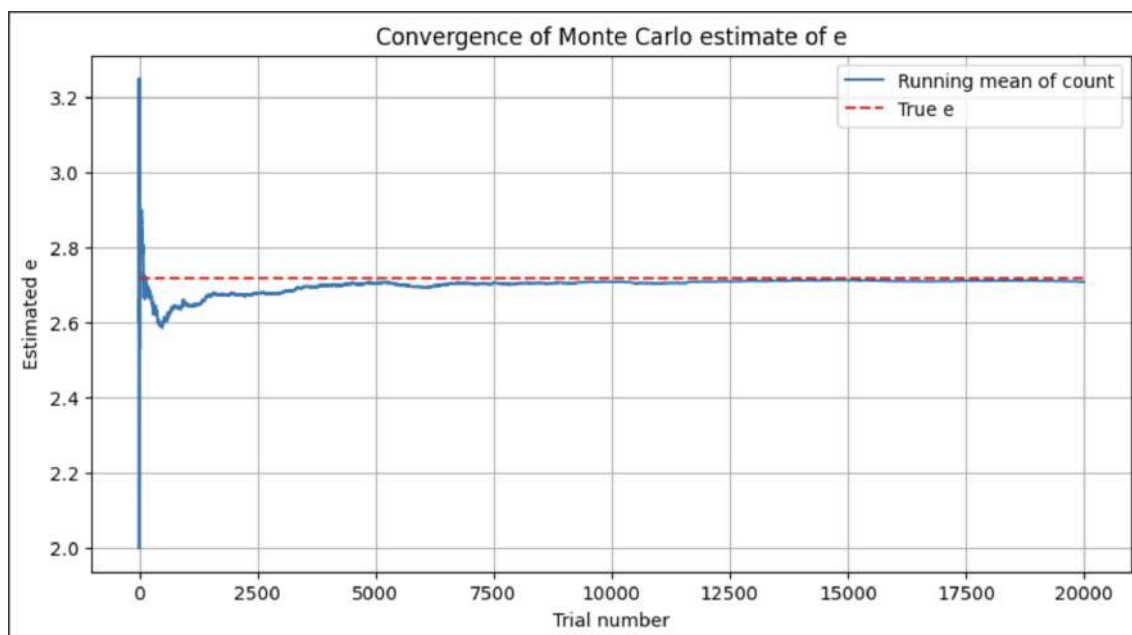
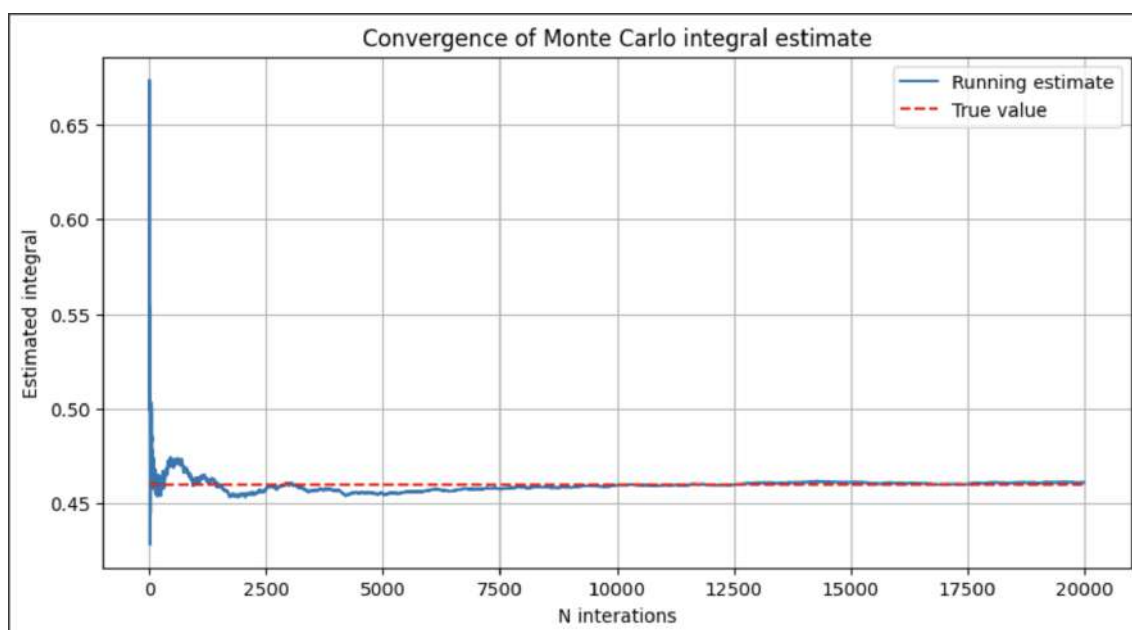
Figure 12.6.1: Convergence of Monte Carlo estimate of e with increase in trials

Figure 12.6.2: Convergence of Monte Carlo integral Estimate

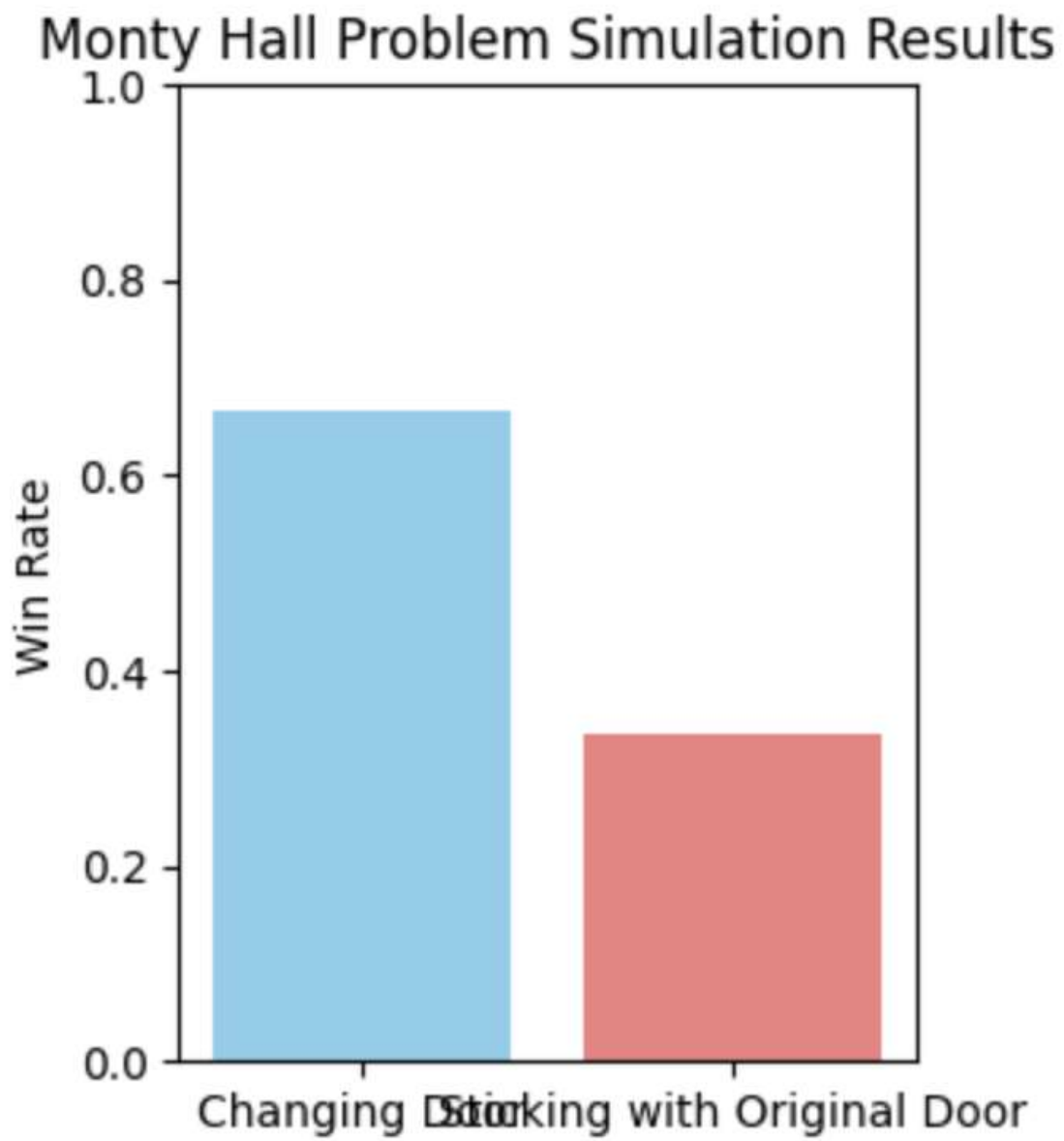
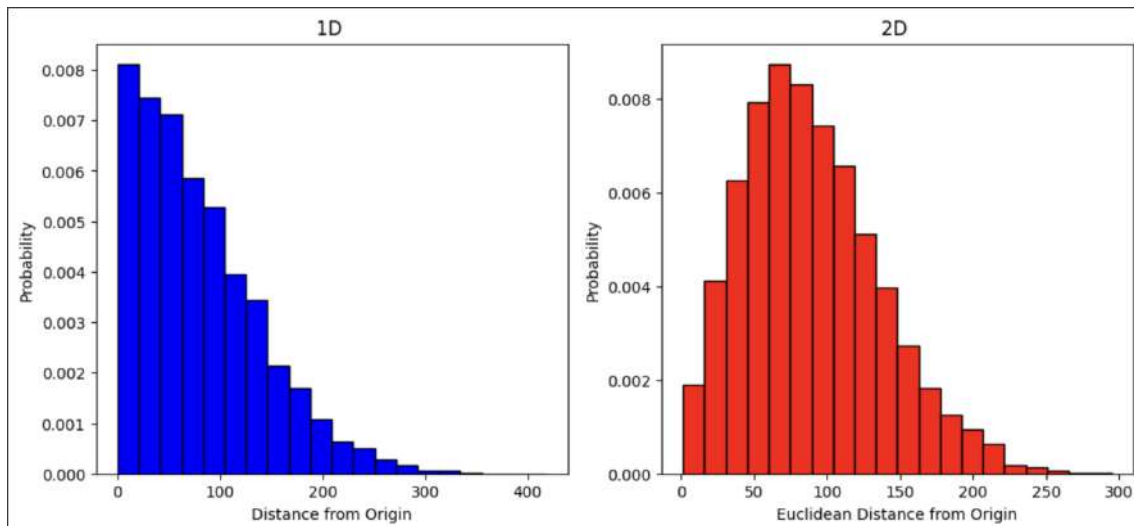


Figure 12.6.3: Monty Hall Problem



Applications:

- **Radiation Transport:** Simulating particle energy loss, scattering, and nuclear reactions (e.g., using GEANT4 or FLUKA).
- **Energy Deposition Maps:** Calculating where and how much energy particles deposit in sensitive electronics.
- **Secondary Particle Production:** Modeling cascades of daughter particles from primary cosmic rays.

12.6.5 Monte Carlo for Genetic Mutation via Cosmic Rays

Mechanism

Ionizing particles from cosmic rays can penetrate biological cells, breaking DNA strands and triggering mutations. Monte Carlo models this probabilistically:

- Simulate ionization events inside cell nuclei
- Model stochastic DNA strand breaks
- Calculate mutation probabilities based on dose and hit distributions

Applications

- **Astronaut Risk Assessment:** Forecasting DNA damage from long-term space travel.
- **Cancer Prediction:** Probabilistic models of mutation accumulation from chronic radiation exposure.
- **Synthetic Biology:** Understanding how high-energy environments might alter cellular genomes.

12.7 Conclusion

Monte Carlo simulations are indispensable tools in both the physical and biological sciences. Their power lies in the ability to represent uncertainty and randomness explicitly—whether estimating π , simulating drunkard walks, or predicting DNA mutations from cosmic rays. In the domain of space exploration, Monte Carlo methods bridge fundamental physics with real-world consequences, helping scientists design safer missions and understand the invisible risks of the cosmos.

13. Electronics Effect

13.1 Single Event Upset (SEU)

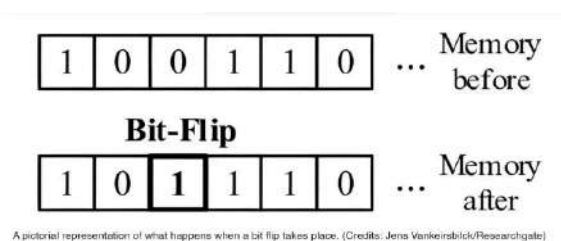
At its core, a Single Event Upset (SEU) is a transient, radiation-induced error in a microelectronic device. Imagine the intricate circuits within a computer chip, where information is stored as tiny electrical charges representing binary 0s and 1s. An SEU occurs when a high-energy particle, often referred to as an ionizing particle, strikes a sensitive region of this circuit. This impact creates a momentary burst of electrical current, which can be strong enough to alter the state of a memory cell or a logic gate, effectively flipping a bit from its intended value (e.g., a 0 becomes a 1, or vice versa). This is why SEUs are frequently called bit-flips.

The Origin of These Energetic Particles

- **Galactic Cosmic Rays (GCRs):** These are extremely high-energy atomic nuclei (like protons, helium, and heavier elements) that originate from outside our solar system, traveling across vast interstellar distances. They are a continuous threat, present throughout space.
- **Solar Cosmic Rays (SCRs):** These particles are emitted by the Sun, particularly during intense solar flares and coronal mass ejections (CMEs). While less energetic than GCRs, their flux (the number of particles passing through a given area) can increase dramatically during solar events, posing a significant, albeit intermittent, threat.

What is a Bit-flip?

Bit flips are a type of unintentional changing of memory data. Computers store data in the form of bits as 0s and 1s. When a piece of data gets bit-flipped, the value of this memory data changes or flips: a 0 becomes 1, and a 1 becomes 0.



The Physics Behind a 'Bit-Flip'

When an ionizing particle penetrates the silicon material of a semiconductor device, it leaves a trail of electron-hole pairs along its path. These charge carriers are then rapidly collected by the electric fields present in the device's sensitive regions (like the depletion region of a transistor). This collection of charge creates a transient current pulse. If this current pulse is sufficiently large and lasts long enough, it can overcome the device's internal electrical stability, causing a change in its logical state. For instance, in a memory cell, this means the stored data bit is inadvertently flipped.

13.2 Consequences of SEUs for Spacecraft

Even a single bit-flip can have cascading and potentially catastrophic effects on a spacecraft's operations. The consequences can range from minor glitches to mission-critical failures:

- **Data Corruption:** SEUs can alter data stored in onboard memory, leading to incorrect calculations, faulty sensor readings, or corrupted command sequences. This can compromise scientific data or navigational information.
- **Software Malfunctions:** The software controlling the spacecraft's various systems can be affected, leading to unexpected program behavior, crashes, or infinite loops. Reloading software from the ground might be necessary, consuming valuable mission time and resources.
- **Central Processing Unit (CPU) Halts:** An SEU can cause the CPU to stop executing instructions, effectively freezing the spacecraft's brain. This results in a loss of operational time and can disrupt critical functions like attitude control or data acquisition.
- **Corruption of Critical Data Tables:** Spacecraft rely on critical data tables for various functions, such as thruster firing sequences or instrument calibration. An SEU can permanently damage these tables, leading to severe operational impacts or even rendering certain systems unusable.
- **Unplanned Events and Mission Termination:** In the worst-case scenarios, SEUs can trigger unintended commands (e.g., turning off vital systems), cause flight software to execute at the wrong time, or even lead to the premature termination of an entire mission.

13.3 Microprocessor Technology and SEU Vulnerability

The relentless march of technological progress has brought us microprocessors that are incredibly powerful, compact, and energy-efficient. However, these very advancements have inadvertently made them more susceptible to SEUs.

Older microchips were relatively large, and their internal components required a substantial amount of electrical charge to switch between states. This made them inherently more robust against the small charge depositions caused by single particles. As technology has advanced, the size of transistors and other components on a chip has shrunk dramatically. This miniaturization means that:

- **Lower Critical Charge:** Modern devices require significantly less electrical charge to change their logical state. This 'critical charge' is the minimum amount of charge needed to flip a bit. A lower critical charge means that even a less energetic particle can deposit enough charge to cause an SEU.
- **Increased Density:** With smaller components, more transistors can be packed onto a single chip. This increases the probability of a particle striking a sensitive area.

Because there is a much higher flux of lower-energy cosmic rays in space compared to very high-energy ones, the increased sensitivity of smaller devices to these more prevalent particles directly translates to a higher likelihood of experiencing an SEU.

The Role of Critical Charge (Q_c)

The critical charge (Q_c) is a fundamental parameter in determining a device's SEU susceptibility. It's directly related to the physical dimensions of the sensitive region within the device. $Q_c = 0.023 \cdot L^2$. Where L represents the longest pathlength through the vulnerable region. This equation highlights that as the physical size (L) of the device's sensitive areas decreases, the critical charge (Q_c) also decreases, making the device more prone to SEUs.

Linear Energy Transfer (LET)

To precisely characterize the impact of an energetic particle, scientists use the concept of Linear Energy Transfer (LET). LET is a measure of the rate at which an ionizing particle loses energy as it travels through a material. It's expressed in units like MeV cm²/mg (Mega-electron Volts per square centimeter per milligram), indicating the energy deposited per unit path length per unit density of the material.

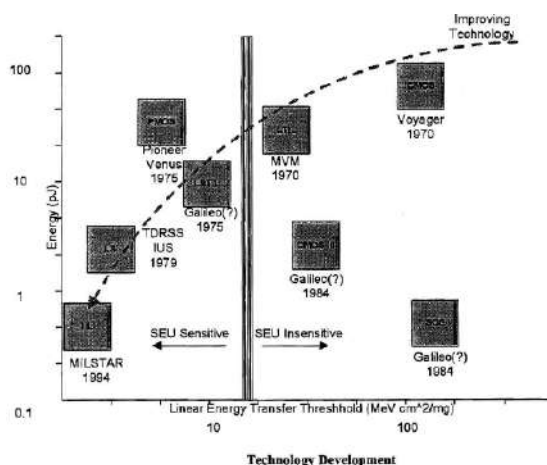
LET and Charge Deposition

The amount of charge deposited by a particle in a semiconductor device is directly proportional to its LET. The relationship is given by $LET = \frac{22.5 \cdot Q}{D_p}$, where 'Q' is the charge deposited (in picocoulombs) and 'Dp' is the particle's pathlength through the sensitive region. This means that particles with higher LET values will deposit more charge, making them more likely to induce an SEU.

Threshold LET

Every device has a 'threshold LET', which is the minimum LET a particle must have to deposit enough charge to cause an SEU. As microprocessors become smaller and their critical charge decreases, their threshold LET also decreases. This means they become vulnerable to a wider range of particles, including the more abundant lower-energy cosmic rays.

This image illustrates the evolution in memory and microprocessors. The y axis represents the energy required to operate the device, and the x axis is the particle energy threshold for SEU effects.



13.4 A Spectrum of Radiation Effects: Single Event Effects (SEE)

While SEU is the most commonly discussed single event phenomenon, it's part of a broader category called Single Event Effects (SEE). These effects represent a spectrum of radiation-induced events, ranging in severity:

- **Single Event Upset (SEU):** As discussed, this is a non-destructive, temporary change in a memory bit or logic state. The device itself is not physically damaged, and the error can

often be corrected by rewriting the data or resetting the affected circuit.

- **Single Hard Error (SHE):** This is a more severe form of SEU where the bit-flip becomes permanent, often due to physical damage to the memory cell. For example, a memory bit might get stuck at a 0 or a 1, regardless of attempts to write new data.
- **Single Event Latchup (SEL):** This is a potentially destructive condition. When an energetic particle strikes, it can trigger a parasitic silicon-controlled rectifier (SCR) structure within the device, leading to a low-impedance path between power and ground. This results in a massive surge of current, which can cause the device to malfunction or even be permanently damaged if not quickly addressed (usually by cycling the power).
- **Single Event Burnout (SEB):** This is a destructive failure mode primarily affecting power transistors. A high-energy particle can cause a localized breakdown, leading to excessive current and thermal runaway, resulting in the physical destruction of the device.
- **Single Event Gate Rupture (SEGR):** This destructive effect occurs in MOSFETs (Metal-Oxide-Semiconductor Field-Effect Transistors). An energetic particle can create a conducting path through the gate oxide, leading to a permanent short circuit and device failure.

Understanding this hierarchy is crucial because an increased SEU rate often implies an increased probability of these more severe and destructive SEE events.

Forecasting the Threat: SEU Rate Calculation

Predicting the rate at which SEUs will occur is a complex but vital task for spacecraft designers. These calculations rely on sophisticated models that integrate various factors:

- **Particle Environment:** This includes the type, energy, and flux of cosmic rays and solar particles in a specific orbital environment (e.g., Low Earth Orbit, Geosynchronous Orbit).
- **Device Characteristics:** This involves the vulnerable cross-section (the effective area a particle can hit to cause an upset), the pathlength distribution (how particles travel through the sensitive region), and the critical charge of the microprocessor.

Models like the Cosmic Ray Effects on Microelectronics (CREME) and CRRES/SPACERAD Heavy Ion Model of the Environment (CHIME) are used to simulate these interactions. The paper highlights that earlier models sometimes overestimated upset rates, and continuous updates based on new measurements of the space radiation environment are necessary for accuracy. Crucially, these models predict that SEU rates will significantly increase with further device miniaturization and during periods of intense solar activity. For instance, a major solar flare can increase particle flux by orders of magnitude, leading to hundreds of upsets per device per day in modern microprocessors.

SEU Mitigation Techniques

Given the increasing vulnerability of microprocessors, engineers employ a range of mitigation techniques to protect spacecraft systems. These methods vary in complexity, cost, and effectiveness:

Mass Shielding

This is the most straightforward approach: surrounding sensitive electronics with dense materials to block or attenuate incoming radiation. However, it comes with significant drawbacks. Shielding adds considerable weight to the spacecraft, increasing launch costs. Furthermore, while it blocks many particles, it can sometimes slow down very high-energy particles just enough for them to deposit more charge in the device, paradoxically increasing the SEU rate for those specific particles.

Software-Based Solutions (Error Detecting and Correcting - EDAC Code)

These techniques involve using software algorithms to detect and correct errors. A common approach includes:

- **Parity Checking:** Adding extra bits (parity bits) to data that allow for the detection of

single-bit errors.

- **Memory Scrubbing:** Regularly reading through memory, checking for errors (using parity or other error-correcting codes), and correcting any detected bit-flips by rewriting the correct data from a redundant source. While effective, frequent scrubbing can consume a significant portion of the processor's time, impacting the performance of other flight software.

Hardware-Based Solutions

These involve designing redundancy directly into the hardware:

- **Lockstepping:** Two identical processors operate in parallel, executing the same instructions simultaneously. Their outputs are constantly compared. If a discrepancy is detected, it indicates an SEU in one of the processors, and the system can either re-execute the instruction or switch to a known good state. This requires precise synchronization between the processors.
- **Redundant CPUs with Voting Logic:** This technique typically uses three (or more) identical processors. All processors perform the same task, and their outputs are fed into a voting circuit. The output that receives the majority vote is considered correct. This approach is highly robust against single-point failures caused by SEUs.

Radiation Hardening (Rad-Hard)

This is the most comprehensive and often most expensive solution. It involves designing and manufacturing microchips specifically to be resistant to radiation effects. This can include:

- **Process Modifications:** Using specialized fabrication processes that create devices less susceptible to charge collection.
- **Layout Techniques:** Designing the physical layout of the chip to minimize sensitive areas or to make them less vulnerable to radiation strikes.
- **Material Choices:** Incorporating radiation-resistant materials. While highly effective, radiation-hardened components are typically more costly, have lower performance compared to commercial off-the-shelf (COTS) parts, and are not always readily available.

13.5 The Crucial Importance of SEU Awareness in Space Mission Design

The insights gained from studying SEUs are not merely academic; they are fundamental to the success and longevity of any space mission. SEU prediction and mitigation techniques are integrated into every phase of mission design and operation:

- **Component Selection:** Choosing microprocessors and memory devices that are either inherently more resistant to radiation or can be effectively protected.
- **Orbit Design:** Understanding the radiation environment of a chosen orbit helps in predicting SEU rates and designing appropriate mitigation strategies.
- **Shielding Evaluation:** Determining the optimal amount and type of shielding to balance protection with weight constraints.
- **Recovery Techniques:** Developing robust software and hardware mechanisms to detect, correct, and recover from SEUs, minimizing their impact on mission objectives.
- **Anomaly Resolution:** When unexpected events occur in space, understanding SEU susceptibility helps engineers quickly diagnose and resolve issues.

Modern small satellites, with their emphasis on compact size and high performance, are particularly susceptible to SEUs. Therefore, a thorough understanding of these phenomena and the implementation of effective mitigation strategies are paramount to ensuring the reliability and success of future space endeavors.

13.6 Limitations in SEU Modeling

Despite significant advancements, predicting SEU rates accurately still faces challenges:

- **Difficulty in Measuring Vulnerable Cross-Sections:** Precisely determining the exact dimensions of the sensitive regions within a complex microchip can be challenging, leading to uncertainties in models.
- **Material Dependence:** Many models, like the original CREME, were developed primarily for silicon-based devices. As new materials like Gallium Arsenide (GaAs) are used in microelectronics, new models or adaptations are required to account for their different material properties.
- **Approximations in LET Spectrum Modeling:** To simplify calculations, the continuous spectrum of particle energies (LET) is often approximated as a step function. While this speeds up computation, it can lead to conservative (overestimated) upset rates. More detailed models require significantly more computational power.

Despite these limitations, continuous research and the use of historical flight data and ground-based testing (though expensive) are constantly refining our understanding and predictive capabilities for SEUs.

13.7 The Impact of Space Weather on Computer Systems

Introduction

This aims to explain, in simple and accessible language, how phenomena originating from space, often referred to as space weather, can influence and sometimes disrupt our everyday electronic devices, such as smartphones and computers. While we commonly attribute device malfunctions to software glitches or hardware failures, there's a fascinating and less-known factor at play: cosmic radiation.

Understanding Space Weather and Cosmic Rays

Our Sun is a dynamic star that continuously emits a stream of tiny particles. Under normal circumstances, these emissions are largely benign. However, during periods of heightened solar activity, such as solar flares or coronal mass ejections (CMEs), the Sun can unleash powerful bursts of highly energetic particles. These particles, along with others originating from beyond our solar system, are collectively known as cosmic rays. Think of it as the Sun having a powerful sneeze that sends particles hurtling towards Earth.

The Mechanism of Disruption: Single-Event Upset

When these high-energy cosmic rays encounter our electronic devices, they can penetrate the protective layers and strike the microscopic components within microchips, particularly those responsible for storing data. This impact can induce a momentary electrical disturbance, leading to a phenomenon known as a single event upset (SEU). An SEU can cause a fundamental change in a data bit – for instance, flipping a '1' to a '0' or vice versa. To illustrate, imagine a critical instruction manual for your device where a single letter is suddenly altered. If that altered letter changes the meaning of an instruction, your device might execute an unintended command, behave erratically, or even cease to function temporarily.

Documented Instances of Space Weather Impact

The impact of space weather on electronics is not merely theoretical; there have been documented cases suggesting its influence:

- **Pacemaker Malfunction:** A notable incident involved a cyber-security researcher whose implanted pacemaker began to malfunction during a flight. Subsequent investigation revealed

corrupted data within the device's tiny computer. While definitive proof is elusive, cosmic radiation was considered a highly probable cause for this bit flip.

- **Aircraft Incident:** In another widely discussed event, a commercial airliner experienced sudden, unexplained drops in altitude. Investigations pointed to erroneous computer data as the trigger for these dangerous maneuvers. Although not conclusively proven, cosmic rays were identified as a potential contributing factor.

Increasing Vulnerability in Modern Technology

The ongoing trend towards miniaturization in electronics, where components become ever smaller and more densely packed, inadvertently increases their susceptibility to cosmic ray interference. As the physical size of transistors and other microchip elements shrinks, the amount of electrical charge required to represent a data bit also decreases. Consequently, a cosmic ray impact needs less energy to induce a bit flip, making modern, compact devices potentially more vulnerable to these space-borne disruptions.

Conclusion

While the concept of space weather affecting our electronics might seem like something out of science fiction, it is a recognized phenomenon. Although such incidents remain relatively rare, and Earth's atmosphere provides a significant protective shield, the continuous advancement in microchip technology means our devices could become more sensitive to these cosmic influences. Nevertheless, engineers and designers are aware of these challenges and incorporate safeguards into electronic systems. This serves as a compelling reminder of the intricate connections between our technological world and the vast, dynamic universe beyond.

13.8 Impact of cosmic rays on satellite communication

Introduction

Satellite communication is a foundational technology that supports global broadcasting, internet, navigation, and emergency response systems. It allows uninterrupted transmission of data, voice, and video signals over large distances, especially in areas lacking physical infrastructure. However, satellites operate in a high-radiation environment where they are continuously exposed to cosmic rays—energetic particles originating from space.

When these particles interact with the Earth's atmosphere, they produce secondary radiation (such as neutrons, protons, and gamma rays), which can penetrate spacecraft shielding and interfere with onboard electronics. This leads to single-event effects (SEEs), ranging from benign data bit flips to severe component failures. In turn, these effects can degrade data integrity, disrupt communication links, and affect signal quality.

Objectives

The purpose of the research topic, *The Impact of Cosmic Rays on Satellite Communication: A One-Year Analysis using Satellite and Observatory Data* is to investigate and analyse the effects of cosmic rays on satellite communication systems over the course of one year. This research aims to provide insights into the extent and nature of cosmic ray-induced disruptions and their implications for satellite communication reliability and performance. The objectives of this research include:

1. Quantifying the frequency and severity of cosmic ray events affecting satellite communication systems during the one-year study period.
2. Assessing the correlation between cosmic ray activity and disruptions in satellite communication links.

Data Sources and Methodology

To conduct this analysis, the study draws upon two primary sources of data:

1. DSCOVR Satellite (Deep Space Climate Observatory):

- Positioned at Lagrange Point 1, about 1.5 million kilometers from Earth.
- Monitors solar wind and space weather, including Earth's radiant energy using its onboard NISTAR instrument.
- Signal strength was recorded continuously at a receiving ground station in Pennsylvania (latitude 39.9°N).
- Monthly averages of signal strength (measured in decibels, dB) were calculated.

2. Pierre Auger Observatory:

- Located in Pampa Amarilla, Argentina (35°12'24"S, 69°18'57"W).
- Designed to detect and analyze extremely high-energy cosmic rays using an array of surface detectors.
- Provided monthly average values of cosmic ray activity throughout 2022, expressed in TCHU (Total Cosmic Ray Units).

The two datasets—cosmic ray activity and satellite signal strength—were time-synchronized to enable precise correlation analysis. Additional attention was given to identifying months with increased radiation activity and correlating them to months with reduced signal quality or increased transmission errors.

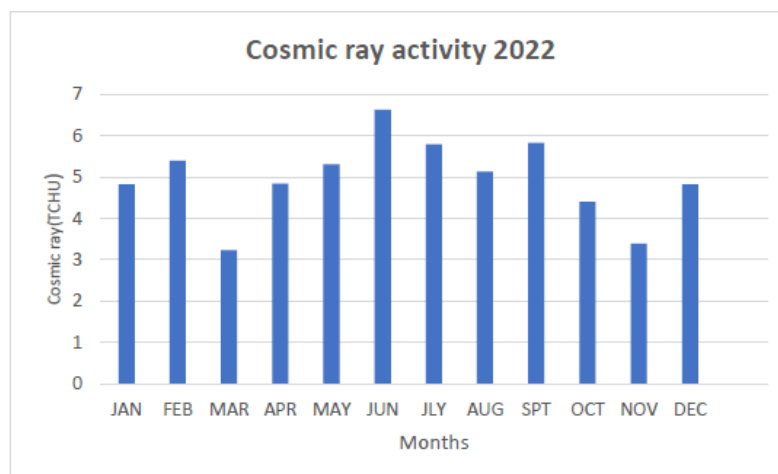
Technical Metrics Used

- **Signal Strength:** Measured in decibels (dB), a logarithmic unit that indicates the power level of satellite transmissions.
- **Total Electron Content (TEC):** Used as an auxiliary indicator of space weather activity. TEC quantifies electron density in the ionosphere and influences how radio signals propagate through space.

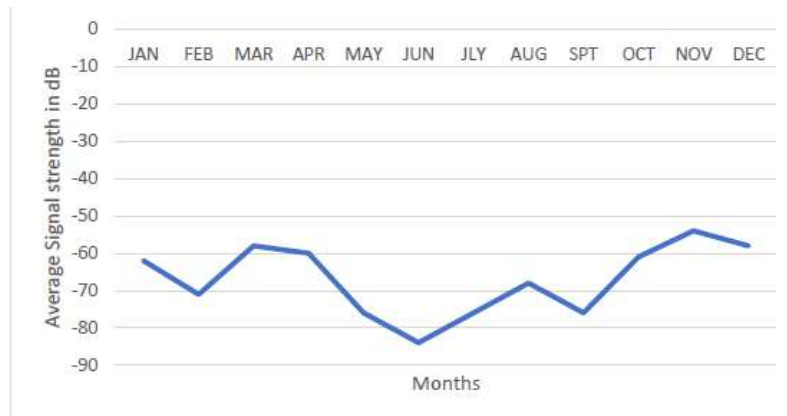
Results

Satellite signal strength exhibited noticeable fluctuations throughout 2022, with the lowest average signal strength recorded in June and July—the months that coincided with peaks in cosmic ray activity.

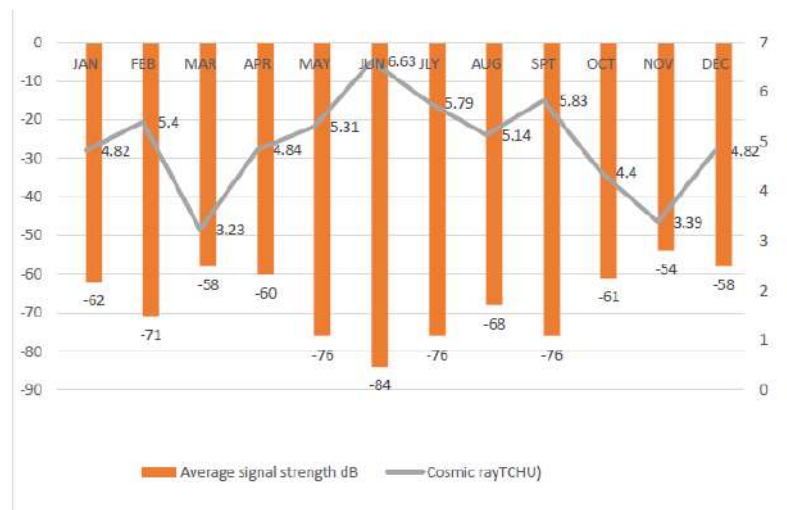
- This image shows a steady monthly variation in cosmic ray flux as measured by the Pierre Auger Observatory.



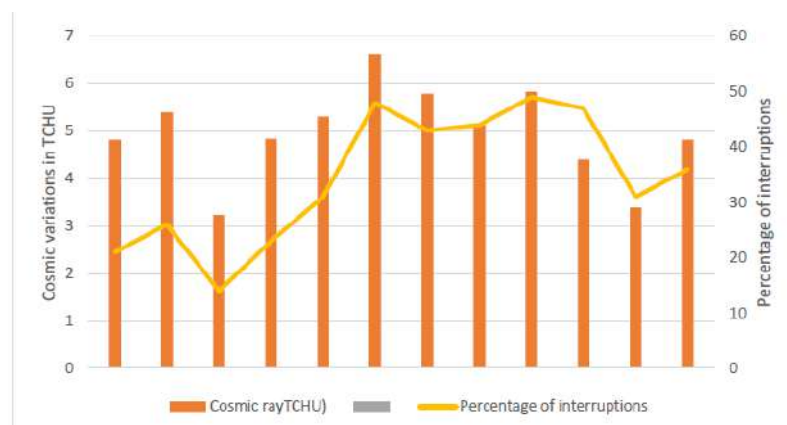
- this illustrates corresponding dips in average satellite signal strength, based on DSCOVR's recorded data.



- This image overlays both datasets and reveals a strong inverse relationship: as cosmic ray activity increases, satellite signal strength tends to decrease.



- This further shows that the percentage of communication interruptions is highest during months of elevated cosmic ray activity.



Discussion

The analysis confirms that cosmic ray variations have a measurable and statistically significant impact on satellite communication quality. These high-energy particles, especially when arriving in elevated bursts during solar storms or galactic events, disrupt the electronic integrity of communica-

tion satellites and induce errors in signal reception.

This correlation emphasizes the role of cosmic weather monitoring in satellite operations. Real-time awareness of cosmic ray conditions could allow operators to switch to redundant systems, reorient sensitive modules, or activate error-correction protocols.

Mitigation Strategies Highlighted

To protect satellite communication from the influence of cosmic radiation, the paper outlines several mitigation strategies that are either currently in use or under research:

- Use of radiation-hardened components and circuits.
- Shielding of sensitive hardware (e.g., flight computers and transponders).
- Implementation of error detection and correction codes.
- Ground-based monitoring of space weather to guide operational adjustments in satellite behavior.
- Building redundancy into communication networks to reroute data when one path is affected.

Conclusion

The study concludes that fluctuations in cosmic ray activity have a clear, direct, and quantifiable impact on satellite signal strength and communication reliability. The findings highlight the necessity of continuous monitoring of space weather phenomena and the development of resilient satellite architectures. By understanding the interaction between cosmic rays and satellite systems, engineers and researchers can better anticipate potential failures and design more robust systems for uninterrupted global connectivity.

13.9 Bit Flips And How Are Spacecraft Protected From Them?

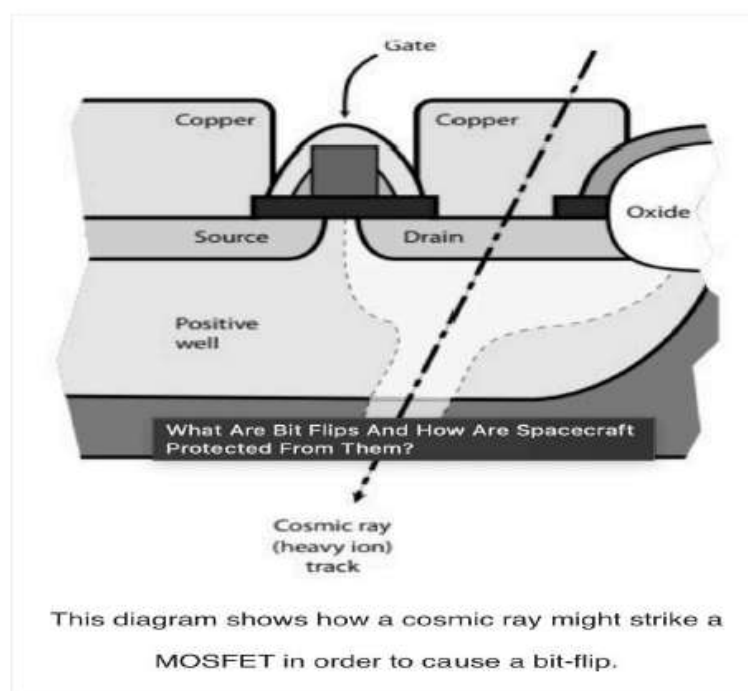
Bit flips fall under the category of 'soft errors'. When a soft error occurs, we can make the required rectification using codes to rewrite the bit value at the place where the fault occurred and get its correct value back. This is different from a hard error, which is usually the result of faulty or damaged hardware. When a hard error happens, the hardware itself needs to change.

Computers on the surface of the Earth are predominantly safe from cosmic rays, since most of them usually end up as muons by the time they reach the Earth's surface. Therefore, computers on the ground don't usually get bit-flipped, but this is not the case with spacecraft traveling in outer space. They are bombarded with cosmic rays, without the disruptive effect of Earth's atmosphere, making them quite vulnerable to bit-flips.

Rectifying Bit Flips

While avoiding cosmic rays can be somewhat impossible for spacecraft once they've left Earth's atmosphere, there are other things we can do to correct a bit flip once it has occurred. Sometimes, rebooting can indirectly clear up the bit-flipped data, resetting it to its original value via memory refresh and reinitialization. However, this technique does not always work, and more robust techniques may be needed.

Sometimes, we use error-correction codes (ECCs) to fix the errors created by bit flips. The codes can detect when a bit flip has taken place, which is usually done by determining the number of 0s or 1s the data contains (provided by the user). If the software detects that there is a mismatch in the number of 0s or 1s from what it received and what the user has provided, it detects the error.



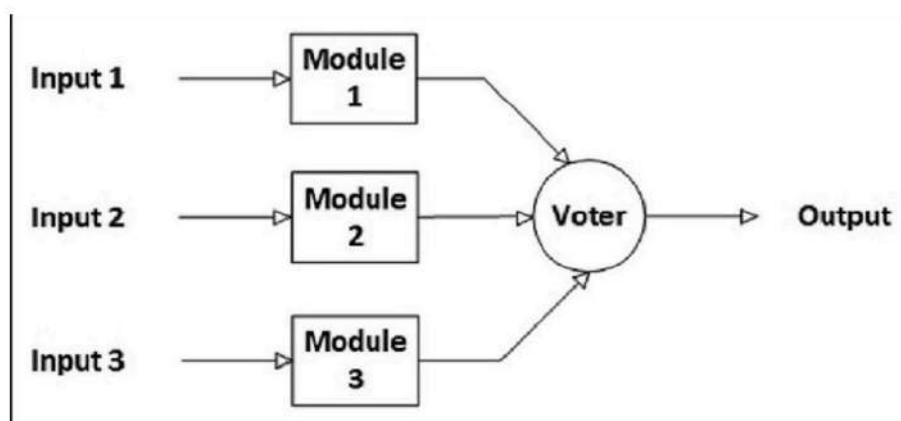
More sophisticated ECCs, like Hamming codes, are also used to rectify the errors caused by bit flips.

Another way that bit flips are handled and amended is by a technique called modular redundancy. Here, the correction is done by repeating the process from where we obtained the data and then conducting a majority vote.

For example, if the data obtained is '1,' then by repeating it three times, we should get '111.' However, suppose a bit-flip occurred, and the data obtained is '110' instead. Since the '1' is still the majority, modular redundancy would tell us that '1' is the correct data for the bit.

Modular redundancy that uses three repetitions is called 3-way modular redundancy or triple modular redundancy.

The computers used in the Shuttle program used five repetitions and were called 5-way modular redundancy. While effective, modular redundancy requires significant mass and power, making its implementation difficult.

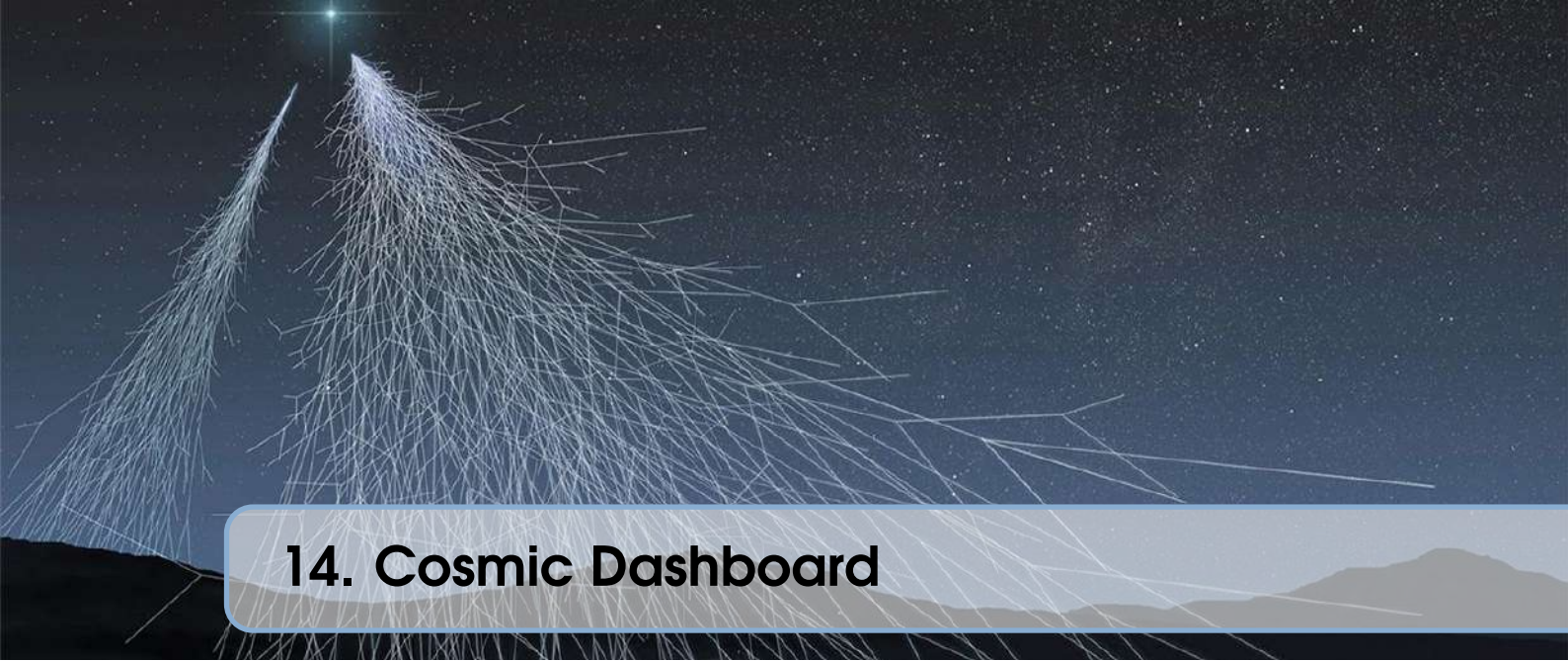


This is a schematic representation of how 3-way modular redundancy works. Here, three inputs are fed (as a result of three repetitions) and the majority value is picked by the 'voter.' (Credits: Arslan Ahmed Amin/Sage Journals)

A Final Word

Outer space is harsh on computers. Just as the atmosphere protects plants and animals on Earth from dangerous rays coming from outer space, it also protects computers and other electronic instruments. However, with the number of space missions increasing and with significant ones like the Mars mission on the books, cosmic ray bit-flipping is an area that must be accounted for very seriously. Outer space is not somewhere you want computers to develop errors and crash at unexpected times.

The good news is that we have found some clever ways to overcome these bit flip issues. Of course, errors in satellites, spacecraft, space telescopes and space stations due to bit-flips do not make any prominent headlines, so while they inevitably happen, we can correct them using appropriate techniques. This way, we can be sure that our space missions are safe, while respecting the undeniable fact that outer space is quite unforgiving to nearly everything!



14. Cosmic Dashboard

14.1 Radiation Risk Calculator

Space exploration, while inspiring and scientifically invaluable, exposes astronauts to significant health risks, chief among them being radiation. Unlike the protected environment on Earth, spacecraft venturing beyond its atmosphere are subject to a constant barrage of energetic particles from cosmic and solar sources. Understanding, predicting, and mitigating this radiation exposure is paramount for ensuring astronaut safety on long-duration missions to the Moon, Mars, and beyond. The Radiation Risk Calculator is an interactive tool designed to help users estimate the total radiation dose an astronaut might receive during a mission and its associated cancer risk. It incorporates live space weather data and allows for the exploration of different mission durations and shielding strategies.

14.1.1 The Space Radiation Environment

Space radiation primarily consists of two major components:

- **Galactic Cosmic Rays (GCRs):** These are high-energy particles (mostly protons and heavier atomic nuclei) originating from outside our solar system, likely from supernovae and other violent astrophysical events. They are pervasive and penetrate spacecraft relatively easily due to their high energies.
- **Solar Particle Events (SPEs):** These are sudden, intense bursts of protons and other energetic particles ejected from the Sun during solar flares and coronal mass ejections (CMEs). SPEs can deliver very high doses in a short period, posing an immediate and severe threat.

Exposure to these forms of radiation can lead to various health effects, including increased lifetime cancer risk, damage to the central nervous system, degenerative tissue effects, and acute radiation sickness during high-dose events.

14.1.2 Key Factors in Radiation Risk Calculation

The calculator takes into account several critical parameters to estimate radiation risk:

14.1.2.1 Mission Duration:

- **Input:** Users define the mission length in "days" using a slider.
- **Impact:** The total radiation dose is directly proportional to the time spent in the radiation environment. Longer missions inherently accumulate higher doses.

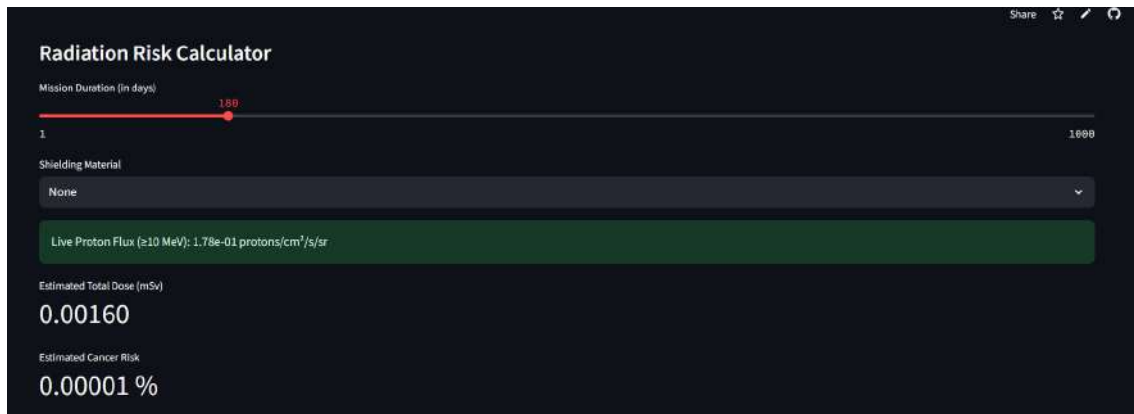


Figure 14.1.1: Radiation Risk Calculator Tab in Cosmic Dashboard

14.1.2.2 Live Proton Flux Data:

- **Source:** The calculator integrates real-time data from the National Oceanic and Atmospheric Administration (NOAA) Space Weather Prediction Center (SWPC), specifically utilizing proton flux measurements from the GOES (Geostationary Operational Environmental Satellite) satellites. These satellites, positioned in geosynchronous orbit, continuously monitor the space environment, including energetic particle fluxes.
- **Metric:** The integral-protons-3-day.json endpoint provides the latest proton flux data (specifically for protons with energies ≥ 10 MeV). These protons are a significant component of both GCRs and SPEs and are crucial for dose estimation.
- **Dynamic Calculation:** By fetching live data, the calculator provides a risk estimate based on current space weather conditions. In cases where live data cannot be retrieved, a default flux value is used to ensure functionality, though a warning is displayed.
- **Base Dose Rate:** The live proton flux is converted into a `base_dose_per_day` (mSv/day) using a predefined conversion factor, representing the unshielded daily dose in the given environment.

14.1.2.3 Shielding Material:

- **Input:** Users can select from various common spacecraft shielding materials: "None," "Aluminum," "Polyethylene," "Lead," "Water," "Titanium," "Carbon Fiber," and "Hydrogen-rich Plastic."
- **Shielding Factors:** Each material is associated with a `shield_factor`, representing its effectiveness in reducing radiation. A factor of 1.0 means no reduction (e.g., "None"), while lower factors (e.g., 0.3 for "Lead") indicate greater attenuation.
- **Mechanism:** Shielding works by absorbing or scattering incoming radiation. Materials rich in hydrogen (like Polyethylene or Water) are particularly effective against protons and GCRs because their light atomic nuclei efficiently scatter and slow down energetic charged particles. Denser materials like Lead are more effective against X-rays and gamma rays but can sometimes generate harmful secondary radiation when hit by very high-energy particles.

14.1.3 How the Calculator Estimates Risk

The calculator performs the following steps to provide its estimates:

- **Daily Dose Calculation:** The `base_dose_per_day` is multiplied by the `shield_factor` of the selected material to determine the `daily_dose` (mSv/day) with shielding.
- **Total Dose Accumulation:** The `daily_dose` is then multiplied by the `mission_days` to calculate the `total_dose` accumulated over the entire mission duration, measured in milliSieverts

(mSv). The Sievert (Sv) is the standard unit for effective dose, accounting for the biological effectiveness of different types of radiation.

- **Estimated Cancer Risk:** A simplified risk_percent is derived from the total_dose. This calculation is based on a general model where, for every 1000 mSv (1 Sv) of exposure, the estimated increase in lifetime fatal cancer risk is approximately 5%. It's important to note that this is a simplified, statistical estimate for illustrative purposes and actual individual risks can vary significantly based on many biological factors.

Visualizations for Enhanced Understanding

The tab includes two crucial plots generated using Matplotlib to help visualize the risk:

- **Dose Accumulation Over Time:** This line plot shows the linear increase of the Cumulative Dose (mSv) over Days of the mission. It visually emphasizes how radiation exposure is a cumulative problem, with longer missions directly leading to higher total doses.
- **Monte Carlo Simulation (1000 Astronauts):** This histogram displays a simulated distribution of Total Dose (mSv) across 1000 hypothetical astronauts for the given mission parameters. While the total_dose is a single estimated value, real-world dose exposure can vary slightly due to minor fluctuations in the radiation environment or individual shielding effectiveness. The Monte Carlo simulation illustrates this statistical spread around the average, showing a range of potential outcomes. It provides information on the probabilistic nature of radiation exposure.

Conclusion

The Radiation Risk Calculator serves as a valuable educational and preliminary planning tool, illustrating the critical interplay between mission duration, shielding, and the dynamic space radiation environment. By providing live data integration and interactive parameters, users can quickly grasp the implications of various choices on astronaut safety. Although a simplified model for complex biological and physical processes, it effectively highlights the ongoing challenges in protecting human explorers from the pervasive and dangerous effects of space radiation. Continuous research and development in advanced shielding and predictive space weather modeling remain essential to enable future long-duration human spaceflight.

14.2 Live Cosmic Ray Shower Map

The Live Cosmic Ray Shower Map is a critical feature of the Cosmic Radiation Research Dashboard, providing near real-time visualization of high-energy particle flux as recorded by the worldwide neutron monitor network (NMDB) [nmdb2010].

This interactive tool integrates several components:

14.2.1 Solar Imaging Integration

The dashboard fetches and displays the latest solar images from the **NOAA Space Weather Prediction Center (SWPC)**. We utilize the *LASCO-C2* and *LASCO-C3* coronagraph frames to visualize coronal mass ejections (CMEs) and solar eruptions, which are primary drivers of solar energetic particle (SEP) events. The dashboard auto-refreshes every 5 minutes to ensure the solar activity feed remains current.

- **LASCO-C2**: Focuses on the inner solar corona (2–6 $R_{\odot}$).
- **LASCO-C3**: Captures the extended corona (up to 30 $R_{\odot}$) for tracking CMEs.

These images provide an early indication of space weather events that may result in enhanced cosmic ray showers on Earth.

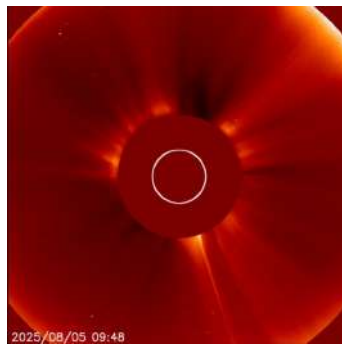


Figure 14.2.1: LASCO-C2



Figure 14.2.2: LASCO-C3

14.2.2 International Space Station (ISS) Tracking

To complement solar activity visualization, the dashboard incorporates a live map of the **International Space Station (ISS)** location. The ISS position is updated using the Open Notify API, allowing researchers to correlate geomagnetic events with spacecraft exposure.



Figure 14.2.3: ISS Live Tracker

14.2.3 Cosmic Ray Shower Visualization

The central feature of this tab is the **Cosmic Ray Shower Map**. It plots neutron count data from multiple ground-based observatories, sourced from the **NMDB (Neutron Monitor Database)**, including stations like:

- OULU (Finland)
- CALG (Canada)
- THUL (Greenland)
- INVK (Canada)
- JUNG (Switzerland)
- SOPB / SOPO (South Pole)

Intensity Classification Algorithm:

1. Count rates < 100 cps $\rightarrow$ **Low Intensity**
2. 100–200 cps $\rightarrow$ **Moderate Intensity**
3. > 200 cps $\rightarrow$ **High Intensity**

Color-coded circular markers are placed on the global map to reflect the detected event intensity in near real-time, enabling fast visual identification of geomagnetically significant events.

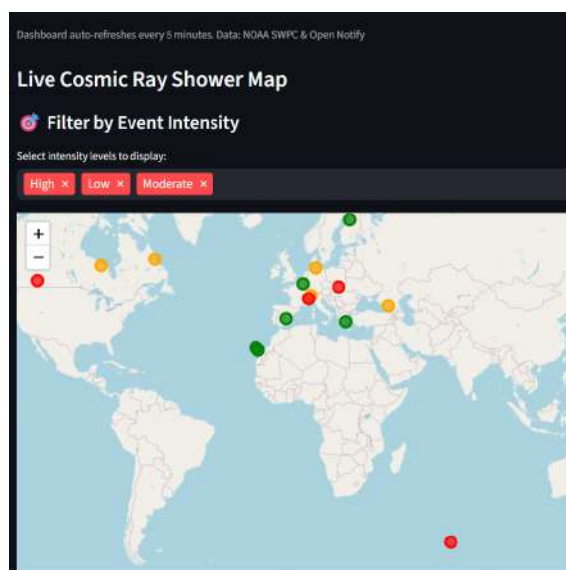


Figure 14.2.4: Live Cosmic Ray Showers

14.2.4 Acknowledgment of Data Sources

The following data sources are integrated in this module:

- **NOAA SWPC:** LASCO imagery and real-time solar wind alerts.
- **Open Notify API:** ISS position tracking.
- **NMDB Database:** Ground-based cosmic ray and neutron monitor counts.

This integration enables researchers to monitor both space weather drivers and their near-Earth effects in a single interactive interface.

```
# Age and Gender Inputs
age = st.slider("Select Age (Years)", 0, 100, 30)
gender = st.selectbox("Select Gender", ["Male", "Female", "Prefer not to say"])

# Duration Input (days)
days = st.slider("Select Duration (Days)", 0, 36500, 30)
```

Figure 14.3.1: user-defined parameters

14.3 Biological Effects Visualizer

This tab estimates the **biological impact of cosmic rays** on the human body, based on customizable parameters. It enables an interactive simulation of radiation exposure and its physiological consequences through graphical visualizations and dose-specific biological risk levels.

14.3.1 User-Defined Parameters

Users begin by specifying key characteristics of the human subject exposed to cosmic radiation:

- **Age (0–100 years):**
This parameter takes into account the fact that radiation sensitivity changes with age.
 - **Infants and children** are more vulnerable because their cells divide more rapidly and their immune and repair systems are not fully developed. Radiation exposure during growth stages can also lead to long-term developmental and genetic effects.
 - **Young adults** are more resilient with stronger repair systems.
 - **Older adults** may face increased damage due to declining cellular recovery and immune response.
- **Gender (Male / Female / NA):**
Gender influences biological effects of radiation due to differences in organ structure, hormone levels, and tissue composition.
 - DNA repair efficiency, body fat distribution, and hormonal cycles can also affect how radiation is absorbed and how damage is managed.
 - Females tend to be at a slightly higher risk of certain radiation-induced effects, particularly in breast tissue and reproductive organs, which are more radiosensitive.
 - Radiation risk models (like those from NASA and ICRP) often apply gender-specific weighting factors for organ damage and cancer risk.
- **Exposure Duration (0–365,000 days):**
The total time for which the subject is exposed to cosmic radiation, covering both short-term and long-term mission profiles.

These inputs together determine the cumulative radiation dose (in mSv) and help predict potential biological effects.

14.3.2 Cumulative Dose Calculation

Once the user sets the parameters (age, gender, and exposure duration), the app calculates the cumulative radiation dose the person would receive over the selected time. This is expressed in millisieverts (mSv) — a unit that reflects how much biological damage ionizing radiation can cause.

The app combines:

- **Cosmic ray flux data**(based on current or simulated space weather conditions)
- **Exposure duration** (number of days)
- **Radiation dose conversion factors**, to estimate the total biological dose received by the human model.

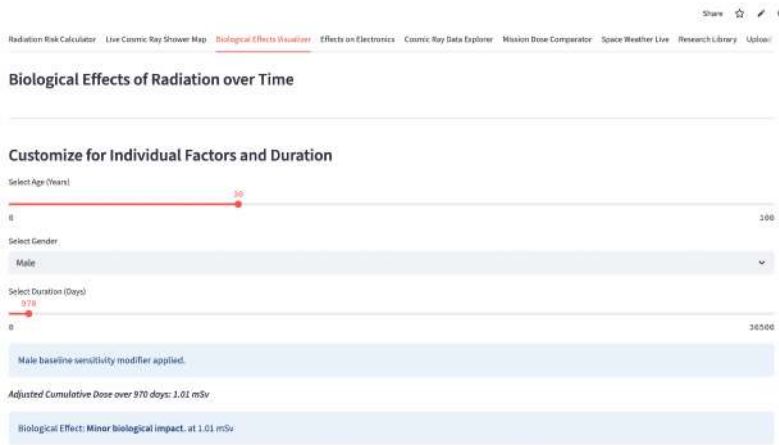


Figure 14.3.2: Dashboard Glimpse

Dose(mSv)	Biological Effect
<50	No Observable Effect
50 - 250	Slight blood changes possible
250 - 1000	Mild symptoms of ARS
1000 - 4000	Moderate to Severe ARS; medical attention required
>4000	Life threatening ARS; high fatality rate without immediate treatment

Table 14.3.1: Radiation Dosage vs Corresponding Biological Effect

This cumulative dose is then used to determine the severity of biological effects in the following sections.

14.3.3 Biological Effects Summary

Once the cumulative dose is calculated, the corresponding health effect severity is estimated. This is categorized using established thresholds: **What is ARS?** Acute Radiation Syndrome (ARS) is a collection of health effects that occur within hours to weeks of exposure to high doses of ionizing radiation. It affects rapidly dividing cells—primarily in the bone marrow, digestive system, and skin—and symptoms include nausea, fatigue, hair loss, bleeding, infection, and death at extreme exposures.

14.3.4 Visual Impact on Human Body

A **human body visualizer** is dynamically updated based on the input parameters: The redness or coloring intensity increases with higher dose levels, visually indicating biological stress. This allows an intuitive understanding of the radiation impact severity at a glance.

```
# Image loading: ensure correct path
script_dir = Path(__file__).parent
image_dir = script_dir / "images"
image_path = image_dir / img_file

try:
    st.image(str(image_path), caption=f"Impact Visualization ({adjusted_dose:.0f} mSv)", use_container_width=True)
    st.caption("Disclaimer: Conceptual illustration only.")
except Exception as e:
    st.error(f"Could not load image: {e}\nCheck that 'images' folder exists alongside this script and contains {img_file}.")

st.subheader("Interactive Risk Severity Chart")
```

Figure 14.3.3: Visual Impact

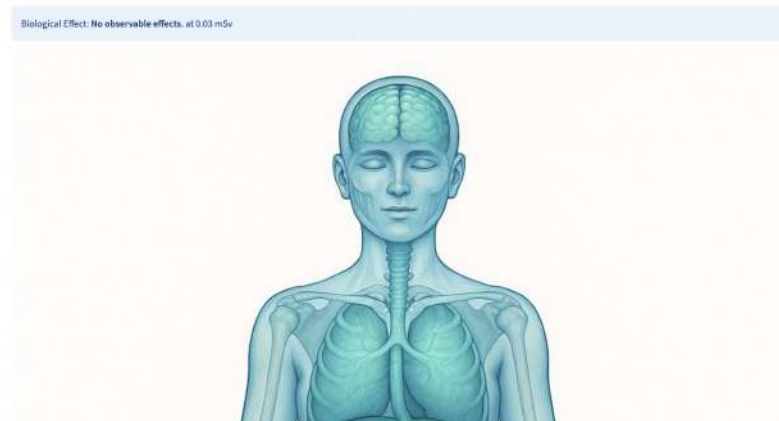


Figure 14.3.4: Visualizing the Effect

```
fig.add_trace(go.Scatter(
    x=[raw_dose], y=[0.5], mode='markers+text', name='Raw Dose',
    marker=dict(size=12), text=['Raw'], textposition='bottom center'
))
fig.add_trace(go.Scatter(
    x=[adjusted_dose], y=[0.5], mode='markers+text', name='Adjusted Dose',
    marker=dict(size=12), text=['Adjusted'], textposition='top center'
))

fig.update_layout(
    xaxis=dict(title="Dose (mSv)", range=[0, thresholds[-1]]),
    yaxis=dict(visible=False),
    title="Radiation Dose vs. Biological Risk",
    height=300, margin=dict(t=40, b=40), showlegend=True
)

st.plotly_chart(fig, use_container_width=True)
```

Figure 14.3.5: Radiation Dose vs Biological Risk Chart

14.3.5 Dose vs Biological Risk Graph

A visual chart plots radiation dose against biological risk level, with two reference markers:

- Raw Dose (unmodified exposure)
- Adjusted Dose (after applying biological sensitivity factors)

Chart Features:

- Interactive tooltips and legends
- Zoom and pan support
- Download as PNG
- Colored background zones indicating risk severity:
 - Green: None
 - Yellow: Minor
 - Orange: Mild ARS
 - Dark Orange: Severe ARS
 - Red: Lethal/Fatal

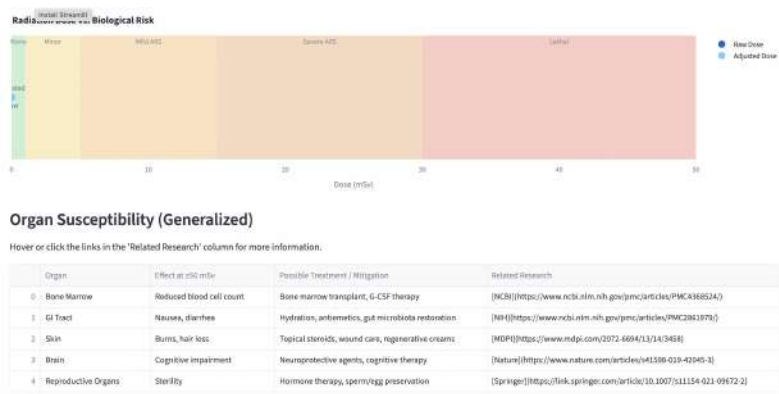


Figure 14.3.6: Chart

14.3.6 Organ-specific Susceptibility Table

This Section displays a reference table outlining key organs, such as the bone marrow and nervous system, affected by **high-dose cosmic radiation (>50 mSv)**, associated symptoms, treatments, and supporting scientific resources. Clickable links in the “Research” column provide access to current scientific literature and studies for users who want to explore the biological mechanisms and treatment options in detail.

14.4 Effect on Electronics

14.4.1 Overview

This Streamlit app estimates and visualizes the impact of cosmic radiation on electronic devices during space missions. It simulates **Single Event Upsets (SEUs)**—radiation-induced bit flips—based on mission parameters such as environment, duration, shielding, and electronics sensitivity. The tool is useful for spacecraft engineers, mission planners, and students to assess potential failure risks in electronics exposed to cosmic radiation.

14.4.2 Key Features

- **Mission Environment:** Choose the space region (e.g., ISS, Lunar Orbit, Mars Transit, Deep Space).
- **Mission Duration:** Specify how long the electronics are exposed to radiation (1 to 1000 days).
- **Shielding Level:** Determine the protection level—None, Light, or Heavy.
- **Electronics Sensitivity:** Reflects how vulnerable or radiation-hardened the components are.

14.4.3 Backend Calculations

Each mission profile has a base SEU rate, modified by electronics sensitivity and shielding effectiveness.

Adjusted SEU Rate = Base Rate $\times$ Sensitivity Factor $\times$ Shielding Factor

Total SEUs = Adjusted Rate $\times$ Mission Duration

These formulas provide an estimate of how many radiation-induced errors may occur over the mission.

14.4.4 Risk Analysis

Based on total SEUs:

- **Low Risk:** Fewer than 1 SEU
- **Moderate Risk:** Between 1 and 5 SEUs
- **High Risk:** More than 5 SEUs

These levels help assess whether the system is acceptable for mission-critical operations.

14.4.5 Visualizations

- **Bar Chart (SEU vs Shielding):** Shows impact of shielding levels.
- **Monte Carlo Simulation:** Displays variation in SEU outcomes across 1000 simulated devices.
- **Line Plot:** Illustrates how SEUs accumulate over time, day by day, throughout the mission.

14.4.6 Output Summary

Displays selected parameters, applied modifiers, estimated SEUs, and risk level. This summary acts as a quick-reference report for evaluating the survivability of electronics in space.

14.4.7 Use Cases

- Satellite and spacecraft electronics design evaluation
- CubeSat and student mission reliability planning
- Educational demonstrations of radiation effects in space environments
- Mission planning for Mars, Moon, or Deep Space operations

14.5 Cosmic Ray Data Explorer

The Cosmic Ray Data Explorer is an interactive tool designed to allow users to visualize and gain insights into the typical characteristics of cosmic ray spectra for various particle types and observational contexts, primarily using mock data to illustrate key principles.

14.5.1 Cosmic Ray Spectra: A Definition

A cosmic ray spectrum quantifies the flux of cosmic ray particles as a function of their energy. In simpler terms, it tells us how many particles of a given type arrive per unit area, per unit time, per solid angle, within a specific energy interval. These spectra are typically plotted on a log-log scale due to the vast range of energies and fluxes involved, often revealing power-law relationships, where flux decreases with increasing energy.

14.5.2 Factors Influencing Cosmic Ray Spectra

The observed spectrum of cosmic rays is influenced by several critical factors:

14.5.2.1 1. Particle Type:

- **Protons:** Being the most abundant cosmic ray particles, their spectrum often serves as a baseline. Their interaction with interstellar gas and magnetic fields shapes their observed distribution.
- **Helium Nuclei:** As the next most abundant, Helium nuclei (alpha particles) have distinct spectral shapes that can provide clues about different acceleration processes or propagation histories compared to protons. **Iron Nuclei:** Heavier nuclei like Iron are less abundant but are crucial tracers of nucleosynthesis in their source regions (e.g., supernovae). Their spectra can exhibit different slopes, indicating varied acceleration or transport efficiencies due to their higher charge-to-mass ratios.

2. Source and Location of Measurement:

- **AMS-02 (Alpha Magnetic Spectrometer-02):** A real-world example of an instrument operating in Low Earth Orbit (LEO) on the International Space Station. Data from such missions are affected by the Earth's geomagnetic field and the solar modulation (influence of the Sun's magnetic field and solar wind), which filter lower-energy cosmic rays.
- **Voyager 1:** A groundbreaking spacecraft that has entered interstellar space. Its measurements of cosmic rays are less affected by solar modulation, providing a more pristine view of the galactic cosmic ray spectrum beyond the heliosphere.

For demonstration purposes, the explorer utilizes simulated data that approximates the general shapes and trends observed in real cosmic ray spectra. This allows for immediate visualization without the complexities of processing large real datasets. In future developments, the integration of actual data from missions like AMS-02, Voyager, and comprehensive Cosmic Ray Databases (CRDB) is envisioned to provide more precise and up-to-date insights.

14.5.3 How the Data Explorer Tab Works

The Cosmic Ray Data Explorer tab in the Streamlit application provides an interactive interface to explore these concepts:

- **Select Data Source:** Users can choose between simulated data approximating observations from "AMS-02" (representing near-Earth space), "Voyager 1" (representing interstellar space), or generic "Mock Data".
- **Select Particle Type:** Users can select "Protons," "Helium Nuclei," or "Iron Nuclei" to see their corresponding mock spectra.
- **Dynamic Spectrum Generation:**

- The energy range is defined logarithmically, typical for cosmic ray analysis.
- The "flux" (number of particles per unit area per unit time per steradian per energy) is calculated based on simplified power-law functions, with different exponents applied for each particle type to reflect their characteristic spectral slopes.
- When "Mock Data" is selected, a small amount of random noise is added to the calculated flux. This simulates the inherent variability and measurement uncertainties present in real experimental data, making the simulation more realistic.
- **Log-Log Plotting:** The generated energy and flux values are plotted on a log-log scale. This standard representation effectively displays the wide dynamic range of cosmic ray phenomena and highlights the power-law nature of their spectra. The plot includes clear labels for "Energy (MeV)" and "Flux (particles/m²-s-sr-MeV)," a title indicating the selected data source, a grid for readability, and a legend.

14.5.4 Interpreting the Visualizations

The generated plots help users understand:

- **Spectral Shape:** Observe how the slope of the spectrum changes for different particles. Flatter slopes at lower energies or steeper drops can indicate different acceleration or propagation conditions.
- **Relative Abundance:** While mock data, the pre-factors in the flux calculation roughly reflect the decreasing abundance of heavier elements (e.g., protons are more abundant than helium, which are more abundant than iron).
- **Environmental Influence (Mocked):** The selection of "AMS-02" versus "Voyager 1" subtly implies the effects of solar modulation - typically, lower energy particles are more suppressed near Earth due to the Sun's magnetic field. The mock data aims to illustrate this general trend.

14.5.5 Conclusion

The Cosmic Ray Data Explorer serves as an accessible educational tool to visualize the fundamental characteristics of cosmic ray spectra. By interactively selecting different particle types and (mocked) data sources, users can develop an intuitive understanding of how these high-energy particles are distributed across energy ranges and the various factors that influence their observed properties. As future enhancements integrate real experimental data, this tool will offer even deeper insights into the complex and dynamic cosmic ray environment.

14.6 Mission Dose Comparator

The **Mission Dose Comparator** tab is designed to allow users to compare space radiation doses by integrating shielding methods for custom mission scenarios. It also benchmarks these against real-world cases, such as historical missions like the *ISS* or *Apollo*. Additionally, it supports Monte Carlo simulations and provides detailed analysis of organ-specific doses.

14.6.1 Shielding Configuration

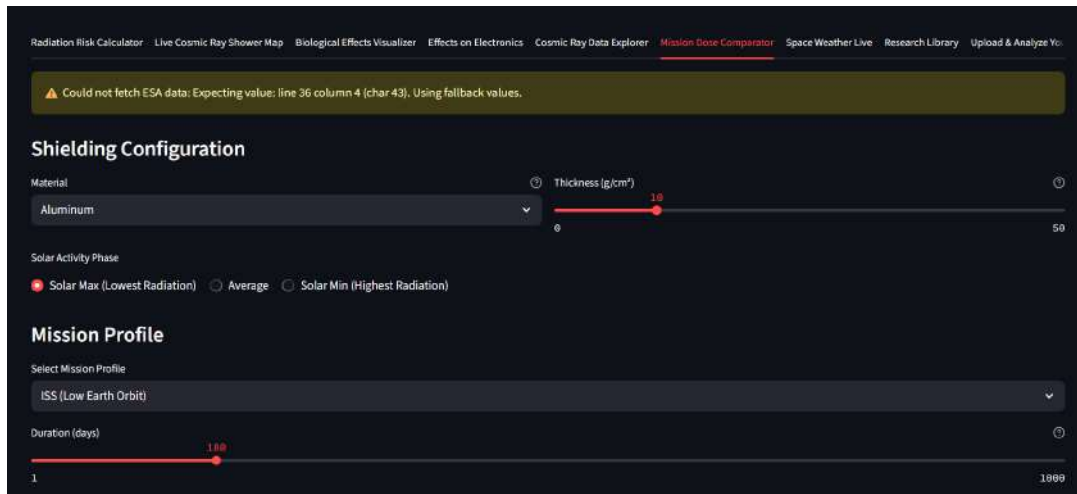


Figure 14.6.1

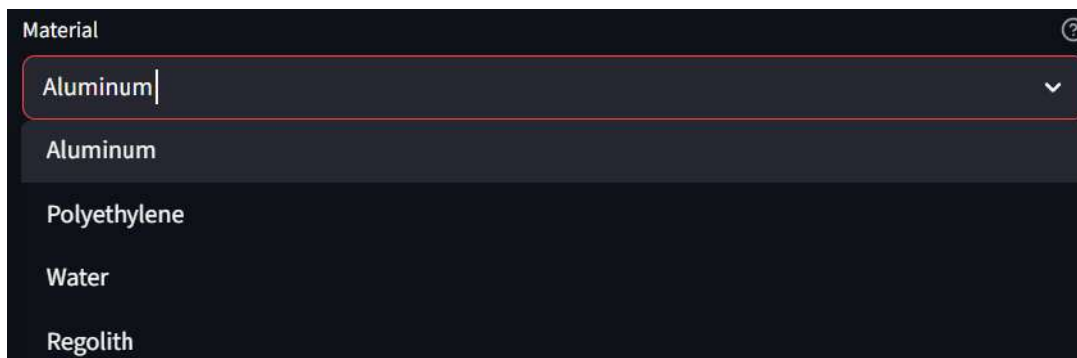


Figure 14.6.2

Shielding configuration directly alters the mission dose by reducing the amount of cosmic radiation that reaches astronauts during a space mission. The code calculates a shielding factor that reduces the base radiation dose depending on the chosen material and its thickness. Each material has an attenuation coefficient representing how effectively it blocks radiation:

Material	Attenuation Coefficient (per g/cm ²)
Aluminum	0.07
Polyethylene	0.05
Water	0.06
Regolith	0.04

The shielding factor is calculated using the exponential decay formula:

$$\text{shielding_factor} = e^{-\text{thickness} \times \text{attenuation_coefficient}}$$

This factor always lies between 0 and 1. Lower values indicate stronger shielding, meaning less radiation passes through the material.

Example

If the user selects Aluminum with a thickness of 10 g/cm² and an attenuation coefficient of 0.07 cm²/g, the calculation is:

$$\text{shielding_factor} = e^{-10 \times 0.07} = e^{-0.7} \approx 0.4966$$

This means that approximately 49.7% of the radiation penetrates through 10 g/cm² of aluminum, compared to 100% penetration with no shielding (i.e., a factor of 1).

14.6.2 Solar Phase Activity

The solar phase activity—such as Solar Maximum, Average, or Solar Minimum—affects the mission dose by altering the intensity of incoming cosmic radiation.

- **Solar Maximum (Solar Max):** During this phase, the Sun is more active. Ironically, the increased solar wind acts as a stronger shield, allowing fewer galactic cosmic rays to reach the spacecraft. As a result, astronauts receive a lower radiation dose. In the code, this is represented by a dose rate multiplier (e.g., 0.7).
- **Solar Minimum (Solar Min):** In this phase, solar activity is low, and the Sun's magnetic field is weaker, allowing more cosmic rays to penetrate the solar system. This leads to a higher radiation dose. The code uses a multiplier like 1.3.
- **Average:** This is an intermediate state with moderate cosmic ray exposure, represented by a multiplier of 1.0.

14.6.3 Mission Profile

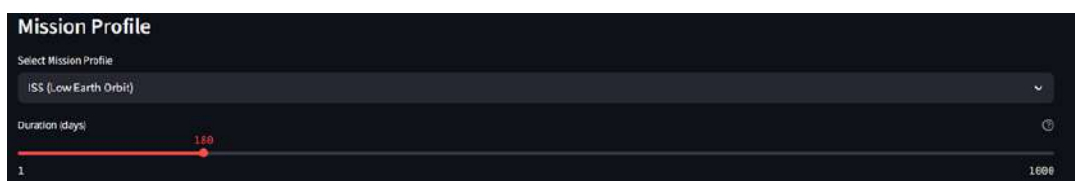


Figure 14.6.3

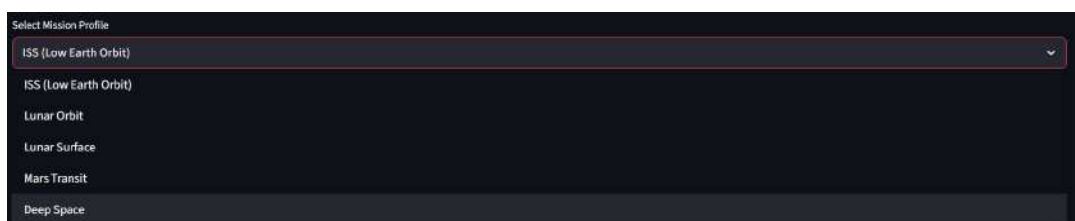


Figure 14.6.4

The code uses base dose rates for typical mission environments (in mSv/day):

Mission Profile	Base Dose Rate (mSv/day)
ISS (Low Earth Orbit)	<code>radiation_data["iss"]</code> (0.3)
Lunar Orbit	<code>radiation_data["lunar"]</code> (0.5)
Lunar Surface	$1.2 \times \text{radiation_data["lunar"]}$ (0.6)
Mars Transit	<code>radiation_data["mars_transit"]</code> (1.8)
Deep Space	<code>radiation_data["deep_space"]</code> (2.5)

Final dose rate and total dose calculation

The adjusted dose rate for a mission is calculated as:

$$\text{adjusted_dose_rate} = \text{base_dose_rate} \times \text{shielding_factor} \times \text{solar_modifier}$$

14.6.4 Organ-Specific Radiation Exposure

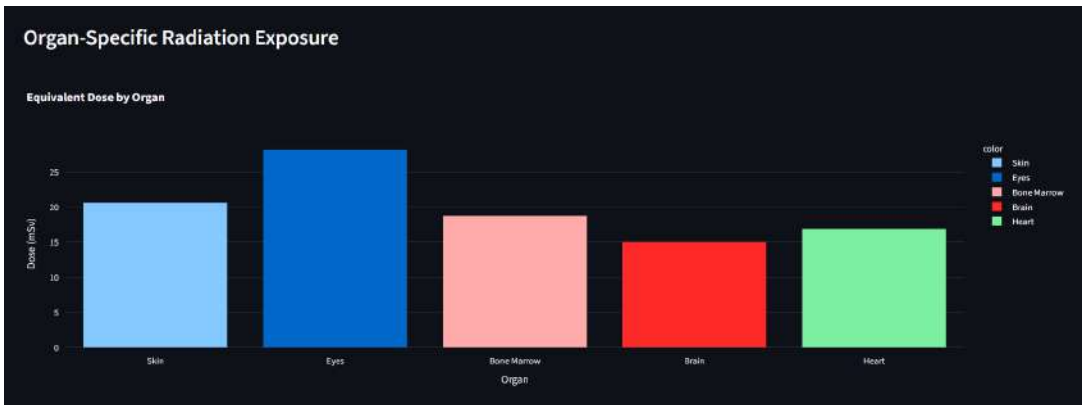


Figure 14.6.5

- The chart displays the radiation dose (in mSv) received by different organs during a simulated space mission.
- The X-axis lists organs: Skin, Eyes, Bone Marrow, Brain, and Heart.
- The Y-axis shows the equivalent radiation dose, indicating how much each organ is exposed to.
- This plot is generated based on user-selected mission parameters, such as mission environment, shielding, solar activity, and duration.
- The different dose levels for each organ are calculated using multipliers. Some organs—such as the Eyes and Skin—receive more radiation due to higher exposure or sensitivity.
- For example, in the chart, the Eyes receive the highest dose, followed by the Skin, Bone Marrow, Heart, and Brain.
- These values change dynamically if mission settings are adjusted (e.g., thicker shielding, different materials, longer or shorter durations, etc.).

14.6.5 Comparison with Real Missions and Risk Assessment

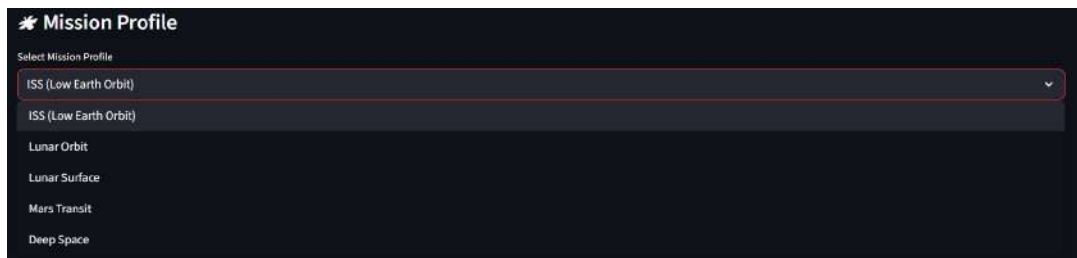


Figure 14.6.6

The dose values are compared with real missions that are pre-set in the system:

- **ISS:** ≈ 90 mSv
- **Apollo 14:** ≈ 1.1 mSv
- **Mars Curiosity:** ≈ 1200 mSv

Risk Assessment: Displayed as **SAFE**, since the value of 61.0 mSv is within permissible spaceflight limits (i.e., below 250 mSv for short missions).

14.6.6 Dose Uncertainty Simulation



Figure 14.6.7

This histogram shows the results of a Monte Carlo simulation built into the app to model variability in space weather conditions.

What the Code Does

- It simulates 1000 possible missions by randomly varying radiation dose inputs (e.g., daily dose or shielding response) using a normal distribution.
- The final dose for each simulated mission is calculated as:

$$\text{total_dose} = \text{simulated_daily_dose} \times \text{duration} \times \text{shielding_factor}$$

14.7 Space Weather Live

This tab displays real-time space weather. It takes live data from the National Oceanic and Atmospheric Administration (NOAA) and generates graphs for easy interpretation. It shows:

14.7.1 Real-Time Solar System Monitor

This feature lets us see our solar system in an interactive 3D view, powered by NASA's "Eyes on the Solar System." It shows the real-time positions of planets, moons, and spacecraft. You can navigate around to see different celestial bodies and their current locations. This is a live feed directly from NASA's official tool, providing accurate space data.

14.7.2 Proton Flux (≥ 10 MeV)

Proton flux refers to high-energy particles emitted by the Sun, particularly associated with solar flares or coronal mass ejections. The app takes the last 3 days of proton flux data for energy ≥ 10 MeV from NOAA's GOES (Geostationary Operational Environmental Satellite) primary satellite. It shows a line graph of proton flux over time. The Y-axis represents the flux, and the X-axis shows the time. Following the graph, one of two messages is displayed: Proton flux is at normal background levels – Flux is in normal safe ranges.

Elevated proton flux — possible solar event in progress – This message is displayed when flux exceeds 100 protons/cm²·s·sr, indicating a potentially significant solar event.

14.7.3 X-Ray Flux (Solar Flares)

X-ray flux measures the intensity of X-rays emitted by the Sun. Powerful bursts of X-rays are known as solar flares. The app uses the latest 3 days of X-ray flux data from NOAA's GOES primary satellite. The graph is a log-scale line graph of flux versus time. The Y-axis shows the flux, and the X-axis shows the time. Following the graph, one of two messages is displayed: No significant X-ray activity – The Sun's X-ray output is at normal levels.

Possible solar flare detected! – This warning is displayed if the X-ray flux exceeds 10^{-5} , indicating a potential solar flare.

14.7.4 Kp Index (Geomagnetic Activity)

The Kp index quantifies disturbances in Earth's magnetic field caused by solar wind. This app uses recent NOAA planetary K-index data. A line graph displays Kp index versus time. The Y-axis shows the Kp value, and the X-axis shows the time. This also shows one of two messages: Geomagnetic field is quiet (Kp = [latest Kp value]) – Displayed when Earth's magnetic field is relatively undisturbed.

Geomagnetic storm likely (Kp = [latest Kp value]) – This warning occurs when the Kp index is greater than 5, indicating a geomagnetic storm.

14.7.5 Solar Flare Activity Map

Real-time precise modeling of solar flare impact is complex and involves sophisticated models. So, this section provides a mock map to show the potential impact of solar flares. It displays a global map with several locations plotted using different colors — red, orange, and green dots representing high, medium, and low intensities of solar flares, respectively. Clicking on a dot reveals its intensity.

14.8 Research Library

This tab serves as a centralized repository of important scientific literature related to cosmic rays, biological effects of radiation, satellite system vulnerabilities, and atmospheric interactions. It is designed to give users quick access to essential papers and brief summaries which we used during

our summer project. The data used in this tab was manually collected from peer-reviewed scientific papers and technical reports available on reputable sources including:

- **arXiv**
- **ResearchGate**
- **Physics Journal India**
- **IOSR Journals**
- **K-Labs Avionics References**

Each entry includes the paper title, authors, publication year, a subject tag, and a summary.

Feature	Description
<code>st.subheader("Research Paper Library")</code>	Sets the section heading
<code>st.markdown()</code>	Provides a short intro about the tab's purpose
<code>pd.DataFrame({...})</code>	Contains metadata (Title, Author, Year, etc.) of papers
<code>st.selectbox("Filter by Tag", [...])</code>	Lets the user filter papers by topic (Atmosphere, Biology or Electronics)
<code>st.dataframe(filtered)</code>	Displays the filtered list of papers in a scrollable table
<code>filtered.iterrows()</code>	Shows full metadata: title, authors, year, summary, and link to the paper
<code>st.download_button()</code>	Simulates paper download functionality with a placeholder PDF

Table 14.8.1: Description of Streamlit Features Used in the Research Library Tab

```

tag = st.selectbox("Filter by Tag", ["All", "Atmosphere", "Biology", "Electronics"])
if tag != "All":
    filtered = papers[papers["tags"] == tag]
else:
    filtered = papers

st.dataframe(filtered)

st.markdown("### Paper Summaries")
for _, row in filtered.iterrows():
    st.write(f"***{row['Title']}***")
    st.write(f"Authors: {row['Authors']}")
    st.write(f"Year: {row['Year']}")
    st.write(f"Summary: {row['Summary']}")
    st.write(f"[Read Paper]({row['Link']})")
    st.write("----")

st.markdown("### Example Paper Download")
st.download_button(
    "Download Example Paper (PDF)",
    data=b"%PDF-1.4 ... (fake content)",
    file_name="example_paper.pdf",
    mime="application/pdf"
)

```

Figure 14.8.1

Visible Features on the Tab

1. Filter by Tag Dropdown:

- A selectbox lets the user filter papers by category.
- Options include: All, Atmosphere, Biology, Electronics.
- Helps focus the list on specific themes (e.g., only biological impact papers).

2. Interactive Data Table (Research Papers):

- Displays research paper metadata using a Pandas DataFrame inside `st.dataframe()`.
- Columns include:
 - **Title:** Full title of the research paper.
 - **Authors:** All authors in a compact format.
 - **Link:** Clickable URL pointing to the original paper or publication site.
 - **Year:** Year of publication.
 - **Tags:** Categorical tag to classify the research (Atmosphere, Biology, Electronics).

3. Dynamic Display: Paper Summaries Section

- Below the table, there's a dynamically updating section showing:
 - Title of the paper
 - Authors
 - Year of publication
 - Summary

14.9 Upload and Analyze Your Data

This tab allows us to upload a csv file and analyse the summary of cosmic ray or radiation data by displaying a summary statistics. Also it would allow to visualise the selected columns at the end of this tab.

Lets understand it by uploading a real time csv file data.

Taking five different temperatures of the star to analyse the data and variations in parameters are as follows -

- 1.) Luminosity(L) - It shows a direct relation with temperature as it is directly proportional to it and decreases with decrease in temperature.
- 2.) Radius(R) - A star's temperature and radius are directly related, with hotter stars generally having larger radii. This relationship is described by the Stefan-Boltzmann law, which states that a star's luminosity (energy output) is proportional to its surface area (related to radius squared) and the fourth power of its temperature.
- 3.) Absolute magnitude - Generally, hotter stars are more luminous (brighter) and thus have a higher absolute magnitude compared to cooler stars of similar size. This relationship is a fundamental aspect of [stellar classification](#) and is often visualized on the [Hertzsprung-Russell diagram](#).
- 4.) Spectral class - A star's spectral class is a classification based on its surface temperature and is determined by analyzing its spectrum. The main spectral classes, from hottest to coolest, are O, B, A, F, G, K, and M, with each class further divided into numbered subclasses (e.g., G0, G1, G2...G9).

With this, we completed analysing our first tab of data preview in a detailed manner.

Then we have tried to show separately, variations of each parameters discussed (in the X axis) with count(in the Y axis) using histograms.

Coming to the ending part of this tab, we can see a custom plot builder which allows to relate among the above parameters by placing it on both X and Y axes as per our choices.

Let's see the code snippet used to build the custom plot builder:

```
[language=Python, caption={Streamlit Custom Plot Builder}]
st.markdown("### Custom Plot Builder")

x_axis = st.selectbox("Select X-axis", options=["None"] + numeric_cols)
y_axis = st.selectbox("Select Y-axis", options=numeric_cols)

if x_axis != "None":
    fig = px.scatter(df, x=x_axis, y=y_axis,
                    title=f"{y_axis} vs {x_axis}")
    st.plotly_chart(fig, use_container_width=True)
else:
    st.line_chart(df[y_axis], use_container_width=True)

else:
    st.warning("No numeric columns found to analyze!")
except Exception as e:
    st.error(f"Error reading file: {e}")
```



```

st.markdown("### Custom Plot Builder")
x_axis = st.selectbox("Select X-axis", options=["None"] + numeric_cols)
y_axis = st.selectbox("Select Y-axis", options=numeric_cols)
if x_axis != "None":
    fig = px.scatter(df, x=x_axis, y=y_axis, title=f"{y_axis} vs {x_axis}")
    st.plotly_chart(fig, use_container_width=True)
else:
    st.line_chart(df[y_axis], use_container_width=True)
else:
    st.warning("No numeric columns found to analyze!")
except Exception as e:
    st.error(f"Error reading file: {e}")

FOOTER
st.markdown(
    """
    <div class="css-1q8dd3e">
        Built by the Impostor Among Us | Last updated: {date}
    </div>
    """.format(date=datetime.datetime.now().strftime("%B %d, %Y")),
    unsafe_allow_html=True

```

Figure 14.9.1: Code Snippet



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